

Classical maximum principles for second elliptic equations

$\Omega \subset \mathbb{R}^n$ is domain (open & connected)

$x \in \mathbb{R}^n$, $x = (x_1, x_2, \dots, x_n)$

$u = u(x)$

consider second order partial differential operator

$$Lu = \sum_{i,j=1}^n a_{ij}(x) \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i(x) \frac{\partial u}{\partial x_i} + c(x)u, \quad x \in \Omega$$

where $(a_{ij}(x))_{n \times n}$ is symmetric (W.L.O.G.)

$$\begin{aligned} a_{12} &= (2x) \cdot \frac{\partial^2 u}{\partial x \partial y} + (e^x) \cdot \frac{\partial^2 u}{\partial y \partial x} \\ &= \frac{2x+e^x}{2} \cdot \frac{\partial^2 u}{\partial x \partial y} + \frac{2x+e^x}{2} \cdot \frac{\partial^2 u}{\partial y \partial x} \end{aligned}$$

Def We say L is elliptic in Ω if matrix $(a_{ij}(x))_{n \times n}$ is positive-definite, $\forall x \in \Omega$

" $(a_{ij}(x))_{n \times n} > 0$ on Ω "

• We say L is strictly/uniformly elliptic on Ω if $\exists$ const. $\lambda_0 > 0$ s.t.

$$(a_{ij}(x))_{n \times n} \geq \lambda_0 I_{n \times n}, \quad \forall x \in \Omega$$

i.e. $(a_{ij}(x))_{n \times n} - \lambda_0 I$ is non-negative-definite

i.e. $\forall \xi = (\xi_1, \dots, \xi_n)^T \in \mathbb{R}^n$

$$\xi^T (a_{ij}(x))_{n \times n} \xi \geq \lambda_0 |\xi|^2 \Leftrightarrow$$

$$\begin{aligned}
 a_{11}(x) &= 1 + u_y^2, \quad a_{12} = a_{21} = -u_x u_y, \quad a_{22} = 1 + u_x^2 \\
 \xi^T (a_{ij})_{2 \times 2} \xi &= (\xi_1, \xi_2) \begin{pmatrix} 1 + u_y^2 & -u_x u_y \\ -u_x u_y & 1 + u_x^2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \\
 &= (1 + u_y^2) \xi_1^2 - 2u_x u_y \xi_1 \xi_2 + (1 + u_x^2) \xi_2^2 \\
 &= |\xi|^2 + (u_y \xi_1 - u_x \xi_2)^2 \\
 &\Rightarrow |\xi|^2 \Rightarrow \odot \text{ is strictly elliptic on } \Omega
 \end{aligned}$$

4. Heat eq. $u(x, t)$ = temperature at x at time t

$$\frac{\partial u}{\partial t} - (k \Delta u = f(x, t))$$

thermal conductivity

Suppose $f(x, t) = f(x)$ $\Rightarrow k \Delta u(x) = f(x)$
 Steady state $u = u(x)$ Poisson eq.

5. Reaction-diffusion eq

$u(x, t)$ = density function of a substance

$$\frac{\partial u}{\partial t} - (k \Delta u = f(x, t))$$

diffusion coeff creation - diffusion rate degradation

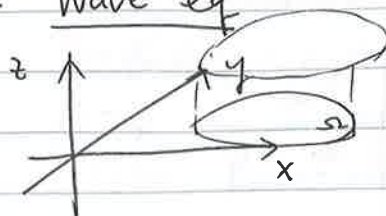
Steady state eq $-k \Delta u = f(x)$
 $u(x, t) \xrightarrow{t \rightarrow \infty}$ steady state
 "usually"

$$\frac{\partial u}{\partial t} - k \Delta u + \nabla \cdot (u \vec{v}) = f(x, t)$$

flux

$\vec{v}$ = velocity vector field. $-k \Delta u + \nabla \cdot (u \vec{v}) = f(x)$

6. Wave eq



$$z = u(x, y)$$

$$u_{tt} - k^2 \Delta u = f(x, t)$$

$$\text{steady state eq} \Rightarrow -k^2 \Delta u = f(x)$$

7. $\vec{v}(x)$ = velocity vector field of a fluid in $\Omega \subset \mathbb{R}^3$ • Suppose fluid irrotational, i.e.

$$\nabla \times \vec{v} = 0 \Rightarrow \vec{v} = \nabla \varphi$$

for some φ (scalar)
potential• Suppose fluid incompressible, i.e.

$$\nabla \cdot \vec{v} = 0$$

$$\Rightarrow \nabla \cdot (\nabla \varphi) = 0$$

$$\Delta \varphi = 0 \quad \text{on } \Omega$$

$$\varphi = 0 \quad \text{on } \partial\Omega$$

$$\frac{\partial u}{\partial n} = 0 \quad \text{on } \partial\Omega$$

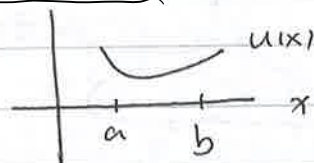
The weak maximum principle

The baby example $Lu = u''$, $\Omega = (a, b)$

Suppose $Lu \geq 0$ on (a, b)

$\Downarrow$

$$\max_{\bar{\Omega}} u(x) = \max_{\partial\Omega} u(x)$$



Weak max principle ($c \equiv 0$)

$$Lu = a_{ij}(x) \frac{\partial^2 u}{\partial x_i \partial x_j} + b_i(x) \frac{\partial u}{\partial x_i} + c(x)u, \quad x \in \Omega \subset \mathbb{R}^n$$

Assume

• Ω is bdd

• L is strictly elliptic on Ω

• b_i 's bdd on Ω

✓ If $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfies $Lu \geq 0$ in Ω

then $\max_{\bar{\Omega}} u = \max_{\partial\Omega} u$

RK If " $Lu \geq 0$ " is replaced by " $Lu \leq 0$ "

$$\Rightarrow \min_{\bar{\Omega}} u = \min_{\partial\Omega} u$$

Proof WMP ($c \equiv 0$)

~~Ω open~~

Special case $Lu > 0$ on Ω

• If $\max_{\bar{\Omega}} u$ is achieved on $\partial\Omega$, then done.

• So assume $\max_{\bar{\Omega}} u$ is attained at $x_0 \in \Omega$

$$\text{then } \begin{cases} \nabla u(x_0) = \vec{0} \end{cases}$$

first derivative test

$$H \leftarrow - \left(u_{x_i x_j}(x_0) \right)_{n \times n} \leq 0 \quad \text{2nd, } \dots$$

Hessian

Now observe:

$$0 \leq Lu(x_0) = a_{11}(x_0) u_{x_1} x_1(x_0) + b_1(x_0) u_{x_1}(x_0)$$

$$= \text{trace}(AH)$$

Take $P_{n \times n}$ orthogonal s.t.

$$P^T A P = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$\lambda_i > 0$ b/c A is > 0

$$= \text{trace}(P^T A H P) \leq 0 \quad \text{b/c } H \leq 0$$

$$= \text{trace}(P^T A P P^T H P) \quad \tilde{H} \quad \tilde{H}_{ii} \leq 0$$

$$= \text{trace} \left(\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \tilde{H} \right)$$

$$= \text{trace} \left(\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} \tilde{h}_{11} & \tilde{h}_{12} & \dots & \tilde{h}_{1n} \\ \vdots & & & \vdots \\ \tilde{h}_{n1} & \dots & \dots & \tilde{h}_{nn} \end{pmatrix} \right)$$

$$= \lambda_1 \tilde{h}_{11} + \lambda_2 \tilde{h}_{22} + \dots + \lambda_n \tilde{h}_{nn} \leq 0$$

$\Rightarrow$ contradiction!

So $\max_{\bar{\Omega}} u$ must be attained on $\partial\Omega$

$\Rightarrow$ done for special case

Now general case,

Let $v(x) = e^{rx_1}$, r const. to be chosen

$$Lv(x) = a_{11}(x) r^2 e^{rx_1} + b_1 r e^{rx_1}$$

$$= (a_{11}(x) r^2 + b_1 r) e^{rx_1}$$

$$\geq (\lambda_0 r^2 - m r) e^{rx_1} > 0 \quad \text{on } \Omega \quad \text{if } r \text{ large}$$

(b/c b_1 is bdd)

$\forall \epsilon > 0$, let $w = u + \epsilon v$

$$\Rightarrow Lw = Lu + \epsilon Lv \geq \epsilon Lv > 0 \quad \text{on } \Omega$$

By special case, $\max_{\bar{\Omega}} w = \max_{\partial\Omega} w$

Sending $\epsilon \rightarrow 0$, we are done.

2.17

Weak maximum principle ($c \equiv 0$)bdd
domain: open
connected

$$Lu = a_{ij}(x) \frac{\partial^2 u}{\partial x_i \partial x_j} + b_i(x) \frac{\partial u}{\partial x_i} + c(x)u, \quad x \in \Omega \subset \mathbb{R}^n$$

(Einstein convention)

 $\lambda_0 > 0$ const.

Suppose L is strictly elliptic on Ω ($(a_{ij}(x))_{n \times n} \geq \lambda_0 I$)
& b_i 's are bdd on Ω .
in Ω

Let $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfy $Lu \geq 0$ in Ω .

Then $\max_{\bar{\Omega}} u = \max_{\partial\Omega} u$

$$\left(\text{if } Lu \leq 0 \Rightarrow \min_{\bar{\Omega}} u = \min_{\partial\Omega} u \right)$$

WMP ($c \leq 0$) "good"

Assume the same condition on L & b_i 's & u

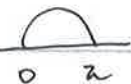
Then $\max_{\bar{\Omega}} u \leq \max_{\partial\Omega} u^+$

$$u^+ = \max(0, u) \geq 0$$

$$u^- = \min(0, u) \leq 0$$

Ⓚ1 This is Not true if " $c \leq 0$ " is not satisfied

$$Lu = u'' + u, \quad \Omega = (0, \pi)$$



$$u = \sin x \Rightarrow Lu = 0$$

$$\max_{\bar{\Omega}} u = 1, \quad \max_{\partial\Omega} u^+ = 0$$

Ⓚ2 Can't replace " u^+ " by " u "

" $\leq$ " can't be replaced by " $=$ "

$$Lu = u'' - u, \quad \Omega = (-1, 1) \quad \text{let } u = -x^2 + 100$$

$$(c = -1)$$

$$\Rightarrow Lu = -2 + x^2 + 100 \geq 98 > 0 \text{ on } \Omega$$

$$\text{But } \max_{\bar{\Omega}} u = -100, \quad \max_{\partial\Omega} u = -101, \quad \max_{\partial\Omega} u^+ = 0$$

RK3 If " $Lu \geq 0$ " is replaced by " $Lu \leq 0$ "
then let $v = -u \Rightarrow Lv \geq 0 \Rightarrow$ WMP ($c \leq 0$)

$$\begin{aligned} \max_{\bar{\Omega}} u &\leq \max_{\partial\Omega} u^+ \\ &\parallel \\ -\min_{\bar{\Omega}} v &\leq \max_{\partial\Omega} (-u^-) = -\min_{\partial\Omega} u^- \end{aligned}$$

$$\min_{\bar{\Omega}} u \geq \min_{\partial\Omega} u^-$$

(RK4) If $Lu = 0$ in Ω , then
 $\max_{\bar{\Omega}} |u| = \max_{\partial\Omega} |u|$

why : case 1 $\max_{\bar{\Omega}} |u| = \max_{\bar{\Omega}} u$ $\downarrow$ WMP ($c \leq 0$)

$$\max_{\partial\Omega} |u| \leq \max_{\bar{\Omega}} |u| = \max_{\bar{\Omega}} u \leq \max_{\partial\Omega} u^+ \leq \max_{\partial\Omega} |u|$$

case 2 $\max_{\bar{\Omega}} |u| = -\min_{\bar{\Omega}} u$

$$\begin{aligned} \max_{\partial\Omega} |u| &\leq \max_{\bar{\Omega}} |u| = -\min_{\bar{\Omega}} u \leq -\min_{\partial\Omega} u^- = \max_{\partial\Omega} (-u^-) \\ &\leq \max_{\partial\Omega} |u| \end{aligned}$$

Proof of WMP ($c \leq 0$)

let $\Omega^+ = \{x \in \Omega \mid u(x) \geq 0\}$

- If Ω^+ is empty, then $u \leq 0$ on $\Omega \Rightarrow$ done
- So suppose $\Omega^+ \neq \emptyset$

Observe on $\bar{\Omega}^+$

$$0 \leq Lu = \underbrace{a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j}}_{Lu} + b_i \frac{\partial u}{\partial x_i} + \cancel{c(x)u} \quad \ominus$$

$$\Rightarrow Lu > 0 \text{ on } \Omega^+ \implies \text{WPM (} c \leq 0 \text{)} \text{ applied to } \Omega^+$$

$$\max_{\bar{\Omega}^+} u = \max_{\partial\Omega^+} u$$

Case 1 $\partial\Omega^+ \subset \Omega \Rightarrow u|_{\partial\Omega^+} \equiv 0 \Rightarrow \max_{\bar{\Omega}^+} u = \max_{\partial\Omega^+} u = 0$



impossible

Case 2 $\partial\Omega^+ \cap \partial\Omega \neq \emptyset \Rightarrow \max_{\partial\Omega^+} u = \max_{\partial\Omega} u$



WPM
($c \leq 0$)

$$\max_{\bar{\Omega}^+} u \geq \max_{\bar{\Omega}} u$$

C.P.

Comparison principle Assume Ω is bdd, L is strictly elliptic on Ω , b_i 's bdd on Ω with $\underline{c} \leq 0$ on Ω
 if $u, v \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfy

$$\begin{cases} Lu \leq Lv & \text{on } \Omega \\ u|_{\partial\Omega} \geq v|_{\partial\Omega} \end{cases}$$

then $u \geq v$ on Ω

Proof $Lv \geq Lu \Rightarrow L(v-u) \geq 0$ on Ω

By WMP ($c \leq 0$), we have $\stackrel{=0}{\Rightarrow}$

$$\max_{\bar{\Omega}} (v-u) \leq \max_{\partial\Omega} (v-u)^+ = 0$$

$$\Rightarrow v-u \leq 0 \quad \text{QED}$$

Corollary Assume all conditions on L , b , & c as in C.P. The the Dirichlet boundary value problem

$$\begin{cases} Lu = f(x) & x \in \Omega \\ u|_{\partial\Omega} = \phi(x) & x \in \partial\Omega \end{cases}$$

has at most one solution $\in C^2(\Omega) \cap C^0(\bar{\Omega})$

Proof: u_1, u_2 two sol
By C.P. $\Rightarrow u_1 = u_2$
QED.

More application of WMP

1. (DBVP) $\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$ bold

where $f \in C^1(\mathbb{R})$ & $\downarrow$ in u , i.e. $f'(u) \leq 0$, $\forall u \in \mathbb{R}$

Then (DBVP) has at most one sol $\in C^2(\Omega) \cap C^0(\bar{\Omega})$

RK if we don't have " $f'(u) \leq 0$ ", then conclusion may not be true $f(u) = u$

$$\begin{cases} u'' + u = 0 & x \in (0, \pi) \\ u(0) = 0 = u(\pi) \end{cases} \quad u(x) = c \sin x$$

Proof: Suppose $u_1, u_2 \in C^2(\Omega) \cap C^0(\bar{\Omega})$ are sol's

$$\Delta u_1 + f(u_1) = 0$$

$$\Delta u_2 + f(u_2) = 0$$

$$\Delta(u_1 - u_2) + \underbrace{f(u_1) - f(u_2)}_{u_1 - u_2} (u_1 - u_2) = 0, \quad u_1 - u_2|_{\partial\Omega} = 0$$

$$"C(x)" = f'(\xi), \quad \xi \in (u_1, u_2)$$

$$C(x) \leq 0$$

$$\text{b/c. } f'(u) \leq 0$$

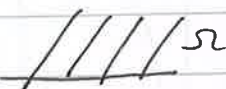
2.22.

Q: What if Ω is unbdd?Complicated: $\Omega = \{(x, y) \in \mathbb{R}_+^2 \mid y > 0\}$

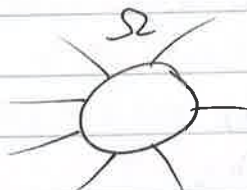
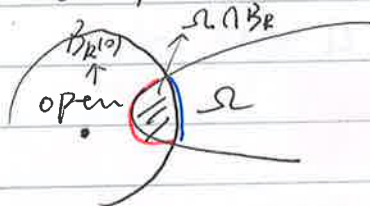
$$u(x, y) = y, \quad \Delta u = 0 \text{ on } \Omega$$

$$\max_{\bar{\Omega}} u = \infty$$

$$\max_{\partial\Omega} u = 0$$



However, if we interpret $\bar{\Omega}$ & $\partial\Omega$ so that they include " ∞ ", then WMP holds.



Theorem

Suppose Ω is unbdd, and $\forall$ large $R > 0$, L is strictly elliptic on $\Omega \cap B_R(0)$ & b's bdd on $\bar{\Omega \cap B_R(0)}$. Assume $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfies $Lu \geq 0$ on Ω & $u(\infty) \stackrel{\text{def}}{=} \lim_{\substack{x \rightarrow \infty \\ x \in \Omega}} u(x)$ exists (may be $\pm \infty$)

Then

$$(i) \quad \sup_{\bar{\Omega}} u = \max(\sup_{\partial\Omega} u, u(\infty)), \text{ if } C \geq 0$$

$$(ii) \quad \sup_{\bar{\Omega}} u \leq \max(\sup_{\partial\Omega} u^+, u(\infty)), \text{ if } C \leq 0 \text{ and}$$

Proof of (i) By WMP ($C \geq 0$) applied to L on $\Omega \cap B_R(0)$

$$\Rightarrow \max_{\bar{\Omega \cap B_R(0)}} u = \max_{\partial(\Omega \cap B_R(0))} u =$$

$$\Rightarrow \max_{\Omega \cap B_R(0)} u = \max \left(\max_{(\partial\Omega) \cap B_R} u, \max_{(\partial B_R) \cap \Omega} u \right)$$

$$\downarrow \text{ send } R \rightarrow \infty \quad \downarrow \quad \downarrow$$

$$\sup_{\bar{\Omega}} u \quad \sup_{\partial\Omega} u \quad u(\infty)$$

done!

3.3) WMP ($c \leq 0$) $\Rightarrow$

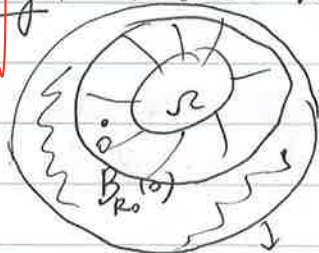
$$\max_{\Omega \cap B_R} u \leq \max_{\partial(\Omega \cap B_R)} u^+ = \max \left(\max_{(\partial\Omega) \cap B_R} u^+, \max_{(\partial B_R) \cap \Omega} u^+ \right)$$

Sending $R \rightarrow \infty$, we have

$$\sup_{\bar{\Omega}} u \leq \max \left(\sup_{\partial\Omega} u^+, (u(\infty))^+ \right)$$



Let Ω be bdd in $\mathbb{R}^n$, $n \geq 3$



Suppose $\Delta u = 0$ in Ω^c

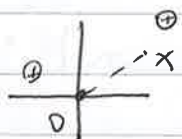
$\lim_{x \rightarrow \infty} u(x) = 0$ 在无穷远处有 decay

Then $|u(x)| \leq C/|x|^{n-2}$, $\forall x \in \Omega^c$ pos. const.

harmonic equ. 的解的 decay rate 和 fundamental sol 一样

$$-\Delta G(x) = \delta(x), \quad x \in \mathbb{R}^n$$

$$G(x) = \frac{1}{|x|^{n-2} C_n}, \quad n \geq 3$$



$$-\Delta u(x) = f(x), \quad x \in \mathbb{R}^n$$

$$u(x) = G * f$$

$$= \int_{\mathbb{R}^n} G(x-y) f(y) dy$$

Proof: claim $\Delta \left(\frac{1}{|x|^{n-2}} \right) = 0$, $x \neq 0$

(Fact: if $u = u(r)$, $r = |x|$, then
 $\Delta u = u_{rr} + \frac{n-1}{r} u_r$)

Now take large R_0 s.t. $B_{R_0} \supset \bar{\Omega}$

let $C = R_0^{n-2} \left(\max_{\partial B_{R_0}} (u(x)) + 1 \right)$

let $w(x) = \pm u(x) + \frac{C}{|x|^{n-2}}$

$\Rightarrow \cdot \Delta w = 0$, $R_0 < |x| \Rightarrow \Delta(-w) = 0$

$\cdot w|_{\partial B_{R_0}} \geq 1$

Now by WMP ($C=0$, unbd'd Ω case) $\rightarrow 0$

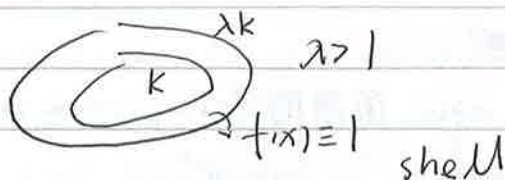
$\Rightarrow \sup_{|x| \geq R_0} (-w) = \max \left(\sup_{|x|=R_0} (-w), \underbrace{(-w)_{\infty}}_{\leq 0} \right) \leq 0$

$\Rightarrow -w \leq 0 \Rightarrow w \geq 0$ for $|x| \geq R_0$

$\Downarrow$
 $\frac{C}{|x|^{n-2}} \geq \mp u(x) \dots \quad |x| \geq R_0$

$|u(x)| \leq \frac{C}{|x|^{n-2}}, |x| \geq R_0$

Still true if $x \in \Omega^c$ & C take large
QED



Define $w(x) = \iint_{\lambda K \setminus K} G(x-y) \cdot 1 \, dy$

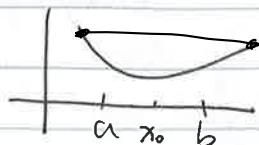
$\Rightarrow u(x) \equiv \text{const. on } K$

The strong max. principle

Baby example

$$Lu = u'', \quad \Omega = (a, b)$$

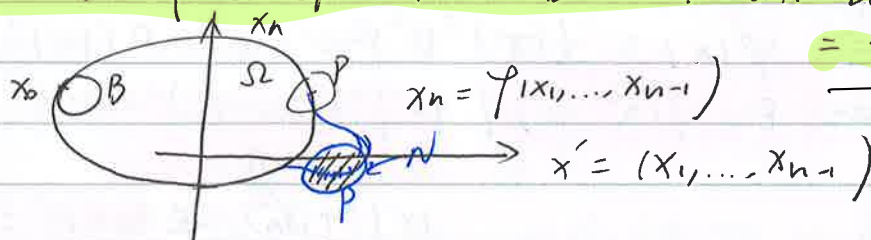
Assume $u'' \geq 0$



$\Rightarrow$ u attains max in (a, b) at x_0
 $u \equiv \text{const.} = u(x_0)$

Smoothness solution on $\partial\Omega$

Def We say Ω satisfies interior sphere condition at $x_0 \in \partial\Omega$ if $\exists$ open ball $B \subset \Omega$ s.t. $\partial B \cap \partial\Omega = \{x_0\}$



Fact: If $\partial\Omega \in C^2$ smooth, then Ω satisfies interior sphere at every point on $\partial\Omega$

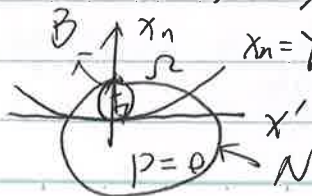
Def: We say $\partial\Omega$ is C^m -smooth ($m \geq 0$), if $\forall p \in \partial\Omega$ $\exists$ neighborhood N of p and a shift and rotation of $(x_1, \dots, x_n)$ coordinate system s.t.

$(\partial\Omega) \cap N$ is on the graph of $x_n = \varphi(x')$ for some $\varphi \in C^m(\mathbb{R}^{n-1})$

$\Omega \cap N = \{(x', x_n) \in N \mid x_n > \varphi(x')\}$

Proof of Fact: WLOG, assume $P = \text{origin}$

x_n -axis is the inner normal direction of $\partial\Omega$ at p



$$\Rightarrow \varphi(0') = 0$$

$$\nabla_{x'} \varphi(0') = 0'$$



Taylor Theorem, for $x' \approx \vec{0}'$

$$\varphi(x') = \varphi(\vec{0}') + x' \cdot \nabla \varphi(\vec{0}') + \frac{1}{2} (x')^T D^2 \varphi(\vec{0}') x' + o(|x'|^2)$$

$$D^2 \varphi(\vec{0}') = \begin{pmatrix} \varphi_{x_1 x_1}, & \varphi_{x_1 x_2}, & \dots & \varphi_{x_1 x_{n-1}} \\ \vdots & & & \vdots \\ \varphi_{x_{n-1} x_1}, & \dots & \dots & \varphi_{x_{n-1} x_{n-1}} \end{pmatrix}$$

Hessian

Eigenvalue of $D^2 \varphi(\vec{0}')$ are called principal curvatures of $\partial \Omega$ at P . Now fix const. $C > \text{principal curvatures of } \partial \Omega \text{ at } P$. $\Rightarrow (x')^T D^2 \varphi(\vec{0}') x' < C |x'|^2$

$$\Rightarrow \varphi(x') = \frac{1}{2} (x')^T D^2 \varphi(\vec{0}') x' + o(|x'|^2) \leq \frac{C}{2} |x'|^2, x' \approx 0$$

$$\text{Let } B = \{(x', x_n) \mid |x'|^2 + (x_n - R)^2 < R^2\}$$

$$\Downarrow \\ |x'|^2 + \underbrace{(x_n)^2}_{\substack{\uparrow \\ R^4}} - 2R x_n + R^2 < R^2 \Rightarrow x_n > \frac{|x'|^2}{2R}$$

If R is small, then

- $B \subset N$
- If $(x', x_n) \in B$, then $x_n > \frac{|x'|^2}{2R} \stackrel{R^4}{\geq} \frac{C}{2} |x'|^2 \geq \varphi(x')$

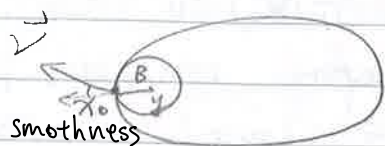
Then $B \subset N$ and B stays above graph of φ , and so $B \subset \Omega$.

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A smoothness assumption on $\partial\Omega$: interior sphere condition

Def Suppose Ω satisfies interior sphere condition at x_0 with y being the center of sphere. A vector v is said to be outward pointing at x_0 if $v \cdot (x_0 - y) > 0$



i.e. $\exists$ nbhd N of x_0 s.t.
 $u(x_0) > u(x), \forall x \in N \cap \Omega$

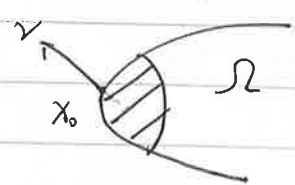
Def. wmp 设置 bdd

Hopf boundary point Lemma

Assume L is strictly elliptic on Ω , any, bc & c bdd on Ω :
 $|a_{ij}|, \sqrt{b_1^2 + \dots + b_n^2}, |c| \leq \text{const } M$ on Ω

Let $u \in C^2(\Omega)$ satisfy $Lu \geq 0$ in Ω ; and assume u has a strictly local max point at $x_0 \in \partial\Omega$, where u is also continuous, Ω satisfies interior sphere condition.

$$u(x_0) = \lim_{x \rightarrow x_0} u(x)$$



Then for any outward pointing vector v at x_0 , we have

$$\frac{\partial u}{\partial \nu}(x_0) > 0 \text{ if it exists,}$$

严格大于号而非大于等于

provided one of the following is satisfied:

- (i) $C \equiv 0$
- (ii) $C \leq 0$ in Ω & $u(x_0) > 0$
- (iii) $u(x_0) = 0$ (regardless of the sign of C)

Rk If $Lu \leq 0$ in Ω & u has a strict loc. min pt.
 $x_0 \in \partial\Omega \Rightarrow \frac{\partial(+u)}{\partial \nu}(x_0) < 0$ (in (ii), replace " $u(x_0) > 0$ " by " $u(x_0) < 0$ ")

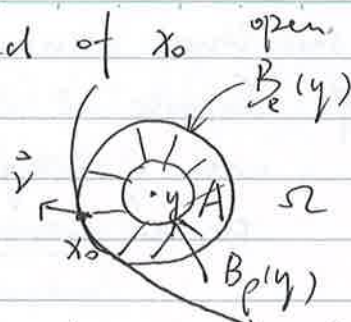
RK2 Only need $Lu > 0$ in a small nbhd of x_0 open

Proof will construct $v \in C^0(\mathbb{R}^n)$ s.t.

(a) $Lu > 0$ in A if $c \leq 0$

(b) $v|_{\partial B_R} = 0$

(c) $\frac{\partial v}{\partial \nu}(x_0) < 0$



Then define $w(x) = u(x) - u(x_0) + \varepsilon v(x)$ $x \in A$, $\varepsilon > 0$ small

$$\Rightarrow \begin{aligned} \Delta w &= Lu - u(x_0)c(x) + \varepsilon \Delta v(x) \\ &\geq 0 &> 0 \text{ by (a)} \end{aligned}$$

$$\geq -u(x_0)c(x) \geq 0 \text{ if either (i) or (ii) occurs}$$

$w|_{\partial B_R} \leq 0$ by (b)

$w|_{\partial B_r} < 0$ if ε small enough

R is taken small enough s.t. $u(x_0) > u(x) \forall x \in B_r(y)$

$$\Rightarrow u(x_0) > u(x) \forall x \in \partial B_r$$

$$\Rightarrow u(x_0) > \max_{\partial B_r} u \quad (\text{在紧集上可以取到最大值})$$

now apply WMP ($c \leq 0$) to w and A

$$\max_{\bar{A}} w \leq \max_{\partial A} w^+ \xrightarrow{0} = 0$$

$$\Rightarrow w \leq 0 \text{ in } A$$

Since $w(x_0) = 0$, we have $\frac{\partial w}{\partial \nu}(x_0) \geq 0$

$$\Rightarrow \frac{\partial u}{\partial \nu}(x_0) + \varepsilon \frac{\partial v}{\partial \nu}(x_0) \geq 0$$

$$\Rightarrow \frac{\partial u}{\partial \nu}(x_0) \geq -\varepsilon \frac{\partial v}{\partial \nu}(x_0) > 0 \text{ by (c)}$$

Done for case (i) & (ii)

Suppose case (iii) occurs, i.e. $u(x_0) = 0 \Rightarrow u(x) < 0$ in $B_r(y)$

Recall $0 \leq Lu = a_{ij} u_{x_i} x_j + b_i u_{x_i} + c(x)u$ $C^- = \min(0, c)$

$$\Rightarrow \underbrace{a_{ij} u_{x_i} x_j + b_i u_{x_i}}_{\Delta u} + C^- u \geq -C^+ u \geq 0 \text{ in } B_r(y)$$

Now apply conclusion for case (i) to $\tilde{L}$ on $B_R(y)$
 $\Rightarrow \frac{\partial u}{\partial \nu}(x_0) > 0$ Done for (i)

Now construct v :

$$v(x) = e^{-\alpha|x-y|^2} - e^{-\alpha R^2}, \quad \alpha > 0 \text{ to be determined}$$

$\Rightarrow$ (b)

$$\frac{\partial v}{\partial \nu}(x_0) = \nabla v(x_0) \cdot \nu = -2\alpha(x_0 - y) e^{-\alpha|x_0-y|^2} \nu < 0 \Rightarrow \text{(c)}$$

$$v_{x_i} = -2\alpha(x_i - y_i) e^{-\alpha|x-y|^2}$$

$$v_{x_i x_j} = -2\alpha \delta_{ij} e^{-\alpha|x-y|^2} + 4\alpha^2(x_i - y_i)(x_j - y_j) e^{-\alpha|x-y|^2}$$

$$2v = -2\alpha a_{ij} b_{ij} e^{-\alpha|x-y|^2} + 4\alpha^2(x_i - y_i)(x_j - y_j) a_{ij} e^{-\alpha|x-y|^2}$$

$$\geq -2\alpha(x_i - y_i) b_i e^{-\alpha|x-y|^2} + c(x)(e^{-\alpha|x-y|^2} - e^{-\alpha R^2})$$

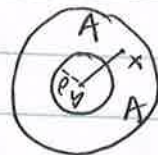
$$= e^{-\alpha|x-y|^2} [-2\alpha a_{ii} + 4\alpha^2(x_i - y_i)(x_j - y_j) a_{ij}] - 2\alpha(x_i - y_i) b_i$$

$$\geq \lambda \cdot |x-y|^2 + c(x) \geq \lambda \cdot \rho^2$$

from below

> 0 if α taken large

QED



Strong maximum principle

Suppose L is strictly elliptic on Ω , a_{ij}, b_i & c bdd on Ω
 $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ satisfies $Lu \geq 0$ in Ω

& $\max_{\bar{\Omega}} u$ is attained at $x_0 \in \Omega$. Then $u \equiv \text{const.} = u(x_0)$
 provided one of the following is satisfied:

(i) $c \equiv 0$

(ii) $c \leq 0$ and $u(x_0) > 0$

(iii) $u(x_0) = 0$

Proof

$$\text{Let } \Omega^- = \{x \in \Omega \mid u(x) < u(x_0)\}$$

If $u(x) \not\equiv u(x_0)$, then Ω is nonempty & open

Claim $(\partial\Omega^-) \cap \Omega \neq \emptyset$

Suppose $(\partial\Omega^-) \cap \Omega = \emptyset$. Let $\Omega_0 = \Omega \setminus \bar{\Omega}^-$

$\Rightarrow$ • Ω_0 is open

$$\bullet \underline{x_0 \in \Omega_0} : \begin{cases} x_0 \notin \partial\Omega^- \\ x_0 \notin \partial\Omega^- \text{ b/c.} \end{cases}$$

$\Downarrow$
 Ω_0 nonempty

$$\begin{aligned} \text{Now } \Omega &= \Omega_0 \cup (\bar{\Omega}^- \cap \Omega) = \Omega_0 \cup ((\Omega \cup \partial\Omega^-) \cap \Omega) \\ &= \Omega_0 \cup (\Omega^- \cap \Omega) = \underbrace{\Omega_0 \cup \Omega^-}_{\text{open}}, \quad \Omega_0 \cap \Omega^- = \emptyset \end{aligned}$$

But Ω is connected, so we have a contradiction

$\Rightarrow$ claim

Pick $x_1 \in (\partial\Omega^-) \cap \Omega$, $x_2 \in \Omega^-$ s.t.

$$|x_1 - x_2| < \frac{1}{2} \text{dist}(x_1, \partial\Omega) \Rightarrow$$

$$\text{dist}(x_2, \partial\Omega^-) \leq |x_2 - x_1| < \frac{1}{2} \text{dist}(x_1, \partial\Omega)$$

$$\text{dist}(x_1, \partial\Omega) \leq |x_1 - x_2| + \text{dist}(x_2, \partial\Omega)$$

$$\Rightarrow \text{dist}(x_2, \partial\Omega) \geq \text{dist}(x_1, \partial\Omega) - |x_1 - x_2|$$

$$> \frac{1}{2} \text{dist}(x_1, \partial\Omega) \geq \text{dist}(x_2, \partial\Omega^-)$$

Then $\exists$ ball $B_R(x_2)$ s.t. $B_R(x_2) \subset \Omega^-$

& $\partial B_R(x_2)$ touches $\partial\Omega^-$ in Ω

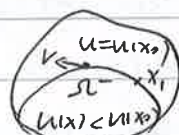
Let $y_0 \in \partial B_R(x_2) \cap (\partial\Omega^-) \cap \Omega$

Note $u(y_0) = u(x_0) > u(x) \quad \forall x \in B_R(x_2)$

Applying ^{Hopf} to L, u on $B_R(x_2)$, we infer

$$\frac{\partial u}{\partial \nu}(y_0) > 0$$

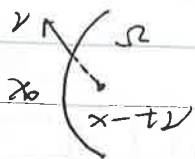
But b/c. u has max at y_0 , $\nabla u(y_0) = \vec{0} \Rightarrow \rightarrow \leftarrow$



2nd B.P.L
 $\nabla u(x^*) = \vec{0}$



$$\frac{\partial u}{\partial \nu}(x_0) = \lim_{t \rightarrow 0^+} \left(\frac{u(x_0 - t\nu) - u(x_0)}{t} \right)$$


 $t > 0$

Applications

1. Separation of Sol's

where

$$a_1(x)u_{x_1x_1} + b_1(x)u_{x_2x_2} = f(x, u), \quad x \in \Omega$$

 L

$$f(x, u) = \text{smooth}(x) + \text{smooth}(x)(u-1)u$$

 e^u
 $\text{arctan } u$

L is strictly elliptic on Ω , a_1, b_1 bdd on Ω .
Suppose $\text{smooth}(x)$ has two solutions $u_1, u_2 \in C^2(\Omega)$ on Ω
with $u_1 > u_2$ on Ω and $\exists x_0 \in \Omega$ s.t. $u_1(x_0) = u_2(x_0)$

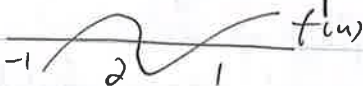
Assume for each fixed $x \in \Omega$, $f(x, u)$ is C^1 -smooth in u

Then $u_1 \equiv u_2$ in Ω

$f_u(x, \cdot)$ bdd for $x \in \Omega$

$$\exists M \text{ s.t. } |f_u(x, \cdot)| \leq M \quad \forall x \in \Omega, \forall u \in [u_2(x), u_1(x)]$$

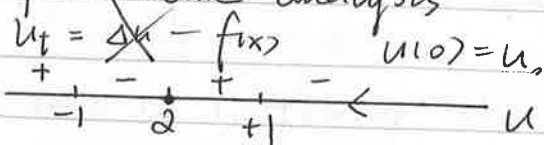
Allen-Cahn eq



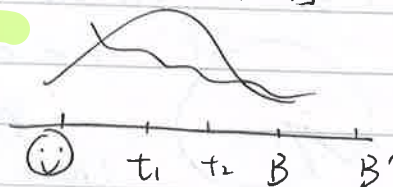
$$\Delta u - f(u) = 0$$

bistable eq

phase-plane analysis



equilibrium



Fact: $\begin{cases} u \in C^1((\cdot), \infty) \\ u \text{ bdd on } ((\cdot), \infty) \end{cases}$

$\Rightarrow \exists t_n \rightarrow \infty$ s.t. $u'(t_n) \rightarrow 0$

$$u(B) - u((\cdot))$$

$B - (\cdot)$

$u'(t_n)$

Proof: $2u_1 = f(x, u_1(x))$, $x \in \Omega$

$$-L u_2 = f(x, u_2(x)), \quad x \in \Omega$$

$$L(u_1, u_2) = f(x, u_1(x)) - f(x, u_2(x))$$

$$= \int_0^1 \frac{dt}{dt} [f(x, tu, x) + (1-t)u_2(x)] dt$$

$$= \int_0^1 f_u(x, t u_1(x) + (1-t) u_2(x)) (u_1(x) - u_2(x)) dt$$

$$= \text{Smiley}(x) (u_1 - u_2)(x)$$

Let $w = u_1 - u_2$

$$\underbrace{Lw - \lambda(x)w}_{=0} = 0 \text{ in } \Omega, \quad w \geq 0 \text{ in } \Omega, \quad w|_{\partial\Omega} = 0$$

ILW Observe $\hat{L}$ is strictly elliptic, a.s. b.i. 😊 b.d.

$$\hat{z}(-w) = 0, \quad \max_{\theta} (-w) = 0, \quad \text{achieved at } x_0 \in \Omega$$

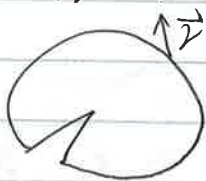
Now $\text{Smp}(u(x)=0) \Rightarrow -w \equiv \text{const} = 0 \Rightarrow w \equiv 0$

$$\Rightarrow u_1 \equiv u_2 \pmod{p} \quad \text{QED}$$

2. Consider Neumann boundary value problem

NBVP $\begin{cases} \Delta u = 0 & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial\Omega \end{cases}$ Ω bdd in $\mathbb{R}^n$
Satisfies strong

Satisfies interior sphere condition



ν - any outward everywhere on $\partial\Omega$

pointing vector field on ∂R

(Not necessarily normal vector field)

$$\int_{\Omega} u \Delta u = 0 = - \int_{\Omega} |u|^2 dx + \int_{\partial \Omega} u \frac{\partial u}{\partial \nu} ds$$

normal vector

Suppose $u \in C^q(\Omega) \cap C^1(\bar{\Omega})$ is a sol of (NBVP)

Then $u \equiv \text{const}$

Proof Case 1 $\max_D u$ is satisfied at some $\lambda \in \Omega$

By SMP (1), $u \equiv \text{const.}$

Case 2 $\max_{\bar{\Omega}} u$ is attained only on $\partial\Omega$ (Not in Ω)

$$\begin{array}{l} x_0 \\ u(x_0) > u(x) \\ \forall x \in \Omega \end{array}$$

Laplace applied to yield
 $\frac{\partial u}{\partial \nu}(x_0) > 0$

contradicted to NBC

3. Comparison principle

old C.P.

$$\begin{cases} Lu \geq Lv & \text{on } \Omega \\ u \leq v & \text{on } \partial\Omega \\ C \leq 0 \end{cases}$$

$$\Rightarrow u \leq v \quad \text{on } \Omega$$

Ω bdd, L strictly elliptic on Ω . Arg. b.i. $C \leq 0$ bdd on Ω . Ω satisfies interior sphere condition everywhere on $\partial\Omega$

Then if $u, v \in C^2(\Omega) \cap C^1(\bar{\Omega})$

Satisfy $\begin{cases} Lu \geq Lv & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} + \beta(x)u \leq \frac{\partial v}{\partial \nu} + \beta(x)v & \text{on } \partial\Omega \end{cases}$

ν - outward pointing vector, $\beta(x) > 0$ on $\partial\Omega$
 field on $\partial\Omega$

$$\Rightarrow u \leq v \quad \text{on } \Omega$$

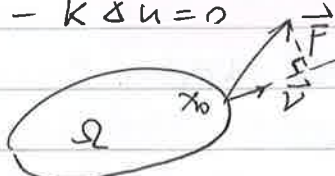
flux of mass

velocity vector. density

Diffusion eq. $u = u(x, t)$ mass density

$$\frac{\partial u}{\partial t} + \nabla \cdot \vec{F}_{\text{diffusion}} = 0$$

$\vec{F}_{\text{diffusion}}$ Fick's Law $-k \nabla u(x, t)$
 $u_t - \nabla \cdot (k \nabla u) = 0$ Fourier's Law k : diffusion coefficient
 $u_t - k \Delta u = 0$


 $\oplus -k \nabla u(x_0, t) \cdot \vec{v} = \underline{\beta(x_0)} (u(x_0, t) - 0.1)$

Proof: Case 1 $\max_{\bar{\Omega}} w$ is attained at some $x_0 \in \Omega$
 By Strong (4.4), we have

Let $w = u - v \Rightarrow$

$\begin{cases} \Delta w \geq 0 \\ \frac{\partial w}{\partial \nu} + \beta(x)w \leq 0 \text{ on } \partial\Omega \end{cases}$

If $\max_{\bar{\Omega}} w \leq 0$, done

So assume $\max_{\bar{\Omega}} w > 0$

$w = \text{const} > 0$

then $\frac{\partial w}{\partial \nu} + \beta(x)w > 0$ on $\partial\Omega$

impossible

Case 2 $\max_{\bar{\Omega}} w$ is attained only on $\partial\Omega$
 at some $x_0 \in \partial\Omega$

Loop (4.4) $\Rightarrow \frac{\partial w}{\partial \nu}(x_0) > 0$ Contradicts B.

Thus, " $\max_{\bar{\Omega}} w > 0$ " is wrong QED

$\Rightarrow$ (Robin BVP) $\begin{cases} Lu = f(x), & x \in \Omega \\ \frac{\partial u}{\partial \nu} + \beta(x)u = g(x), & x \in \partial\Omega \end{cases}$
has at most one sol $\in C^2(\Omega) \cap C^1(\bar{\Omega})$

Gidas - Ni - Nirenberg

$B = B_R(0) \quad f \in C^1(\mathbb{R})$

$\begin{cases} \Delta u = f(u) & \text{in } B \\ u|_{\partial B} = 0 \\ u > 0 & \text{in } B \end{cases}$

$\Rightarrow u = u(r) \quad \& \quad u'(r) < 0$

$r = |x| \quad r \in (0, R)$



Bond

moving plane method

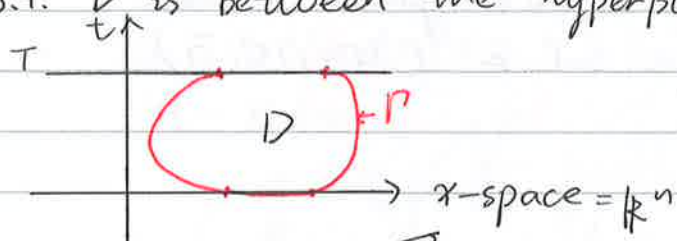
$\begin{cases} u_{rr} + \frac{n-1}{r} u_r = f(u) & (u) \\ u(R) = 0 \end{cases}$

mean curvature = const. $H > 0$

Maximum principle for second order parabolic equation

Let D be $(n+1)$ -dimensional bold domain in x - t space

s.t. D is between the hyperplane $t=0$ & $t=T$



with $\bar{D} \cap \{t=0\} \neq \emptyset$

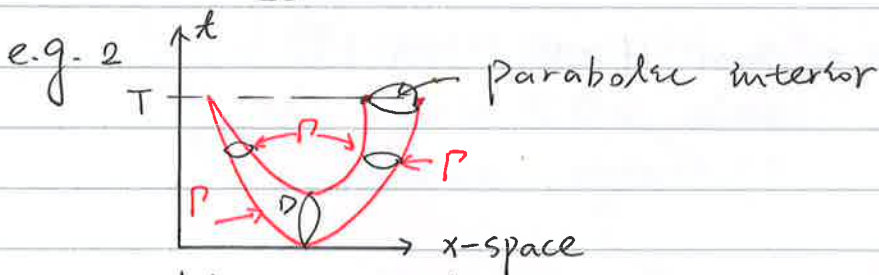
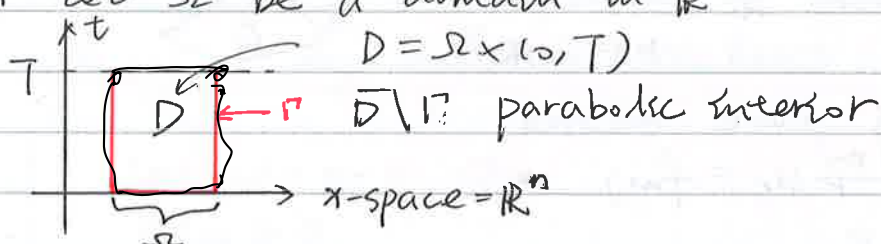
$\bar{D} \cap \{t=T\} \neq \emptyset$

Def. $\partial D \cap \{0 < t < T\}$ is called parabolic boundary of D , denoted by Γ

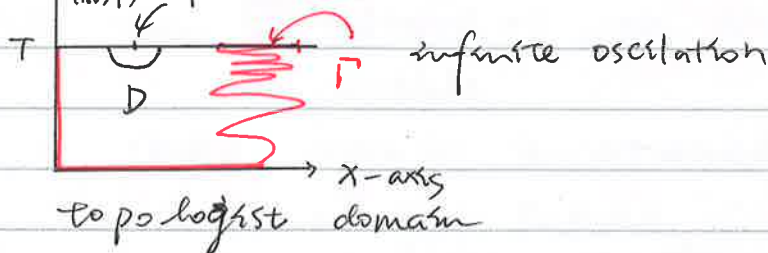
$\bar{D} \setminus \Gamma$ is called parabolic interior of D

e.g. 1 let Ω be a domain in $\mathbb{R}^n$

$$D = \Omega \times (0, T)$$



e.g. 3. (x_0, T) parabolic interior



topologist domain

$$B_{\delta}(x_0, T) \cap \{t < T\}$$

Fact Suppose $(x_0, T) \in \text{parabolic interior } \bar{D} \setminus \Gamma$

Then $\exists \delta > 0$, s.t. open ball $B_{\delta}(x_0, T)$ with

$$\underbrace{B_{\delta}(x_0, T) \cap \{x \in \Gamma\}}_{\subset \mathbb{R}^n \times B_{\delta}} = B_{\delta}(x_0, T) \cap D$$

Proof. $\because (x_0, T) \notin \Gamma$ & Γ is cpt.

nonoverlapping

$\therefore \exists$ small $\delta > 0$, s.t. $B_{\delta}(x_0, T) \cap \Gamma = \emptyset$ $\rightarrow$ open,

$$B_{\delta}^- = (B_{\delta}^- \cap D) \cup (B_{\delta}^- \cap D^c) = (B_{\delta}^- \cap D) \cup (B_{\delta}^- \cap \bar{D}^c)$$

$$\overline{B_{\delta}^- \cap D^c} = B_{\delta}^- \cap (\bar{D}^c \cup \Gamma \cup (\partial D \cap \{t = T\})) \supseteq B_{\delta}^- \cap \bar{D}^c$$

Since the half ball B_{δ}^- is connected

one of these open sets on RHS must be empty

Since $(x_0, T) \in \partial D$, $B_{\delta}(x_0, T) \cap D \neq \emptyset$

moreover, $D \subset \{t < T\} \Rightarrow B_{\delta}^- \neq \emptyset$ 勘誤 $B_{\delta}^- \neq \emptyset$

$$\Rightarrow B_{\delta}^- \cap \bar{D}^c = \emptyset \Rightarrow B_{\delta}^- = B_{\delta}^- \cap D = B_{\delta}(x_0, T) \cap D$$

QED.

Consider operator:

$$Lu = \frac{\partial u}{\partial t} - a_{ij}(x, t) u_{x_i x_j} + b_i(x, t) u_{x_i} + c(x, t) u$$

$$u = u(x, t)$$

$$(x, t) \in D$$

e.g. $Lu = \frac{\partial u}{\partial t} - \kappa \Delta u$

Def We say L is strictly parabolic in D if $\exists$ positive const. λ_0 s.t. Symmetric

$$(a_{ij}(x, t))_{n \times n} \geq \lambda_0 I_{n \times n}, \quad (x, t) \in \bar{D} \setminus \Gamma$$

i.e. $\forall \xi = (\xi_1, \xi_2, \dots, \xi_n) \in \mathbb{R}^n$

$$\sum_{i,j=1}^n a_{ij}(x, t) \xi_i \xi_j \geq \lambda_0 |\xi|^2, \quad \forall (x, t) \in \bar{D} \setminus \Gamma$$

$\uparrow$ parabolic interior

$\cap$

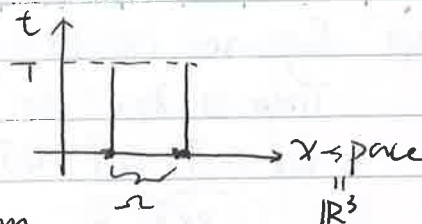
$\subset \leq 0$

Weak max principle

Intuition

$\Omega = \text{冰箱}$

$$D = \Omega \times (0, T)$$



$$u_t - \Delta u = f(x, t) \leq 0 \quad \text{源 Source term}$$

Q: Where is $\max u$ achieved?

A: $\max_{\bar{D}} u$ must be achieved on P
 i.e. $\max_{\bar{D}} u = \max_P u$

WMP ($CC \equiv 0$) Suppose L is strictly parabolic on D
 $u \in C^{2,1}(\bar{D} \setminus P) \cap C^0(\bar{D})$ satisfies $Lu \leq 0$ in $\bar{D} \setminus P$
 Then $\max_{\bar{D}} u = \max_P u$

Proof: Special Case ($Lu < 0$ in $\bar{D} \setminus P$)

Let $\max_{\bar{D}} u$ be attained at $(x_0, t_0) \in \bar{D}$

Case 1 $(x_0, t_0) \in P$, obvious

Case 2 $(x_0, t_0) \in \bar{D} \setminus P$ non positive-definite

$$\Rightarrow D_x^2 u(x_0, t_0)_{n \times n} \leq 0$$

$$D_x u(x_0, t_0) = \vec{0}$$

$$0 > Lu(x_0, t_0) = \frac{\partial u}{\partial t}(x_0, t_0) - \underbrace{a_{ij}(x_0, t_0) u_{x_i x_j}(x_0, t_0)}_{\leq 0} + \underbrace{b_i(x_0, t_0) u_{x_i}}_{\geq 0}$$

$\Rightarrow \rightarrow \leftarrow \Rightarrow$ case 2 does not occur done!

General case Let $v = u - \epsilon t$, $\epsilon > 0$

$$Lv = -\epsilon + Lu \leq -\epsilon < 0 \text{ in } \bar{D} \setminus P$$

Special case

$$\max_{\bar{D}} v_\epsilon = \max_P v_\epsilon$$

Sending $\varepsilon \rightarrow 0^+$, $\max_{\bar{D}} u = \max_{\Gamma} u$ QED

~~⊗~~ In the proof, we only need $(a_{ij}(x,t))_{n \times n} \geq 0$ in $\bar{D} \setminus \Gamma$
In particular, when $a_{ij} \equiv 0$, we still have WMP ($c \geq 0$)

WMP ($c \geq 0$) Let L be strictly parabolic on D .

Suppose $u \in C^{2,1}(\bar{D} \setminus \Gamma) \cap C^0(\bar{D})$ satisfies

$Lu \leq 0$ in $\bar{D} \setminus \Gamma$ and $c \geq 0$ in $\bar{D} \setminus \Gamma$

Then $\max_{\bar{D}} u \leq \max_{\Gamma} u^+$

Proof. case 1, $\max_{\bar{D}} u \leq 0$. Obvious

case 2, $\max_{\bar{D}} u > 0$. Repeating the same proof, we are done.

RK " $(a_{ij})_{n \times n} \geq 0$ " is sufficient for WMP ($c \geq 0$)
degenerate parabolicity condition
strictly parabolic is not needed.

Theorem Suppose L is strictly parabolic on D & c is bdd from below on $\bar{D} \setminus \Gamma$

Let $u \in C^{2,1}(\bar{D} \setminus \Gamma) \cap C^0(\bar{D})$ satisfy $Lu \leq 0$ in $\bar{D} \setminus \Gamma$
Then $u \leq 0$ on $\Gamma \Rightarrow u \leq 0$ in $\bar{D}$

Proof. Suppose $c(x,t) \geq \text{const. } c_0$ in $\bar{D} \setminus \Gamma$

Let $v = e^{c_0 t} u$. W.T.S. $v \leq 0$ in $\bar{D}$

$$\Delta v = \underbrace{v_t - a_{ij} v_{ij} + b_i v_i}_{\sum v} + (c - c_0) v \leq 0 \text{ in } \bar{D} \setminus \Gamma$$

Handwritten signature

$$(u' + cu)e^{ct} = (ue^{ct})'$$

$$(u' + cu)e^{ct} = \odot e^{ct}$$

$$e^{cot} (u_t - a_1 u_{1t} + b_1 u_{2t} + cu) \leq 0$$

$$(e^{cot} u)_t - a_1 (e^{cot} u)_{1t} + b_1 (e^{cot} u)_{2t} + (c - c_0) e^{cot} u \leq 0$$

By WMP ($c > 0$) applied to $\tilde{L}, v \Rightarrow$

$$\max_{\bar{D}} v \leq \max_{\Gamma} v^+ = 0 \Rightarrow v \leq 0 \text{ Done.}$$

Rk Degenerate parabolicity is allowed

Comparison principle Same assumptions on L & C

Suppose $u, v \in C^{2,1}(\bar{D} \setminus \Gamma) \cap C^0(\bar{D})$

Satisfy

$$\begin{cases} Lu \geq Lv & \text{in } \bar{D} \setminus \Gamma \\ u|_{\Gamma} \geq v|_{\Gamma} \end{cases}$$

$$\Rightarrow u \geq v \text{ in } \bar{D}$$

$$L(u-v) \leq 0$$

$$\Rightarrow L(u-v) \leq 0$$

$$v - u|_{\Gamma} \leq 0$$

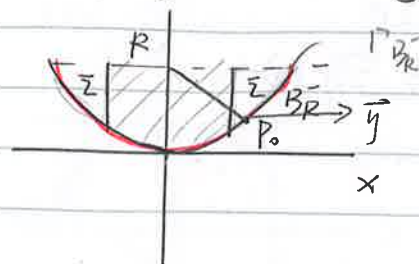
$\Downarrow$ Then

$$v - u \leq 0 \text{ in } \bar{D}$$

$$-\Delta u + \lambda u = f$$

$$u_{rr} + \frac{n-1}{r} u_r$$

Baby Hopf boundary point Lemma 1



$$\text{Let } B_R^- = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid |x|^2 + (t - R)^2 < R^2, 0 < t \leq R\}$$

Suppose L is strictly parabolic on B_R^- & a, b, c bdd on B_R^-

Assume $u \in C^{2,1}(B_R^-) \cap C^0(\bar{B}_R^-)$ satisfies $Lu \leq 0$ on B_R^-

$\exists P_0 = (x_0, t_0) \in \Gamma_{B_R^-}$ s.t. $u(P_0) > u(x, t)$,

$\forall (x, t) \in \bar{B}_R^- \setminus \{P_0\}$. P_0 is not at south pole, i.e. $x_0 \neq 0$

Then for any outward pointing vector $\vec{y}$ at P_0 , we have

$\frac{\partial u}{\partial \vec{y}}(P_0) > 0$ (if it exists), provided one of the following holds:

(i) $c \equiv 0$

(ii) $c \geq 0$ in B_R^- , and $u(P_0) \geq 0$

(iii) $u(P_0) = 0$

Proof: We will construct $v \in C^\infty(\mathbb{R}^{n+1})$ s.t.

(a) $2v < 0$ on $\Sigma = \{(x, t) \in B_R^- \mid |x| > \frac{|x_0|}{2}\}$ if $c \geq 0$ on

(b) $v|_{\Gamma_{B_R^-}} = 0$

(c) $\frac{\partial v}{\partial \vec{y}}(P_0) < 0$

either (i), or (ii), occurs

Then define

$$w(x, t) = u(x, t) - u(P_0) + \varepsilon v(x, t) \quad \varepsilon > 0 \text{ small}$$

Going to apply WMP ($c \geq 0$) on Σ to w

$Lw = Lu - c(x, t)u(P_0) + \varepsilon Lv < 0$ on Σ

$w|_{\Gamma_\Sigma \cap \{|x|^2 + (t - R)^2 = R^2\}} \leq 0$

$w|_{\Gamma_\Sigma \cap B_R^-} < 0$ if ε small enough $u(P_0) > u(x, t)$
 $\forall (x, t) \in \bar{B}_R^-$

If either (i) or (ii) occurs, by WMP ($c > 0$) applied to L , w on Σ , we have $w \leq 0$ on $\bar{\Sigma}$. Note $w(p_0) = 0$
 $\Rightarrow \frac{\partial w}{\partial \eta}(p_0) > 0 \Rightarrow \frac{\partial u}{\partial \eta}(p_0) + c \frac{\partial v}{\partial \eta}(p_0) > 0 \Rightarrow \frac{\partial u}{\partial \eta}(p_0) > 0$

If (iii) holds, then $u < 0$ on $\bar{B}_R \setminus \{p_0\}$

$$0 \geq Lu \geq \underbrace{u_t - a_{ij} u_{ij} + b_i u_i}_{\tilde{L}u} + (c^+ + \delta)u \quad \text{Apply (iii) to } \tilde{L} \Rightarrow \frac{\partial u}{\partial \eta}(p_0) > 0$$

Now construction of v :

$$v(x, t) = e^{-\alpha(|x|^2 + (t-R)^2)} - e^{-\alpha R^2} \quad (\alpha > 0 \text{ to be chosen})$$

$\Rightarrow$ (b)

$$(c): \frac{\partial v}{\partial \eta}(p_0) = \nabla_{(x,t)} v(p_0) \cdot \vec{\eta} = e^{-\alpha} [-2\alpha(x_0, t_0 - R)] \cdot \vec{\eta} < 0$$

$$(a): v_{x_i} = e^{-\alpha} (-2\alpha x_i)$$

$$v_{x_i x_j} = e^{-\alpha} (4\alpha^2 x_i x_j) + e^{-\alpha} (-2\alpha \delta_{ij})$$

$$Lv \leq e^{-\alpha} (-2\alpha(t-R)) - a_{ij} v_{ij} + b_i (e^{-\alpha} (-2\alpha x_i)) + c = -ce$$

$$\leq e^{-\alpha} (-4\alpha^2 a_{ij} x_i x_j - 2\alpha(-a_{ii} + t - R + b_i x_i) + c|x|, t) \\ \leq e^{-\alpha} [-2\alpha^2 \lambda_0 |x_0|^2 + O(\alpha) + O(1)] < 0$$

If α large enough

RR: Same results hold

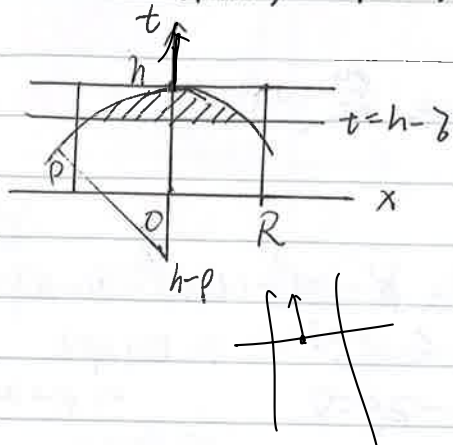
- If $\bar{B}_R$ is shifted (but not
- If $\bar{B}_R$ is replaced by B_R^+

Corollary not all can

all the conditions in this Lemma cannot happen at the same time

Baby Hopf boundary point lemma 2.

Let $Q(R, h) = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid |x| < R, 0 < t \leq h\}$



Suppose L is strictly parabolic on $Q(R, h)$ and a, b, c bdd. on $\overline{Q(R, h)}$

Assume $u \in C^{2,1}(Q(R, h)) \cap C^0(\overline{Q(R, h)})$

Satisfies $Lu \leq 0$ on $Q(R, h)$

$u(0, h) > u(x, t), \forall |x| < R, t \in (0, h)$

Then $\frac{\partial u}{\partial t}(0, h) > 0$,

Provided one of the following

(i) $c \equiv 0$

(ii) $c > 0$ in $Q(R, h)$ & $u(0, h) \geq 0$

(iii) $u(0, h) = 0$

Proof Take a large $p > 0$ & small $\delta > 0$, s.t.

$\underbrace{B_p(0, h-p) \cap \{t > h-\delta\}}_N \subset Q(R, h)$

Shall construct $v \in C^0(\mathbb{R}^{n+1})$ s.t.

(a) $Lv < 0$ on N

(b) $v|_{\partial B_p(0, h-p)} = 0$

(c) $\frac{\partial v}{\partial t}(0, h) < 0$

Let $w(x, t) = \underbrace{u(x, t)}_{\leq 0} - u(0, h) + \varepsilon \underbrace{v(x, t)}_{< 0}$

$Lw = Lu - u(0, h)c(x, t) + \varepsilon Lv < 0$ on N if (i) or (ii) holds

• on the curved part of ∂N $w < 0$ if ε small enough

$w \leq 0 \Rightarrow$ if (i) or (ii) occurs, we have $w \leq 0$ on $\bar{N}$

Note $w(0, h) = 0 \Rightarrow \frac{\partial w}{\partial t}(0, h) > 0 \Rightarrow \frac{\partial u}{\partial t}(0, h) > 0$

If (iii) occurs, then $u \leq 0$ in $Q(R, h)$

$$0 > Lu > \underbrace{u_t - a_{ij} u_{x_i} x_j + b_i u_{x_i} + (c + \delta) u}_{\tilde{L}u}$$

By (i) applied to $\tilde{L}$, we have $\frac{du}{dt}(0, h) > 0$

construction of v :

$$0 < v(x, t) = \rho^2 - |x|^2 - (t - (h - \rho))^2 = \rho^2 - |x|^2 - (t - h)^2 - 2(t - h) \leq 2\rho^2 \Rightarrow v = O(1)\rho^2$$

$$\Rightarrow (b) \quad \frac{dv}{dt}(0, h) = -2(t - (h - \rho)) \Big|_{t=h} = -2\rho < 0 \quad \text{as } \rho \rightarrow$$

$$\begin{aligned} (m): \quad Lu &= -2(t - h + \rho) + a_{ij} 2\delta_{ij} + b_i(-2x_i) + (c + \delta)u^{(1)} \\ &\leq -2(h - \delta - h + \rho) + 2a_{ii} - 2b_i x_i + O(1)\rho^2 \\ &= -2\rho + O(1) + O(1)\rho^2 \\ &= \rho(-2 + O(1)) + O(1) \\ &< 0 \text{ on } N \text{ if } \rho \text{ large \& } \delta \text{ small} \end{aligned}$$

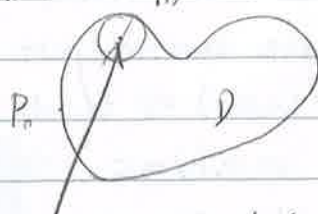
Hopf boundary pt. Lemma (Grown-up version)

Suppose a_{ij}, b_i, c hold on $\bar{D} \setminus P$ & L is strictly parabolic on D . Assume $u \in C^{2,1}(\bar{D} \setminus P) \cap C^0(\bar{D})$ satisfies $Lu \leq 0$ on $\bar{D} \setminus P$, $\exists P_0 \in \partial D$, s.t. $u(P_0) > u(P)$, $\forall P \in D$ & D satisfies interior sphere condition at P_0 with $OP_0 \neq x$ -axis. Then for any outward pointing vector η at P_0 , we have $\frac{\partial u}{\partial \eta}(P_0) > 0$ (if it is provided one of the following holds:

(i) $c \equiv 0$

(ii) $c > 0$ on $\bar{D} \setminus P$, $u(P_0) > 0$

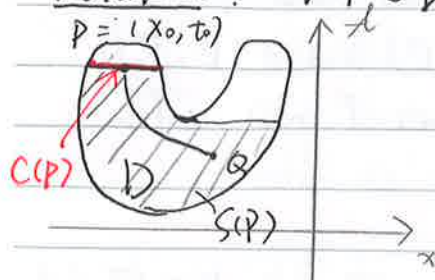
(iii) $u(P_0) = 0$



O = center of ball

Strong maximum principle

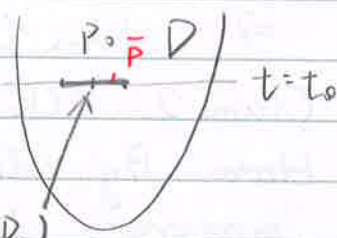
Notation: $\forall P \in \bar{D} \setminus \Gamma$, define



$S(P) = \{Q \in D \mid \exists \text{ continuous curve } \gamma \text{ connecting } P \text{ \& } Q \text{ s.t. } \gamma \subset D, \text{ and when a point travels from } P \text{ to } Q, \text{ its } t\text{-coordinate is non-increasing}\}$
 $C(P) = S(P) \cap \{t = t_0\}$

RK $S(P)$ is ~~open~~ & connected (b/c. it is path-connected)
 $C(P)$ is also path-connected, hence connected

SMP Suppose L is strictly parabolic on D . Ag. br. c bold. on $\bar{D} \setminus \Gamma$, $u \in C^{2,1}(\bar{D} \setminus \Gamma) \cap C^0(\bar{D})$. $Lu \leq 0$ in $\bar{D} \setminus \Gamma$. Assume $\exists P_0 = (x_0, t_0) \in \bar{D} \setminus \Gamma$ s.t. $u(P_0) = \max u$. Then $u \equiv u(P_0)$ on $S(P_0)$, provided one of the following holds: "M"
 i) $C \equiv 0$
 ii) $C \geq 0$ on $\bar{D} \setminus \Gamma$ & $u(P_0) \geq 0$
 iii) $u(P_0) = 0$



Proof: Let $F = \{(x, t) \in \bar{D} \mid u(x, t) = M\}$

want to show: $\forall P \in F$, then $C(P) \subset F$ $B_{d_P/3}(P_0)$

Let $d_P = \text{dist}(P_0, \Gamma) > 0$

Claim 1 $B_{d_P/3}(P_0) \stackrel{\text{def}}{=} \{(x, t) \mid |x - x_0| \leq \frac{d_P}{3}\} \subset F$

otherwise, $\exists \bar{P} = (\bar{x}, t_0) \in B_{d_P/3}(P_0)$ s.t. $u(\bar{P}) < M$

let $b = \frac{1}{2} \text{dist}(\bar{P}, \Gamma)$ Define semi-ellipsoid

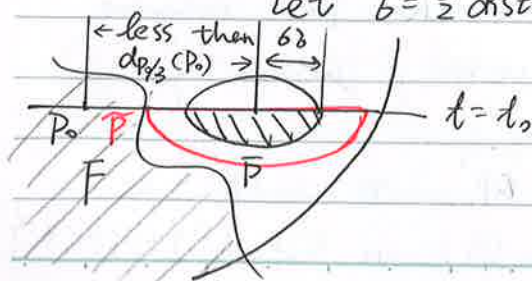
$$E_b = \{(x, t) \mid \frac{|x - \bar{x}|^2}{(6b)^2} + \frac{(t - t_0)^2}{b^2} < 1, t \leq t_0\}$$

$\forall b > 0$

Observe: when $0 < b \leq 1$ ✓

$E_b \cap F = \emptyset$, when $6b > \frac{d_P}{3}$

$E_b \ni P_0$, in particular, $E_b \cap F \neq \emptyset$



Then $\exists \delta_0 > 0$, s.t. E_{δ_0} touches to $\bar{F}$ at some $\tilde{P} \in D$ &
 $F \cap E_{\delta_0} = \emptyset$ (why $\tilde{P} \in D$: b/c $\text{dist}(\bar{P}, P) > \text{dist}(\bar{P}, F)$)
 So E_{δ_0} remains in D & touches F in D before
 it touches P)

$$\begin{aligned} \text{dist}(\bar{P}, P) &\geq \text{dist}(P_0, P) - \text{dist}(P_0, \bar{P}) \\ &\geq d_{P_0} - \frac{dP_0}{3} = \frac{2dP_0}{3} > \frac{dP_0}{3} \geq \text{dist}(P_0, \bar{P}) \geq \text{dist}(\bar{P}, F) \end{aligned}$$

$\tilde{P}$ can't be the lowest pt in E_{δ_0} b/c if so, $|\bar{P} - \tilde{P}| = \delta_0$
 $\stackrel{!}{=} \text{dist}(\bar{P}, F)$

but $\tilde{P} \in F$, so $|\bar{P} - \tilde{P}| \geq \text{dist}(\bar{P}, F)$

impossible

Now by Baby Hopf boundary lemma | or ~~grow up~~ ~~Hopf~~
 $\frac{\partial u}{\partial \eta}(\tilde{P}) > 0$. But $\tilde{P} \in \text{interior } D$

$$\Downarrow \\ \nabla u(\tilde{P}) = \vec{0}$$

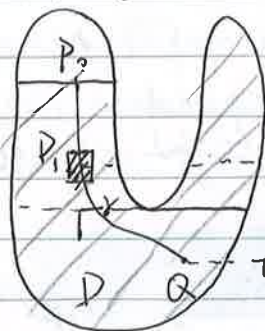
$\Rightarrow$ Claim 1

Claim 2 $C(P_0) \subset F \neq \emptyset$

Claim By Claim 1 $F \cap C(P_0)$ is relatively open in $C(P_0)$
 moreover, $F \cap C(P_0)$ is relatively closed in $C(P_0)$
 Since $C(P_0)$ is connected, $F \cap C(P_0) = C(P_0) \Rightarrow \text{Claim 2}$

$\gamma \subset D$

$= (x_2, t_2)$



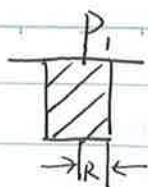
If $\exists Q \in S(P_0)$ s.t. $u(Q) < u(P_0)$

Let $P_1 = (x_1, t_1)$ be the first intersection

$t=t_1$, intersection of γ with F (when going
~~upwards~~ upwards)

Then on arc $\widehat{QP_1}$, $u < u(P_0)$

By Claim 2, $u < u(P_0)$ on $S(P_0) \cap \{t_2 \leq t < t_1\}$



Now construct cylinder

$$Q_{P_1}(R, h) = \{ (x, t) \mid |x - x_1| < R, t_1 - h < t \leq t_1 \}$$

By corollary Baby Hopf 2,
this is impossible!

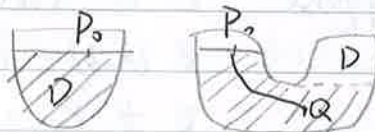
$$Lu = u_t - a_{ij}(x, t) u_{x_i x_j} + b_i(x, t) u_{x_i} + c(x, t) u$$

SMP Suppose L is strictly parabolic on D . Arg. b_i & c bdd $\bar{D} \cap \Gamma$, $u \in C^{2,1}(\bar{D} \cap \Gamma) \cap C^0(\bar{D})$ satisfies $Lu \leq 0$ in $\bar{D} \cap \Gamma$. If $\exists \underline{P}_0 \in \bar{D} \cap \Gamma$ s.t. $u(\underline{P}_0) = \max u$, then $u = u(\underline{P}_0)$ in $S(\underline{P}_0)$, provided one of the following holds:

(i) $C \equiv 0$

(ii) $C \geq 0$ in $\bar{D} \cap \Gamma$ & $u(\underline{P}_0) \geq 0$

(iii) $u(\underline{P}_0) = 0$



Hopf bdry pt lemma (grow-up version)

Proof: Just use Baby Hopf Lemma I on B^- or B^+ (u holds as RK on B^+ , as long as $\underline{P}_0 \neq$ north point) which is guaranteed by $\overrightarrow{OP_0} \nparallel t$ -axis

Corollary of Baby Hopf 2.

All the conditions in this lemma can not happen at the same time



Application

Ω - bdd domain in $\mathbb{R}^n$ with C^2 -smooth $\partial\Omega$

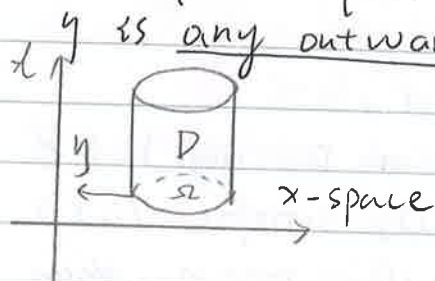
$$D = \Omega \times (0, T)$$

$$Lu = u_t - a_{ij}(x, t) u_{x_i x_j} + b_i(x, t) u_{x_i} + f(x, t, u)$$

nonlinear

$$Bu = \frac{\partial u}{\partial \eta}(x, t) + \beta(x, t)u, \quad (x, t) \in \partial\Omega \times (0, T], \quad \beta \geq 0 \text{ on } \partial\Omega \times (0, T]$$

η is any outward pointing vector field on $\partial\Omega$

Comparison Principal (C.P.)

on $\bar{D}$

Suppose L is strictly parabolic on D & a_{ij}, b_i bdd

$\forall (x, t) \in \bar{D} \setminus \{t=0\}$, $f_u(x, t, u)$ exists for $u \in (-\infty, \infty)$

& $\forall R > 0$, $\exists$ const. M st. $|f_u(x, t, u)| \leq M$,

$\forall (x, t) \in \bar{D} \setminus \{t=0\}$, $u \in (-R, R)$

Assume $\begin{cases} Lu \geq Lv \text{ in } D \\ Bu \geq Bv \text{ on } \partial\Omega \times (0, T] \\ u|_{t=0} \geq v|_{t=0} \text{ on } \Omega \end{cases}$

Then $u \geq v$ in $\bar{D}$, moreover, if $u|_{t=0} \not\equiv v|_{t=0}$

$u > v$ in $\bar{D} \setminus \{t=0\}$

$u(x, 0)$

$v(x, 0)$

Rk if $Bu = u$ (Dirichlet case)

to have $u \geq v$ in $\bar{D}$, we don't need bddness of a_{ij} & b_i , b/c. we only weak maximum principle

Proof: Let $w = v - u$ want to show $w \leq 0$ in D

$$\text{Let } \hat{L}w = w_t - a_{ij}(x, t) w_{x_i x_j} + b_i(x, t) w_{x_i} +$$

$$\frac{e^{mt}(u' + mu)}{e^{mt}} = \frac{(e^{mt}u)'}{e^{mt}}$$

No

Date

$$+ \underbrace{f(x, t, v) - f(x, t, u)}_{\text{"}}$$

$$\int_0^1 \frac{d}{ds} f(x, t, sv + (1-s)u) ds$$

$$\int_0^1 f_u(x, t, sv + (1-s)u) \underbrace{(v-u)}_w ds$$

$$\tilde{L}w = w_t - a_{ij} w_{x_i x_j} + b_i w_{x_i} + c(x, t) w$$

$c(x, t)$ is bdd on $\bar{D} \cap P$, $|c(x, t)| \leq M$

$$\Rightarrow \begin{cases} \tilde{L}w \leq 0 & \text{on } \bar{D} \cap P \\ Bw \leq 0 & \text{on } \partial\Omega \times (0, T] \\ w|_{t=0} \leq 0 & \text{on } \bar{\Omega} \end{cases} \quad h.e$$

$$\text{Let } \tilde{w} = e^{-mt} w \Rightarrow w = e^{mt} \tilde{w} \Rightarrow$$

$$me^{mt} \tilde{w} + e^{mt} \tilde{w}_t - e^{mt} a_{ij} \tilde{w}_{x_i x_j} + e^{mt} b_i \tilde{w}_{x_i} + ce^{mt} \tilde{w} \leq 0 \text{ in } \bar{D} \cap P$$

$$\underbrace{\tilde{w}_t - a_{ij} \tilde{w}_{x_i x_j} + b_i \tilde{w}_{x_i}}_{\tilde{L}\tilde{w}} + \underbrace{(c+M)}_{\geq 0} \tilde{w} \leq 0 \text{ in } \bar{D} \cap P$$

Case 1 $\max_{\bar{D}} \tilde{w} < 0$, Done

Case 2 $\max_{\bar{D}} \tilde{w} \geq 0$

Subcase 1 $\max_{\bar{D}} \tilde{w}$ is attained at $(x_0, t_0) \in \bar{D} \cap P$

By SMP ($c \geq 0$ & $\max \geq 0$).

$\tilde{w} \equiv \text{const.}$ for $0 \leq t \leq t_0$

$\tilde{w}(x_0, t_0) \xrightarrow{\text{"}} v = u|_{t=0} \geq 0$ but $u \geq v$ at $t=0$

$\Rightarrow \tilde{w} \equiv 0$ for $0 \leq t \leq t_0$

$$\Rightarrow \max_{\bar{D}} \tilde{w} = \tilde{w}(x_0, t_0) = 0$$

$$\Rightarrow \tilde{w} \leq 0 \text{ \& hence } w \leq 0 \text{ in } \bar{D}$$

If $u \neq v$ at $t=0$, then subcase 1

D.N.O. (does not occur), so proof is complete

Subcase 2 $\max_{\bar{D}} \tilde{w}$ is attained only on Γ , say, at $(x_0, t_0) \in \Gamma$ b/c. $\tilde{w}|_{t=0} \leq 0$

Subsubcase 1

$(x_0, t_0) \in \bar{\Omega} \times \{0\} \Rightarrow \max_{\bar{D}} \tilde{w} = 0 \Rightarrow w \leq 0$ in $\bar{D}$
 max is Not attained on $\partial\Omega \times (0, T]$ $\Downarrow$
 $\tilde{w} < 0$ if $0 < t \leq t_0$

Subsubcase 2

$$(x_0, t_0) \in \partial\Omega \times (0, T]$$

(max may occur on $\bar{\Omega} \times \{0\}$)

but Not in $\bar{D} \setminus \Gamma$!

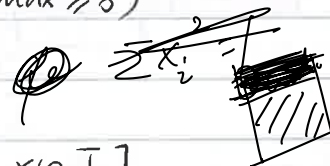
By Baby Hopf 1 ($0 < t_0$, max > 0)

$$\frac{\partial \tilde{w}}{\partial \eta}(x_0, t_0) > 0$$

Recall $B\tilde{w} \leq 0$ on $\partial\Omega \times (0, T]$

$$\underbrace{\frac{\partial \tilde{w}}{\partial \eta}}_{>0} + \underbrace{\beta(x, t)}_{\geq 0} \underbrace{(\tilde{w})}_{\geq 0}(x_0, t_0)$$

$\rightarrow$ QED.



$$\mathcal{D}(\Omega) = \{u \mid u \in C_0^\infty(\Omega), \text{supp } u \subset \Omega\}$$

Distributions

Ω - domain in $\mathbb{R}^n$

$$\mathcal{D}(\Omega) = C_0^\infty(\Omega)$$

$$C. \frac{1}{t-1} \quad 0 \leq t \leq 1$$



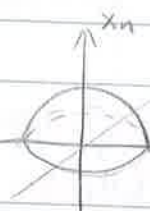
$$\text{Supp } u = \{x \in \Omega \mid u(x) \neq 0\}$$

"compactly supported"

e.g.

$$j(x) = \begin{cases} c e^{\frac{-1}{1-|x|^2}} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

$$\frac{\partial j(x)}{\partial x_1} = c e^{\frac{-1}{1-|x|^2}} \cdot \frac{2x_1}{(1-|x|^2)^2}, \quad |x| < 1$$



$(x_1, \dots, x_{n-1})$

$$\Rightarrow j \in C_0^\infty(\mathbb{R}^n), \text{supp } j = \text{open } (B, 1)$$

Now, $\forall x_0 \in \Omega, \forall \epsilon > 0, \text{s.t. } B_\epsilon(x_0) \subset \Omega$, let

$$j_\epsilon(x) = \frac{1}{\epsilon^n} j\left(\frac{x-x_0}{\epsilon}\right)$$

$$\int_{\mathbb{R}^n} j(x) dx = \int_{\mathbb{R}^n} j_\epsilon(x) dx$$

$\Downarrow$

$$j_\epsilon \in \mathcal{D}(\Omega)$$

$$j \in \mathcal{D}(\Omega)$$

$\Rightarrow \mathcal{D}(\Omega)$ not empty.

After take $c > 0$, s.t. $\int_{\mathbb{R}^n} j(x) dx = 1$

In this case $j(x)$ is called standard mollifier.

Def A distribution is just linear functional f
 $\mathcal{D}(\Omega) \rightarrow \mathbb{R}$

$\phi \rightarrow \langle f, \phi \rangle$ s.t. f is continuous on $\mathcal{D}(\Omega)$

in the following sense: For any $\{\phi_k\} \subset \mathcal{D}(\Omega)$, s.t.

(i) $\text{supp } \phi_k \subset \text{fixed opt } K \subset \Omega$

(ii) $\forall \alpha \in \mathbb{Z}^n, \lim_{k \rightarrow \infty} \|\partial^\alpha \phi_k\|_{L^\infty(\Omega)} = 0$

$$\partial^\alpha \phi = \frac{\partial_1}{\partial x_1} \frac{\partial_2}{\partial x_2} \cdots \frac{\partial_n}{\partial x_n}$$

We have $\langle f, \phi_k \rangle \xrightarrow[k \rightarrow \infty]{} 0$ $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$

Notation $\mathcal{D}'(\Omega) =$ set of all distributions on $\mathcal{D}(\Omega)$

$\Rightarrow \mathcal{D}'(\Omega)$ is linear space

e.g. 1. $\forall f \in L^1_{loc}(\Omega)$ (i.e. $\forall \text{ cpt } K \subset \Omega$,
local we have $f \in L^1(K)$)

Define on $\mathcal{D}(\Omega)$:

$\langle F, \phi \rangle = \int_{\Omega} f(x) \phi(x) dx$ 控制收敛

$\Rightarrow F \in \mathcal{D}'(\Omega)$

e.g. 2. Suppose $f \in \mathcal{D}'(\Omega)$, $g \in C^\infty(\Omega)$

Define $\langle fg, \phi \rangle$ test function

$\stackrel{\text{def}}{=} \langle f, \phi g \rangle \Rightarrow fg \in \mathcal{D}'(\Omega)$

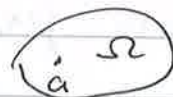
e.g. 3. Fix $a \in \Omega$, define

$\langle \delta_a, \phi \rangle = \phi(a)$, $\forall \phi \in \mathcal{D}(\Omega)$

$\Rightarrow \delta_a \in \mathcal{D}'(\Omega)$

δ_a is called δ -function

Dirac Delta



Tanglang Village:

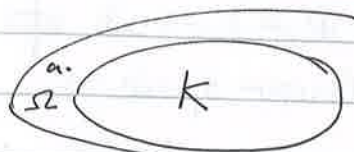
$$\delta_a(x) = \begin{cases} \infty & x = a \\ 0 & x \neq a \end{cases}$$

$$\int_{\mathbb{R}^n} \delta_a(x) dx = 1$$

Dirac 函数可以看成是一个 distribution

Distributions

Ω - domain in $\mathbb{R}^n$, $\mathcal{D}(\Omega) = C_0^\infty(\Omega)$
 $\uparrow$
 nonempty



Standard mollifier
 $j(x) \in \mathcal{D}(\mathbb{R}^n)$

$\text{supp } f \subset K \subset \Omega$

K does not touch bdy

$$j(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2 - 1}\right) & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1 \end{cases}$$

$\text{supp } j = \text{open } B_{1,0}$

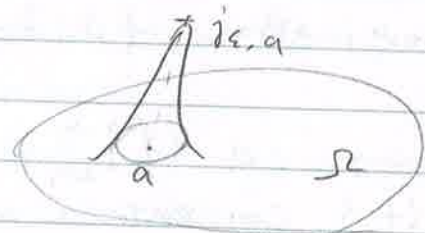
$$\int_{\mathbb{R}^n} j(x) dx = 1$$

$\forall a \in \Omega, \quad \varepsilon > 0$ small

$$j_{\varepsilon,a}(x) = \frac{1}{\varepsilon^n} j\left(\frac{x-a}{\varepsilon}\right)$$

$j_{\varepsilon,a} \in \mathcal{D}(\Omega)$

$$\int_{\Omega} j_{\varepsilon,a}(x) dx = 1$$



Def A distribution is a linear functional f defined on $\mathcal{D}(\Omega)$

$\phi \rightarrow \langle f, \phi \rangle \in \mathbb{R}$, s.t. f is continuous in $\phi \in \mathcal{D}(\Omega)$ the following sense

For any $\{\phi_k\}_{k=1}^\infty \subset \mathcal{D}(\Omega)$, s.t. $\phi_k \xrightarrow{k \rightarrow \infty} 0$ in the sense of

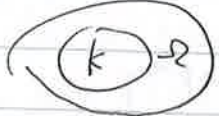
i. $\text{supp } \phi_k \subset \text{cpt } K \subset \Omega, \quad \forall k \geq 1$

ii. $\forall \alpha \in \mathbb{Z}^n$ nonnegative integers

$$(\alpha_1, \dots, \alpha_n) \quad \lim_{k \rightarrow \infty} \|D^\alpha \phi_k\|_{C^0(\Omega)} = 0$$

We have $\langle f, \phi_k \rangle \xrightarrow{k \rightarrow \infty} 0$

$\mathcal{D}'(\Omega)$ = set of all distributions on $\mathcal{D}(\Omega)$
linear space



$\forall \text{ cpt } K \subset \Omega, f \in L^1(K)$

eg. 1 $\forall f \in L^1_{\text{loc}}(\Omega)$

$$\langle f, \phi \rangle = \int_{\Omega} f(x) \phi(x) dx, \quad \forall \phi \in \mathcal{D}(\Omega)$$

eg. 2 $\forall$ fixed $a \in \Omega$, define

$$\langle \delta_a, \phi \rangle = \phi(a)$$

Def Suppose $\{F_k\}_{k=1}^{\infty} \subset \mathcal{D}'(\Omega), F \in \mathcal{D}'(\Omega)$

We say $F_k \rightarrow F$ as $k \rightarrow \infty$ if

$$\forall \phi \in \mathcal{D}(\Omega), \langle F_k, \phi \rangle \xrightarrow{k \rightarrow \infty} \langle F, \phi \rangle$$

Approximation of δ -function by "functions on the line"

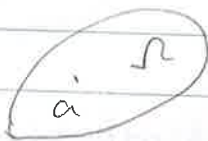
Theorem Let $\{f_k\}_{k=1}^{\infty} \in L^1(\Omega)$ s.t.

(i) $\{f_k\}$ concentrate at $a \in \Omega$: $\forall$ small $\delta > 0$

$$\int_{\Omega \setminus B_{\delta}(a)} |f_k|(x) dx \xrightarrow{k \rightarrow \infty} 0$$

(ii) $\|f_k\|_{L^1(\Omega)} \leq \text{const. } M, \forall k \geq 1$

(iii) $\int_{\Omega} f_k(x) dx \xrightarrow{k \rightarrow \infty} A$



Then f_k (regards as distributions) $\xrightarrow{k \rightarrow \infty} A\delta_a$ in the sense of distribution

Proof: Fix $\phi \in \mathcal{D}(\Omega)$

$$\begin{aligned} \langle f_k, \phi \rangle &= \int_{\Omega} f_k(x) \phi(x) dx = \int_{\Omega} f_k(x) (\phi(x) - \phi(a)) dx \\ &\quad \downarrow \text{w.t.s.} \\ &\quad A\phi(a) \end{aligned}$$

$$+ \int_{\Omega} f_k(x) \phi(a) dx \xrightarrow{k \rightarrow \infty} A\phi(a)$$

since ϕ is contin. at a , $\forall \epsilon > 0$, $\exists \delta > 0$, s.t.
 $|\phi(x) - \phi(a)| < \epsilon$ if $|x - a| < \delta$, $x \in \mathcal{R}$

Now,

$$\begin{aligned} \left| \int_{\mathcal{R}} f_k(x) (\phi(x) - \phi(a)) dx \right| &= \left| \int_{\mathcal{R} \cap B_\delta(a)} + \int_{\mathcal{R} \cap B_\delta^c(a)} \right| \\ &\leq \int_{\mathcal{R} \cap B_\delta(a)} |f_k(x)| |\phi(x) - \phi(a)| dx + \int_{\mathcal{R} \cap B_\delta^c(a)} |f_k(x)| dx \\ &\leq \int_{\mathcal{R} \cap B_\delta(a)} |f_k(x)| dx \cdot 2\|\phi\|_{L^\infty(\mathcal{R})} + \epsilon \int_{\mathcal{R} \cap B_\delta^c(a)} |f_k(x)| dx \quad M \end{aligned}$$

$$\Rightarrow \lim_{k \rightarrow \infty} \left| \langle f_k, \phi \rangle - \int_{\mathcal{R}} f_k(x) \phi(a) dx \right| \leq 0 + \epsilon M \leq 0$$

QED

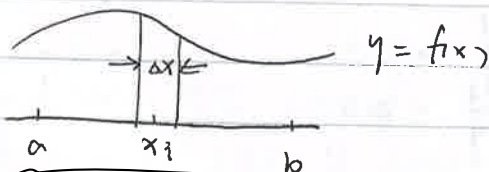
eg. Let f be Riemann integrable on $[a, b]$

Let P_k be a seq of partitions

$$P_k = \{a, x_1^k, x_2^k, \dots, x_{n_k}^k, b\}$$

$$\text{s.t. } \|P_k\| \xrightarrow{k \rightarrow \infty} 0$$

$$(\text{Recall: Then } \sum_{i=1}^{n_k} f(x_i^k) \Delta x_i^k \xrightarrow{k \rightarrow \infty} \int_a^b f(x) dx)$$



$$\sum_{i=1}^{n_k} f(x_i^k) \Delta x_i^k \xrightarrow{k \rightarrow \infty} f(x) \text{ in the sense of distributions.}$$

Proof: $\forall \phi \in \mathcal{D}(a, b)$

$$\langle \text{LHS}, \phi \rangle = \sum_{i=1}^K f(x_i^k) \phi(x_i^k) \Delta x_i^k$$

$\downarrow k \rightarrow \infty$ by calculus I

$$\langle \text{RHS}, \phi \rangle = \int_a^b f(x) \phi(x) dx \quad \text{done.}$$

Derivatives of distributions

Motivation

$$f \in C^1(\Omega) \Rightarrow f, \frac{\partial f}{\partial x_i} \in L^1_{loc}(\Omega)$$

$$\forall \phi \in \mathcal{D}(\Omega)$$

$$\langle \frac{\partial f}{\partial x_i}, \phi \rangle = \int_{\Omega} \frac{\partial f}{\partial x_i} \phi(x) dx$$

integration

by parts

$$= \int_{\Omega} f \frac{\partial \phi}{\partial x_i} dx + \int_{\partial \Omega} f \phi \nu_i dS$$

$\in \mathcal{D}'(\Omega)$
 $\vec{\nu}$ outer normal

$$\Rightarrow \langle \frac{\partial f}{\partial x_i}, \phi \rangle = - \langle f, \frac{\partial \phi}{\partial x_i} \rangle \in \mathcal{D}'(\Omega)$$

Def $\forall f \in \mathcal{D}'(\Omega)$, define distributional derivatives $D_i f$ of f by:

$$\langle D_i f, \phi \rangle = - \langle f, \frac{\partial \phi}{\partial x_i} \rangle$$

Fact: $D_i f \in \mathcal{D}'(\Omega)$

In general, $\forall \alpha \in \mathbb{Z}^n$, define distributional derivative $D^\alpha f$ by:

$$\langle D^\alpha f, \phi \rangle = (-1)^{|\alpha|} \langle f, \partial^\alpha \phi \rangle$$

e.g. Heaviside function

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$



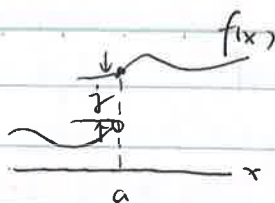
$$DH = ?$$

$$\langle DH, \phi \rangle = - \langle H, \phi' \rangle = - \int_{\mathbb{R}} H(x) \phi'(x) dx$$

$$= -\phi(1) + \phi(0)$$

$$= \phi(0) = \langle \delta_0, \phi \rangle, \forall \phi \in \mathcal{D}(\mathbb{R})$$

$$\Rightarrow DH = \delta_0$$



$$Df(x) = f'(x) + j \delta_a(x)$$

e.g. consider operator:

$$u \in \mathcal{D}'(\Omega) \rightarrow Lu \stackrel{\text{def}}{=} \sum_{|\alpha|=0}^m A_\alpha(x) D^\alpha u, \text{ where}$$

$$A_\alpha \in C^\infty(\Omega) \Rightarrow Lu \in \mathcal{D}'(\Omega)$$

$$\langle Lu, \phi \rangle = \sum_{|\alpha|=0}^m \langle A_\alpha D^\alpha u, \phi \rangle = \sum_{|\alpha|=0}^m \langle D^\alpha u, A_\alpha \phi \rangle$$

$$= \sum_{|\alpha|=0}^m (-1)^{|\alpha|} \langle u, D^\alpha (A_\alpha \phi) \rangle$$

$$= \langle u, \left(\sum_{|\alpha|=0}^m (-1)^{|\alpha|} D^\alpha (A_\alpha \phi) \right) \rangle$$

$$(Au, \phi) = (u, A^* \phi)$$

A^* - adjoint operator

Formal
adjoint
operator of L

denoted by L^*

e.g. $L = \Delta$, $L^* = \Delta$, we say Δ is symmetric
 $A_\alpha(x) = 1$ $\alpha = (2, 2, \dots)$

Def. Consider PDE $Lu = f$, $f \in \mathcal{D}'(\Omega)$

If $v \in \mathcal{D}'(\Omega)$ & satisfy $Lu = f$ in the sense of distribution, i.e. $\langle Lv, \phi \rangle = \langle f, \phi \rangle \quad \forall \phi \in \mathcal{D}(\Omega)$

$$\parallel$$

$$\langle v, L^* \phi \rangle$$

then we say v is a distributional sol of PDE

Rk1 Classical sol must be distributional sol

Rk2 If $f = \delta_y$, $\Omega = \mathbb{R}^n$, then we have fundamental sol
 (f. s.)

~~Notes~~
~~107~~ $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

q:

$-\Delta u = \delta_y$ in $\mathbb{R}^n$, $y \in \mathbb{R}^n$ fixed ~~order~~

No uniqueness of f, s .

$$u(x) = \begin{cases} -\frac{1}{2\pi} \ln |x-y| & n=2 \\ \frac{1}{(n-2)\omega_n |x-y|^{n-2}} & n \geq 3 \end{cases}$$

$\uparrow$
unit sphere's area

$$-\frac{1}{2\pi} \ln |x-y|$$

$$\frac{1}{(n-2)\omega_n |x-y|^{n-2}}$$

On the street:

$$r = |x-y|$$

$$-(u_{rr} + \frac{1}{r} u_r) = \delta_y(x)$$

$$-(r u_{rr} + u_r) = r \delta_y(x)$$

$$-\int_0^{2\pi} \int_0^r (t u_t) \frac{dt d\theta}{t} = \int_0^{2\pi} \int_0^r (t \delta_y(x)) dt d\theta$$

$$-2\pi r u_r - 0 = 1$$

$$u'(r) = -\frac{1}{2\pi r}$$

$$u(r) = -\frac{1}{2\pi} \ln r$$

$dx dy$

Fundamental Sol of

$$-\Delta u = \delta_y(x) = \delta(x-y), \quad x \in \mathbb{R}^n$$

$$\langle u, -\Delta \phi \rangle = \langle \delta_y, \phi \rangle = \phi(y)$$

$$\forall \phi \in \mathcal{D}(\mathbb{R}^n) = C_0^\infty(\mathbb{R}^n)$$

$$u = ?$$

On the street: Since $\delta_y(x)$ is radially symmetric about y in x , we expect u is radially symmetric about y in x , so write $u = u(r)$, $r = |x - y|$

$$\Rightarrow -(u_{rr} + \frac{n-1}{r} u_r) = \delta_y(x)$$

$$-(r^{n-1} u_{rr} + (n-1)r^{n-2} u_r) = r^{n-1} \delta_y(x)$$

$$\int_0^r \int_{|b|=1} (r^{n-1} u_r)_r d\sigma dr = - \int_0^r \int_{|b|=1} r^{n-1} \delta_y(x) d\sigma dr$$

$x = y + r\delta, \quad |\delta| = 1$

$(r^{n-1} d\sigma) dr$

$dv = dx_1 \dots dx_n$

$$\omega_n r^{n-1} u_r \Big|_{u=0}^{u=r} = - \int_{|y-x|<r} \delta_y(x) dx_1 \dots dx_n$$

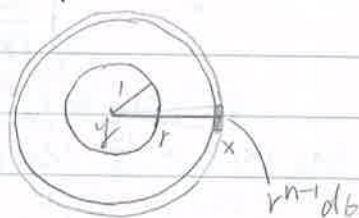
ω_n area of unit sphere in $\mathbb{R}^n$

$$\omega_n u_r r^{n-1} = -1$$

$$n=2: \quad u = -\frac{1}{2\pi} \ln r$$

$$n \geq 3: \quad u_r = \frac{-1}{\omega_n r^{n-1}}$$

$$u = \frac{-1}{\omega_n (n-2)r^{n-2}} + \text{const}$$



In the ivory tower:

$$-\Delta u = \delta_y(x)$$

$$-\Delta u = 0, \quad \text{if } x \neq y$$

Seek sol s.t. $u = u(r)$, $r = |x - y|$

$$\Rightarrow u_{rr} + \frac{n-1}{r} u_r = 0, \quad \text{if } r \neq 0$$

$$(r^{n-1} u_r)_r = 0, \quad r > 0$$

$$r^{n-1} u_r = C'$$

$$u_r = \frac{C}{r^{n-1}} \Rightarrow u = \begin{cases} C \ln r, & n=2 \\ \frac{C}{r^{n-2}}, & n \geq 3 \end{cases}$$

$u \in L'_{loc}(\mathbb{R}^n)$, why:

$$n=2: |u| \leq |C|$$

$$\lim_{r \rightarrow 0^+} \frac{\ln r}{r - \frac{1}{10,000}} = \lim_{r \rightarrow 0} \frac{\frac{1}{r} \cdot 1}{-\frac{1}{10,000} r^{-\frac{1}{10,000}-1}} = 0$$

$$\int_{|x-y|<1} r^{-\frac{1}{10,000}} dx_1 dx_2 < \infty$$

$$n \geq 3: |u| \leq \frac{|C|}{r^{n-2}} \in L'(B_1(y)) \quad r^{n-1} dr$$

$$\int_{|x-y|<1} \frac{1}{r^{n-2}} dx_1 \dots dx_n < \infty$$

Since u chosen above $\in L'_{loc}(\mathbb{R}^n)$

$$\langle u, -\Delta \phi \rangle = - \int_{\mathbb{R}^n} u(x) \Delta \phi(x) dx \stackrel{\text{W.T.S.}}{=} \lim_{\varepsilon \rightarrow 0^+} - \int u_\varepsilon \Delta \phi$$

Now regularize u as

$$u_\varepsilon(x) = \begin{cases} \frac{C \ln \sqrt{r^2 + \varepsilon^2}}{2} & n=2 \\ \frac{C}{(r^2 + \varepsilon^2)^{\frac{n-2}{2}}} & n \geq 3 \end{cases}$$

$$\Rightarrow u_\varepsilon \in C^\infty(\mathbb{R}^n)$$

$$|u_\varepsilon(x)| \leq \frac{|C|}{2} |\ln r^2| \quad \text{if } r < \frac{1}{\varepsilon}$$

" $|u|$ ε small

$$\Rightarrow \forall k > 0, \exists \text{ const. } M > 0 \text{ s.t.}$$

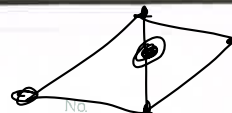
$$n=2: |u_\varepsilon(x)| \leq M |u(x)|, \quad \forall x \in B_R(y)$$

$\forall$ small $\varepsilon > 0$

$$\Rightarrow |u_\varepsilon \Delta \phi| \leq M |u \Delta \phi| \in L'(B_R(y))$$

Now by Lebesgue Dominated Conv. Th.

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} u_\varepsilon \Delta \phi dx = \int_{\mathbb{R}^n} u \Delta \phi dx$$



Date

$$\lim_{\varepsilon \rightarrow 0^+} - \int_{\mathbb{R}^n} u_\varepsilon \Delta \phi \, dx \stackrel{\text{2nd by parts}}{=} - \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} (\Delta u_\varepsilon) \phi \, dx$$

So we just need to show

$-\Delta u_\varepsilon \rightarrow \delta_y$ in the distributional sense

By a previous theorem, it suffices to show

(i) $\int_{\mathbb{R}^n} -\Delta u_\varepsilon(x) \, dx \xrightarrow{\varepsilon \rightarrow 0^+} 1$ (i)

(ii) $-\Delta u_\varepsilon$ concentrates at y as $\varepsilon \rightarrow 0^+$:

$$\forall \delta > 0, \int_{|x-y| \geq \delta} |-\Delta u_\varepsilon|(x) \, dx \xrightarrow{\varepsilon \rightarrow 0^+} 0$$

(iii) $\|-\Delta u_\varepsilon\|_{L^1(\mathbb{R}^n)} \leq \text{const. } C, \quad \forall \varepsilon > 0 \text{ small}$

$n=2$: Have to choose C s.t. $-2\pi C = 1$. (i)

$$\begin{aligned} \Delta u_\varepsilon &= (u_\varepsilon)_{rr} + \frac{1}{r} (u_\varepsilon)_r \\ &= \frac{C}{2} \left[\frac{2(\varepsilon^2 - r^2)}{(r^2 + \varepsilon^2)^2} + \frac{1}{r} \frac{2r}{r^2 + \varepsilon^2} \right] \\ &= \frac{C 2\varepsilon^2}{(r^2 + \varepsilon^2)^2} \int_{\mathbb{R}^2} -\Delta u_\varepsilon \, dx_1 \, dx_2 = -2 \\ &= -2C\varepsilon^2 \int_0^{2\pi} \int_0^\infty \frac{1}{(r^2 + \varepsilon^2)^2} r \, dr \, d\theta \\ &= -2\pi C\varepsilon^2 \int_0^\infty \frac{(2r \, dr)}{(r^2 + \varepsilon^2)^2} \\ &= -2\pi C\varepsilon^2 \int_{\varepsilon^2}^\infty \frac{dz}{2z^2} = \frac{-2\pi(\varepsilon^2 + \delta^2)}{\varepsilon^2 + \delta^2} = -2\pi \end{aligned}$$

Have $C = \frac{1}{2\pi}$

We have shown:

$$\langle u, -\Delta \phi \rangle = \phi(y), \text{ where } u(x) = \begin{cases} \frac{1}{2\pi} \ln|x-y|, & n=2 \\ \frac{1}{(n-2)\omega_n |x-y|^{n-2}}, & n \geq 3 \end{cases}$$

$$\int_{\mathbb{R}^n} u(x) (-\Delta \phi(x)) \, dx$$

$$\int_{\mathbb{R}^n} \Gamma(x-y) (-\Delta \phi(x)) \, dx = \Gamma_* (-\Delta \phi)$$

$$\Gamma(y-x)$$

$$\forall \phi \in C_0^\infty(\mathbb{R}^n)$$

Poisson eq.

$$-\Delta u = f(x)$$

$$u(x) = \int_{\mathbb{R}^n} P(x-y) f(y) dy \quad (\odot)$$

Theorem : $\forall f \in C_0^2(\mathbb{R}^n)$, define $u(x)$ by $(\odot)$

Then $u \in C^2(\mathbb{R}^n)$ & satisfies Poisson eq.

Proof . 1. $u(x)$ is well-defined. True, we have $f \in L^1_{loc}(\mathbb{R}^n)$

$$2. \forall \text{ fixed } x \in \mathbb{R}^n, u(x) = \int_{\mathbb{R}^n} P(y) f(x-y) dy$$

$$\frac{u(x+he_1) - u(x)}{h} = \int_{\mathbb{R}^n} \frac{f(x+he_1-y) - f(x-y)}{h} P(y) dy$$

$$= \int_{\mathbb{R}^n} \underbrace{f_{x+h} (x+se_1-y)}_{s \in (0,h)} P(y) dy$$

$\forall s \in (0,h)$ $\uparrow$ $\underbrace{\text{supp of } f_{x+h}}_{\text{some } C B_{R^1}^1}$ as a function of y

$$\Rightarrow 1. \quad 1 \leq |P(y)| \leq \|f_{x+h}\|_{L^1(\mathbb{R}^n)} \chi_{B_{R^1}^1}(y), \quad \forall y \in \mathbb{R}^n$$

$\uparrow$
 $L^1(\mathbb{R}^n)$? Yes!

Now by L.D.C..

$$\lim_{h \rightarrow 0} \frac{u(x+he_1) - u(x)}{h} = \int_{\mathbb{R}^n} \lim_{h \rightarrow 0} \frac{f(x+se_1-y)}{h} P(y) dy$$

$$= \int_{\mathbb{R}^n} f_{x+h} (x-y) P(y) dy$$

$\Rightarrow \frac{\partial u}{\partial x_1}$ exists in $\mathbb{R}^n$

Similarly all

$$\frac{\partial^2 u}{\partial x_i \partial x_j} \text{ exists} = \int_{\mathbb{R}^n} \int_{x_i x_j} (x-y) P_{ij}(y) dy$$

By L.C.D.T. can show all partials of orders ≤ 2 are cont. in $\mathbb{R}^n$

$$\text{Now } -\Delta u(x) = \int_{\mathbb{R}^n} -\Delta f(x-y) |^2(y) dy$$

$$= \int_{\mathbb{R}^n} |^2(x-y) (-\Delta f(y)) dy$$

Recall
old formula $f(x) = \int_{\mathbb{R}^n} |^2(x-y) (-\Delta f(y)) dy$

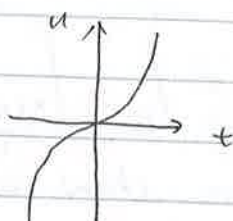
$$y = u(x)$$

$$dy = \left| \det \frac{\partial u}{\partial (x_1, \dots, x_n)} \right| dx$$

$$\left(\begin{array}{cc} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_n} \\ \vdots & \vdots \\ \frac{\partial u_n}{\partial x_1} & \frac{\partial u_n}{\partial x_n} \end{array} \right)$$

ODE

$$\begin{cases} \frac{du}{dt} = f(u) \\ u(0) = u_0 \end{cases}$$

 $\exists!$ SolSteady state

$$0 = f(u), \quad \exists! \text{ Sol}$$

No uniqueness in general

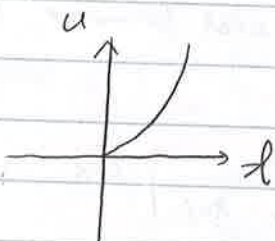
PDE— elliptic operator such as Δu

$$\begin{cases} \frac{\partial u}{\partial t} = Au + f(u) \\ \text{B.C.} \end{cases}$$

$$u(x,0) = \varphi(x)$$

$$\exists! \text{ Sol } u(x,t)$$

$$t > 0$$

Steady state

$$\begin{cases} 0 = Au + f(u) \\ \text{B.C.} \end{cases}$$

No uniqueness in general.

Weak derivatives

Def Suppose $u \in L^1_{loc}(\Omega)$ & the distributional derivative $\partial^\alpha u$ can be realized/regarded as a $L^1_{loc}(\Omega)$ function v , i.e. $\partial^\alpha u = v$ in the distributional sense

i.e. $\forall \varphi \in \mathcal{D}(\Omega) = C^\infty_0(\Omega)$

$$\langle \partial^\alpha u, \varphi \rangle = \int_\Omega v \varphi \, dx$$

$$(-1)^{|\alpha|} \int_\Omega u(x) \partial^\alpha \varphi(x) \, dx$$

Then we say v is 2 -th weak derivative of u
write: $v = \partial^\alpha u$ $|\alpha| = \alpha_1 + \dots + \alpha_n$

RK If $u \in C^{|\alpha|}(\Omega)$, then classical $\partial^\alpha u = \text{weak } \partial^\alpha u$

Def We say $u \in L^1_{loc}(\Omega)$ is k -times weakly differentiable if all weak $\partial^\alpha u$, $|\alpha| \leq k$ exists

Notation $W^k(\Omega) = \text{Set of all such } u\text{'s}$

Linear Space

~~eg. 1~~

$$f(x) = |x|, \quad \Omega = \mathbb{R}$$

$$\text{weak } f'(x) = ?$$

$$\forall \varphi \in \mathcal{D}(\mathbb{R}),$$

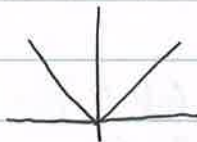
$$\int_{-\infty}^{\infty} f'(x) \varphi(x) \, dx = - \int_{-\infty}^{+\infty} f(x) \varphi'(x) \, dx$$

$$= - \int_0^{\infty} f(x) \varphi'(x) \, dx - \int_{-\infty}^0 f(x) \varphi'(x) \, dx$$

$$= \int_0^{\infty} f'(x) \varphi(x) \, dx - f(x) \varphi(x) \Big|_{x=0}^{x=\infty}$$

$$+ \int_{-\infty}^0 f'(x) \varphi(x) \, dx$$

0



$$\in L'_{loc}(\mathbb{R})$$

$$= \int_{-\infty}^{\infty} \left(\begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases} \right) \varphi(x) dx$$

$$\Rightarrow \text{weak } f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

eg. 2 weak f'' exist? $2\delta_0(x)$

if weak f'' exists, then $f'' \in L'_{loc}(\mathbb{R})$, satisfy

$$\int_{-\infty}^{\infty} f'' \varphi dx = - \int_{-\infty}^{\infty} f'(x) \varphi'(x) dx$$

$$= - \int_0^{\infty} \varphi'(x) dx + \int_{-\infty}^0 \varphi'(x) dx$$

$$\forall \varphi \in \mathcal{D}(\mathbb{R}) = 2\varphi(0)$$

Take $\varphi(x) = j(\frac{x}{\varepsilon})$.

$j(x) = \text{mollifier}$

$$= \begin{cases} C \exp(\frac{1}{|x|^2-1}) & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1 \end{cases}$$

$$\int_{\mathbb{R}^n} j(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} f''(x) j(\frac{x}{\varepsilon}) dx = 2j(0) = 2Ce^{-1} > 0$$

$$|f''(x) j(\frac{x}{\varepsilon})| \leq |f''(x)| \chi_{(-1,1)}(x) j(0)$$

$$\cap L'(\mathbb{R})$$

Now by L.D.C.T

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} f''(x) j(\frac{x}{\varepsilon}) dx = \int_{-\infty}^{\infty} 0 dx = 0 \Rightarrow C=0$$

DNE

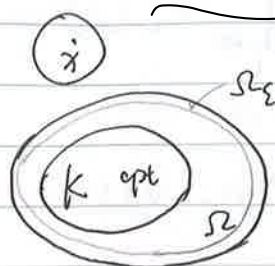
$$\begin{matrix} \in C^\infty \\ u_k \end{matrix} \xrightarrow{\text{in } L^p} u \in L^p$$

$$\partial^\alpha u_k \rightarrow \partial^\alpha u$$

Approximating bad functions by good ones

Def $\forall u \in L^1_{loc}(\Omega)$, the regularization of u is

$$u_\varepsilon(x) = \int_{\Omega} j_\varepsilon(x-y) u(y) dy$$



$$K \subset \Omega$$

$$u \in L^1(K)$$

well-defined for $x \in \Omega_\varepsilon$

$$\Omega_\varepsilon \triangleq \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > \varepsilon\}$$

$\forall u \in L^1(\Omega)$, u_ε well-defined on $\mathbb{R}^n$

$$u_\varepsilon(x) \equiv 0, \quad \forall x \text{ s.t. } \text{dist}(x, \Omega) > \varepsilon$$

$$u \in L^p(\mathbb{R}^n) \Rightarrow$$

$$\|u(\cdot+h) - u(\cdot)\|_{L^p(\mathbb{R}^n)}$$

$\downarrow$ 鲁津定理

0 as $h \rightarrow 0$

Egoroff's
Thm.

$$\int_{\mathbb{R}^n} j_\varepsilon(x-y) dy = 1$$

$$\Downarrow_{\varepsilon \rightarrow 0^+}$$

$$j_\varepsilon(x-y) \rightarrow \delta_0(x-y)$$

Lemma 1 $u_\varepsilon \in C^\infty(\Omega_\varepsilon)$. Moreover, if $u \in L^1(\Omega)$, then $u_\varepsilon \in C^\infty(\mathbb{R}^n)$ ($\in C_0^\infty(\mathbb{R}^n)$ if Ω bdd)

Proof $\varepsilon > 0$ fixed!

$\forall$ fixed $x \in \Omega_\varepsilon$, $\exists \delta > \varepsilon$, s.t. $\overline{B_\delta(x)} \subset \Omega$

if z is closed to x , then $B_\varepsilon(z) \subset B_\delta(x)$

Let $e_i = (0, \dots, 0, 1, 0, \dots, 0)$

$\uparrow$
i-th place

if $s \in \mathbb{R}$ is small, then $B_\varepsilon(x+se_i) \subset B_\delta(x)$

Observe
$$\frac{u_\varepsilon(x+se_i) - u_\varepsilon(x)}{h} = \int_{\mathbb{R}^n} \frac{j_\varepsilon(x+h\frac{-y}{h}) - j_\varepsilon(x-y)}{h} u(y) dy$$



Lemma 2 If $u \in C^0(\Omega)$, then $\forall \Omega' \subset \subset \Omega$ (i.e. $\bar{\Omega}'$ cpt, $\bar{\Omega}' \subset \Omega$)
 $u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0^+} u$ in $C^0(\bar{\Omega}')$
 ($u \in L^p(\Omega)$)

Lemma 3 Let $1 \leq p < \infty$. If $u \in L^p_{loc}(\Omega)$, then
 $u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0^+} u$ in $L^p_{loc}(\Omega)$ (i.e. $\forall \Omega' \subset \subset \Omega$)
 ($L^p(\Omega)$) $u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0^+} u$ in $L^p(\Omega')$

RK. $L^p(\Omega) \cap C^0_b(\Omega)$ is dense in $L^p(\Omega)$ (from above)
 $C^0_b(\Omega)$ is dense in $L^p(\Omega)$ (later)
 $1 \leq p < \infty$ $W^{k,p}$
 $\uparrow$ $u_k \rightarrow u \in L^\infty$ - 一致收敛 $\Rightarrow u$ 光滑

Proof. $\forall x \in \Omega' \subset \subset \Omega$ (cptly contained)

$$u_\varepsilon(x) = \int_{\Omega} j_\varepsilon(x-y) u(y) dy$$

$$\begin{aligned} z = \frac{x-y}{\varepsilon} \\ dz = \frac{1}{\varepsilon^n} dy \end{aligned} \quad \int_{|z| \leq 1} j(z) u(x-\varepsilon z) dz \quad \text{if } \varepsilon \text{ small}$$

$$u(x) = \int_{|z| \leq 1} j(z) u(x) dz$$

$$\|u_\varepsilon - u\|_{L^p(\Omega')} = \left\| \int_{B_{1/2}} (u(\cdot - \varepsilon z) - u(\cdot)) j(z) dz \right\|_{L^p(\Omega')}$$

Minkowski's ineq.

$$\checkmark \|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$$

$$\checkmark \left\| \sum_{i=1}^n f_i \right\|_{L^p} \leq \sum_{i=1}^n \|f_i\|_{L^p}$$

$$\checkmark \left\| \int_{\mathbb{R}^n} f(\cdot - y) dy \right\|_{L^p} \leq \int_{\mathbb{R}^n} \|f(\cdot - y)\|_{L^p} dy$$

$$du = f(z) dz$$

No

Date

$$\|f(z)\|_{L^p} \in \mathbb{C}$$

$$\leq \int_{B_{1/2}} \|u(\cdot - \varepsilon z) - u(\cdot)\|_{L^p(\Omega)} f(z) dz$$

subdomain

$$\left(\int_{\Omega'} |u(x) - u(x - \varepsilon z)|^p dx \right)^{\frac{1}{p}}$$

Take $\Omega'' \subset \subset \Omega$

$$\Omega' \subset \subset \Omega'' \subset \subset \Omega$$

$$\Downarrow$$

$$x - \varepsilon z \in \Omega''$$

$$\forall x \in \Omega'$$

$$\forall z \in B_{1/2}$$

$$\forall \text{ small } \varepsilon > 0$$

Define

$$u_{\text{new}}(x) = \begin{cases} u(x), & x \in \Omega'' \\ 0, & x \notin \Omega'' \end{cases}$$

$$\text{Now, } \|u_\varepsilon - u\|_{L^p(\Omega')} \leq \int_{B_{1/2}} \|u_{\text{new}}(\cdot) - u_{\text{new}}(\cdot - \varepsilon z)\|_{L^p(\mathbb{R}^n)} f(z) dz$$

$$\text{Since } u \in L^p_{\text{loc}}(\Omega), u \in L^p(\Omega'')$$

$$\Rightarrow u_{\text{new}} \in L^p(\mathbb{R}^n)$$

By continuity in L^p -norm,

$$\|u_{\text{new}}(\cdot) - u_{\text{new}}(\cdot - h)\|_{L^p(\mathbb{R}^n)} \rightarrow 0 \text{ as } h \rightarrow 0$$

$$\Rightarrow \rightarrow 0 \text{ as } \varepsilon \rightarrow 0 \text{ uniformly}$$

$$\Rightarrow \|u_\varepsilon - u\|_{L^p(\Omega')} \xrightarrow{\varepsilon \rightarrow 0} 0 \text{ for } z \in B_{1/2}$$

$\forall u \in L^p(\Omega)$, extend $u \equiv 0$ on Ω^c

$$\begin{aligned} u_\varepsilon(x) &= \int_{\Omega} f_\varepsilon(x-y) u(y) dy = \int_{\mathbb{R}^n} f_\varepsilon(x-y) u(y) dy \\ &= \int_{\mathbb{R}^n} f(z) u(x - \varepsilon z) dz \end{aligned}$$

QED

RK $C_0^\infty(\Omega)$ is dense in $L^p(\Omega)$ ($1 \leq p < \infty$)

Proof $\forall u \in L^p(\Omega)$

Define $\Omega_k = \{x \in \Omega \mid |x| < k, \text{dist}(x, \partial\Omega) > \frac{1}{k}\}$

Theorem Let $(u, v) \in L'_{loc}(\Omega)$

Then $v = \text{weak } \partial^2 u$



$\exists \{u_k\}_{k=1}^{\infty} \subset C^{\infty}(\Omega)$ s.t.

$u_k \rightarrow u$ in $L'_{loc}(\Omega)$

classical $\partial^2 u_k \rightarrow v$ in $L'_{loc}(\Omega)$

Proof: " $\Leftarrow$ ". $\forall \varphi \in \mathcal{D}(\Omega)$

$$\int \partial^2 u_k(x) \varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} u_k(x) \partial^2 \varphi(x) dx$$

$$\downarrow \qquad \qquad \qquad \downarrow$$

$$\int_{\Omega} v(x) \varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} u(x) \partial^2 \varphi(x) dx$$

$$\Rightarrow v(x) = \text{weak } \partial^2 u(x)$$

" $\Rightarrow$ " Let $\Omega_k = \{x \in \Omega \mid |x| < k, \text{dist}(x, \partial\Omega) > \frac{1}{k}\}$

For k large,

$$\text{Let } u_k(x) = \int_{\Omega_k} j_{\frac{1}{k}}(x-y) u(y) dy, \quad x \in \Omega_k$$

$$\therefore u \in L'(\Omega_k)$$

$\therefore$ by lemma 1, $u_k \in C^{\infty}(\mathbb{R}^n)$

$\forall \Omega' \subset\subset \Omega, \exists \frac{1}{k_0} \gg 1$ s.t. $\Omega' \subset\subset \Omega_{\frac{1}{k_0}}$ if $k \geq k_0$

$$\forall x \in \Omega', \quad u_k(x) = \int_{\Omega_k} j_{\frac{1}{k}}(x-y) u(y) dy$$

$$= \int_{\Omega_{k_0}} j_{\frac{1}{k}}(x-y) u(y) dy \quad \text{if } k \geq k_0$$

$$\text{By lemma 3, } \frac{u_k \text{ in } L'(\Omega_{k_0})}{\varepsilon \rightarrow 0} \rightarrow u$$

$$\Rightarrow u_k \rightarrow u \text{ in } L'(\Omega')$$

By lemma 4 (with $\Omega = \Omega_{k_0}$),

on Ω' , we have

$$\partial^\alpha (u_k)(x) = (\partial^\alpha u)_k(x), \quad x \in \Omega'$$

$$= (v)_k(x)$$

$\downarrow \leftarrow$ Lemma 3 (with $\Omega = \Omega_{k_0}$)
 $v(x)$ in $L^1(\Omega')$

QED

Theorem Let $u, v \in L^1_{loc}(\Omega)$. Then $v = \text{weak } \partial^\alpha u \Leftrightarrow$
 $\exists \{u_k\}_{k=1}^\infty \subset C^\infty(\Omega)$

s.t. $u_k \rightarrow u$

classical $\partial^\alpha u_k \rightarrow v$ in $L^1_{loc}(\Omega)$

RK1 This implies $C^\infty(\Omega)$ is dense in $W^1(\Omega)$ in some sense

RK2 To have " $\Leftarrow$ ", only need $\{u_k\} \subset C^{|\alpha|}(\Omega)$
 (Same proof)

(Chain Rule) Suppose $f(u) \in C^1(\mathbb{R})$, $f' \in L^\infty(\mathbb{R})$ &
 $u(x) \in W^1(\Omega)$. Then $f'(u(x)) \in W^1(\Omega)$ &
 weak $\partial_{x_i}(f(u(x))) = f'(u(x)) \partial_{x_i} u$, $i=1, \dots, n$

Proof: $\because u \in W^1(\Omega)$

$\therefore$ by previous Th, $\exists \{u_k\}_{k=1}^\infty \subset C^\infty(\Omega)$ s.t.
 $u_k \rightarrow u$ in $L^1_{loc}(\Omega)$

weak $\nabla u_k \rightarrow \nabla u$

consider $\{f(u_k(x))\}_{k=1}^\infty \subset C^1(\Omega)$

W.T.S.: $f(u_k) \xrightarrow{k \rightarrow \infty} f(u)$

$\nabla f(u_k) \xrightarrow{k \rightarrow \infty} f'(u) \nabla u$ in $L^1_{loc}(\Omega)$

$$\forall \Omega' \subset \subset \Omega,$$

$$\int_{\Omega'} |f(u_k(x)) - f(u(x))| dx \leq \int_{\Omega'} \|f'\|_{L^\infty} |u_k(x) - u(x)| dx$$

↓ as $k \rightarrow \infty$
0

$$\text{b.c. } u_k \rightarrow u \text{ in } L^1_{loc}(\Omega')$$

$$\int_{\Omega} |f'(u_k) \nabla u_k - f'(u) \nabla u|(x) dx$$

$$= f'(u_k) \nabla u + f'(u) \nabla u$$

$$\leq \int_{\Omega} \|f'\|_{\infty} |\nabla u_k - \nabla u| dx + \int_{\Omega'} |f'(u_k) - f'(u)| |\nabla u|(x) dx$$

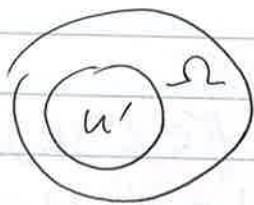
↓
0

$$u_k \rightarrow u \text{ a.e.}$$

$$f'(\cdot) \in C'$$

$\Rightarrow$ 依测度收敛 $\Rightarrow$ 抽子列

a.e. on Ω
↓
after passing
to a subseq

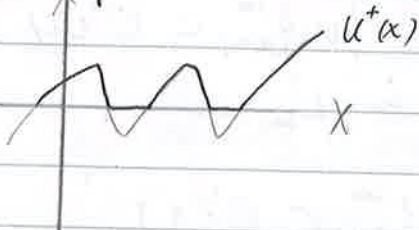


$$\xrightarrow{k \rightarrow \infty} 0 \text{ by L.D.C.T.}$$

QED

Theorem

If $u \in W^1(\Omega)$, then $u^+, u^-, |u| \in W^1(\Omega)$



$$\text{and } \nabla u^+(x) = \begin{cases} \nabla u(x) & \text{if } u(x) > 0 \\ 0 & \text{if } u(x) \leq 0 \end{cases}$$

$$\nabla u^-(x) = \begin{cases} \nabla u(x) & \text{if } u(x) < 0 \\ 0 & \text{if } u(x) \geq 0 \end{cases}$$

$$\nabla |u(x)| = \begin{cases} \nabla u(x) & \text{if } u(x) > 0 \\ -\nabla u(x) & \text{if } u(x) < 0 \\ 0 & \text{if } u(x) = 0 \end{cases}$$

Proof $\forall \varepsilon > 0$, let $f_\varepsilon(u) = \begin{cases} \sqrt{u^2 + \varepsilon^2} - \varepsilon & \text{if } u > 0 \\ 0 & \text{if } u \leq 0 \end{cases}$

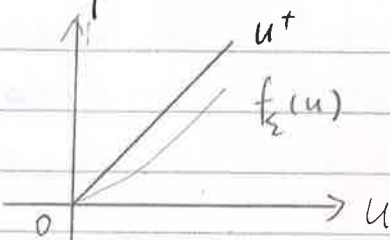
$\Rightarrow \bullet f_\varepsilon \in C^1(\mathbb{R})$

$$\bullet f'_\varepsilon(u) = \frac{u^+}{\sqrt{u^2 + \varepsilon^2}} \leq 1$$

$$\bullet 0 \leq f_\varepsilon(u) \leq u^+, \quad \forall u \in \mathbb{R}$$

$\bullet$ when $\varepsilon \rightarrow 0$, $f_\varepsilon(u) \rightarrow u^+$ pointwise

$$\Rightarrow f_\varepsilon(u(x)) \in W^1(\Omega)$$



By chain rule, $\nabla f_\varepsilon(u(x))$ exists & $= f'_\varepsilon(u(x)) \nabla u$
 $= \frac{u^+ \nabla u}{\sqrt{u^2 + \varepsilon^2}}$

$$\Rightarrow f_\varepsilon(u(x)) \in W^1(\Omega)$$

Now $\forall \varphi \in \mathcal{D}(\Omega)$

def of weak deriv.

opt supp.

$$\int_\Omega \frac{u^+ \nabla u}{\sqrt{u^2 + \varepsilon^2}} \varphi(x) dx$$

$$= - \int_\Omega (f_\varepsilon(u) \nabla \varphi) dx$$

$$\downarrow \begin{matrix} \text{opt supp.} \\ \downarrow \\ \varphi \in L^1(\Omega) \end{matrix}$$

Sending $\varepsilon \rightarrow 0$, we have $\int_\Omega u^+ \nabla \varphi(x) dx$ by LDC

$$\int_\Omega \chi_{\{u > 0\}} \nabla u(x) \varphi(x) dx$$

$$\int_\Omega u^+ \nabla \varphi(x) dx \text{ by LDC}$$

$$\Rightarrow \text{weak } \nabla u^+ = \chi_{\{u > 0\}}(x) \nabla u(x) = \begin{cases} \nabla u & \text{if } u > 0 \\ 0 & \text{if } u \leq 0 \end{cases}$$

$$\nabla u^- \text{ similar} \quad |u| = u^+ - u^- \in W^1(\Omega)$$

$$|u| = \underbrace{u^+}_{\in W'(\Omega)} - \underbrace{u^-}_{\in W'(\Omega)} \stackrel{\text{So}}{\in} W'(\Omega)$$

$$\nabla |u| = \nabla u^+ - \nabla u^- = \begin{cases} \nabla u(x) & \text{if } u(x) > 0 \\ 0 & \text{if } u(x) \leq 0 \end{cases} - \begin{cases} \nabla u(x) & \text{if } u(x) < 0 \\ 0 & \text{if } u(x) \geq 0 \end{cases}$$

$$= \begin{cases} \nabla u(x) & \text{if } u(x) > 0 \\ -\nabla u(x) & \text{if } u(x) < 0 \\ 0 & u(x) = 0 \end{cases}$$

↑ 七姑八大姨的生日都行

Corollary $u \in W'(\Omega) \Rightarrow \nabla u \equiv 0$ on any set where $u \equiv \text{const. } C$

Proof $\nabla u = \nabla(u - C)$
 $= \nabla[(u - C)^+ + (u - C)^-]$
 $= 0 \quad \text{if } (u - C)(x) = 0$

Sobolev space $W^{k,p}(\Omega)$

$\forall 1 \leq p < \infty$, integer $k \geq 0$

$$W^{k,p}(\Omega) = \left\{ u \in L^p(\Omega) \mid \text{weak } \partial^\alpha u \in L^p(\Omega) \quad \forall \alpha \text{ s.t. } |\alpha| \leq k \right\}$$

Linear space

Norm $\|u\|_{W^{k,p}(\Omega)} = \left(\int_{\Omega} \left(\sum_{|\alpha| \leq k} |\partial^\alpha u|^p \right) dx \right)^{\frac{1}{p}}$

Another equivalent norm $\|u\|_{W^{k,p}(\Omega)} = \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^p(\Omega)}$

$\|u\|_{W^{k,p}(\Omega)}^p \geq \|u\|^p \quad \left((\sum a_i)^p \geq \sum a_i^p \right)$

$\|u\|_{W^{k,p}(\Omega)}^p \geq \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^p(\Omega)}^p = C \|u\|^p \quad \left| \sum a_i \right| \leq \left(\sum |a_i|^p \right)^{\frac{1}{p}} \left(\sum 1 \right)^{\frac{p-1}{p}}$

choose
More Notation: $C_0^\infty(\Omega) \subset W^{k,p}(\Omega)$

Take closure of $C_0^\infty(\Omega)$ with respect to $W^{k,p}$ -norm
 $\Rightarrow W_0^{k,p}(\Omega)$

$$\begin{cases} \Delta u = f(x) \\ u|_{\partial\Omega} = 0 \end{cases}$$

$u \in W_0^{k,p}(\Omega)$ u 在弱意义下等于 0

when $p=2$, $W^{k,p}(\Omega)$ is denoted as $H^k(\Omega)$

In $H^k(\Omega)$, introduce inner product

$$(u, v)_{H^k(\Omega)} = \sum_{|\alpha| \leq k} \int_{\Omega} \partial^\alpha u(x) \partial^\alpha v(x) dx$$

Hilbert

Theorem $\forall 1 \leq p \leq \infty$, $W^{k,p}(\Omega)$ is Banach space

Proof $p=\infty$ easier

Suppose $1 \leq p < \infty$, check $\|u\|_{W^{k,p}(\Omega)}$ is a norm

$$\bullet \|cu\|_{W^{k,p}} = |c| \cdot \|u\|_{W^{k,p}}$$

$$\bullet \|u\|_{W^{k,p}} = 0 \Rightarrow \|u\|_{L^p} = 0 \Rightarrow u = 0 \text{ a.e. on } \Omega$$

$$\bullet \|u+v\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \int_{\Omega} |\partial^\alpha u + \partial^\alpha v|^p dx \right)^{\frac{1}{p}} \Rightarrow u=0$$

$$= \left(\int_{\Omega} \left[\left(\sum_{|\alpha| \leq k} |\partial^\alpha u + \partial^\alpha v|^p \right)^{\frac{1}{p}} \right]^p dx \right)^{\frac{1}{p}}$$

$$\stackrel{\text{discrete}}{\leq} \left(\left[\left(\sum_{|\alpha| \leq k} |\partial^\alpha u|^p \right)^{\frac{1}{p}} \right]^p + \left[\left(\sum_{|\alpha| \leq k} |\partial^\alpha v|^p \right)^{\frac{1}{p}} \right]^p \right)^{\frac{1}{p}}$$

$$\stackrel{\text{Minkowski}}{\leq} \int_{\Omega} \left(\sum_{|\alpha| \leq k} |\partial^\alpha u|^p \right)^{\frac{1}{p}} + \left(\sum_{|\alpha| \leq k} |\partial^\alpha v|^p \right)^{\frac{1}{p}} dx$$

$$= \| \text{☺} + \text{☹} \|_{L^p(\Omega)}$$

Minkowski

$$\leq \| \text{☺} \|_{L^p} + \| \text{☹} \|_{L^p(\Omega)}$$

$$= \|u\|_{W^{k,p}} + \|v\|_{W^{k,p}}$$

Completeness: Suppose u_k is Cauchy in $W^{k,p}(\Omega)$
 $\Rightarrow \{u_k\}$ is Cauchy in $L^p(\Omega) \Rightarrow u_k \rightarrow$ some u_∞ in $L^p(\Omega)$
 $\partial^\alpha u_k$ is Cauchy in $L^p(\Omega)$, $|\alpha| \leq k \Rightarrow \partial^\alpha u_k \rightarrow v_\alpha$ in $L^p(\Omega)$

W.T.S. $v_\alpha = \text{weak } \partial^\alpha u_\infty$

why: $\forall \varphi \in \mathcal{D}(\Omega)$

$$\int_{\Omega} \partial^\alpha u_k(x) \varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} u_k(x) \partial^\alpha \varphi(x) dx$$

$$\downarrow \qquad \qquad \downarrow$$

$$\int_{\Omega} v_\alpha(x) \varphi(x) dx \stackrel{S_n}{=} (-1)^{|\alpha|} \left(\int_{\Omega} u_\infty(x) \partial^\alpha \varphi(x) dx \right)$$

$\Rightarrow v_\alpha = \text{weak } \partial^\alpha u_\infty$

Q.E.D.

$k \geq 0$ integers : $1 \leq p < \infty$

$W^{k,p}(\Omega) \leftarrow$ Sobolev space

$$= \{ u \in L^p(\Omega) \mid \text{weak } \partial^\alpha u \in L^p(\Omega), \forall \alpha \text{ s.t. } |\alpha| \leq k \}$$

$$\|u\|_{W^{k,p}(\Omega)} = \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^p(\Omega)}$$

Theorem : $W^{k,p}(\Omega)$ is a Banach space.

Theorem If $1 < p < \infty$, the $W^{k,p}(\Omega)$ is separable & reflexive

Proof : Define mapping $T: W^{k,p}(\Omega) \rightarrow \prod_{|\alpha|=0}^k L^p(\Omega)$

$$Tu = \{\partial^\alpha u\}_{|\alpha|=0}^k$$

$$\Rightarrow \|Tu\|_{\prod_{|\alpha|=0}^k L^p(\Omega)} = \|u\|_{W^{k,p}(\Omega)}$$

So T is an isometry & linear

Since $W^{k,p}$ is Banach, $T(W^{k,p})$ is closed subspace of $\prod_{|\alpha|=0}^k L^p(\Omega)$

$M'' \leftarrow$ denote

On the other hand, $\prod L^p(\Omega)$ is separable & reflexive

By Functional Analysis, M must be "↓"
 $\Rightarrow W^{k,p}$ is also separable & reflexive.

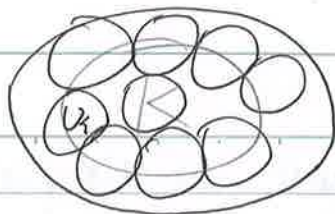
Partition of unity Def.

Let $\{U_i\}_{i=1}^\infty$ be bdd & open sets in Ω , s.t.

(i) $\bar{U}_i \subset \Omega$

(ii) every cpt K in Ω intersects only finitely many U_i 's

$$\{U_i\} \quad \bigcup_{i=1}^{\infty} U_i = \Omega.$$



A partition of unity subordinate to the covering $\{U_i\}$ of Ω is $\{\phi_i\}_{i=1}^{\infty} \subset C_0^{\infty}(\Omega)$, s.t.

1. $\phi_i \geq 0$;
2. $\text{supp } \phi_i \subset U_i$;
3. $\sum_{i=1}^{\infty} \phi_i(x) = 1, \quad \forall x \in \Omega$

Theorem partition of unity exists

Proof Step 1 construct a new open covering $\{V_i\}_{i=1}^{\infty}$

$$\text{s.t. } V_i \subset U_i, \quad \bigcup_{i=1}^{\infty} V_i = \Omega$$

$$\text{Let } F_1 = \bar{U}_1 \setminus \bigcup_{i=2}^{\infty} U_i \Rightarrow F_1 \text{ is closed}$$

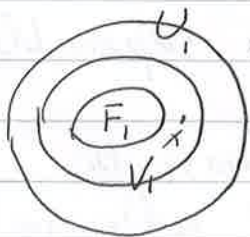
$$F_1 \cup \left(\bigcup_{i=2}^{\infty} U_i \right) = \Omega$$

$$\text{Since } \bar{U}_1 \subset \Omega, \quad \partial U_1 \subset \Omega = \bigcup_{i=2}^{\infty} U_i \Rightarrow$$

$$\partial U_1 \subset \bigcup_{i=2}^{\infty} U_i \Rightarrow F_1 \subset V_1 \quad (\bar{V}_1 = U_1 \cup \partial V_1)$$

$$\Rightarrow \text{dist}(F_1, \partial U_1) > 0$$

$$\text{Define } V_1 = \left\{ x \in U_1 \mid \text{dist}(x, F_1) < \frac{1}{2} \text{dist}(F_1, \partial U_1) \right\}$$



$$\Rightarrow V_1 \text{ open } F_1 \subset V_1 \subset \bar{V}_1 \subset U_1$$

$$\bigcup_{i=1}^{\infty} V_i = \Omega$$

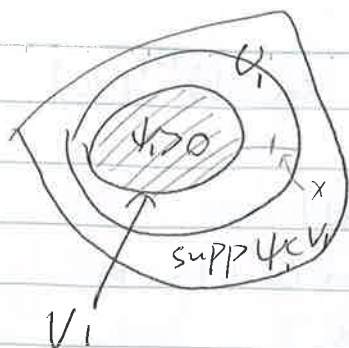
$$\text{Let } F_2 = \bar{U}_2 \setminus \left(\bigcup_{i=3}^{\infty} U_i \cup V_1 \right)$$

$$\Rightarrow F_2 \text{ closed}$$

$$F_2 \cup \left(\bigcup_{i=3}^{\infty} U_i \cup V_1 \right) = \Omega$$

$$F_2 \subset U_2 \Rightarrow \exists V_2 \text{ open s.t. } F_2 \subset V_2 \subset \bar{V}_2 \subset U_2$$

$$\Rightarrow \bigcup_{i=1}^{\infty} V_i = \Omega$$



Step 2. Construct $\psi_i \in C_0^\infty(U_i)$ s.t. $\psi_i \geq 0$
 $\psi_i(x) > 0, \forall x \in \bar{V}_i$
 Let $\psi_i(x) = \int_{V_i} \chi_\varepsilon(x-y) \chi_{V_i}(y) dy$
 把特征函数磨光.

$$(\chi_{V_i})_\varepsilon(x) \Big|_{\text{supp } \chi_\varepsilon(x)}$$

$$\Rightarrow \psi_i \in C^\infty(\mathbb{R}^n)$$

Take $\varepsilon < \frac{1}{2} \text{dist}(\bar{V}_i, \partial U_i)$

• Then if $x \in V_i$, $\text{dist}(x, \bar{V}_i) > \frac{1}{2} \text{dist}(\bar{V}_i, \partial U_i)$

We have $\psi_i(x) = 0 \Rightarrow \psi_i \in C_0^\infty(U_i)$

• If $x \in \bar{V}_i$, then $|B_\varepsilon(x) \cap V_i| > 0 \Rightarrow \psi_i(x) > 0$

Step 3 Let $\psi(x) = \sum_{i=1}^N \psi_i(x)$

This is actually a finite sum for each fixed $x \in \Omega$

Since $\bigcup_{i=1}^N V_i = \Omega$ & $\psi_i > 0$ on $\bar{V}_i$, $\psi(x) > 0$

$\forall$ fixed $x_0 \in \Omega$, take tiny ball $\bar{B}_\delta(x_0) \subset \Omega$

By assumption, only finitely many U_i 's intersect $\bar{B}_\delta(x_0)$

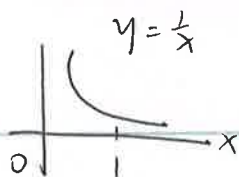
$\Rightarrow$ in $B_\delta(x_0)$, $\sum \psi_i$ is a finite sum $\Rightarrow \psi \in C^\infty(B_\delta(x_0))$
 $\Rightarrow \psi \in C^\infty(\Omega)$

$$1 = \frac{\psi(x)}{\psi(x)} = \sum_{i=1}^N \left(\frac{\psi_i(x)}{\psi(x)} \right) \phi_i(x) \quad \phi_i \in C_0^\infty(V_i)$$

RE If Ω is bdd & $\bar{\Omega} \subset \bigcup_{i=1}^k U_i$ open & bdd

then $\exists \psi_i \in C_0^\infty(U_i)_{i=1}^k$ s.t. $0 \leq \psi_i \leq 1$

$$\sum_{i=1}^k \psi_i(x) = 1, \quad \forall x \in \bar{\Omega}$$



属于 C^∞ , 不属于 $W^{k,p}(\Omega)$

Density

Theorem

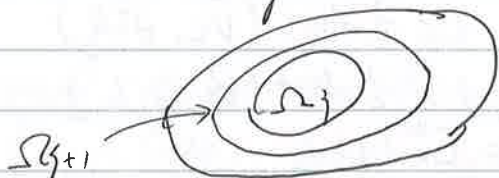
$C^\infty(\Omega) \cap W^{k,p}(\Omega)$ is dense in $W^{k,p}(\Omega)$ if $1 \leq p < \infty$

(在 $p = \infty$ 时, C^∞ 不是 dense 的)

RK. If Ω is bdd & $\partial\Omega \in C^1$, then $C^\infty(\bar{\Omega})$ is dense in $W^{k,p}$ $1 \leq p < \infty$

Proof: Take large $R > 0$, s.t. $B_R(0) \cap \Omega \neq \emptyset$

Let $\Omega_j = \{x \in \Omega \mid |x| < R+j, \text{dist}(x, \partial\Omega) > \frac{1}{j}\}$



Then $\Omega_j \subset \subset \Omega_{j+1} \subset \subset \Omega$

$\Omega_j \uparrow$

$\Rightarrow \{\underbrace{\Omega_{j+1} \setminus \bar{\Omega}_{j-1}}_{\text{open}}\}_{j=0}^\infty$ covers Ω (let $\Omega_{-1} = \Omega_0 = \emptyset$)

Satisfying: $\forall$ cpt $K \subset \Omega$, only finitely many $\Omega_{j+1} \setminus \bar{\Omega}_{j-1}$ intersect K

Now by partition of unity, $\exists \{\varphi_j\}_{j=0}^\infty$ s.t. $\varphi_j \geq 0$, $\text{supp } \varphi_j \subset \Omega_{j+1} \setminus \bar{\Omega}_{j-1}$, $\sum_{j=0}^\infty \varphi_j = 1$ on Ω , $\varphi_j \in C^\infty$

$\forall u \in W^{k,p}(\Omega)$, $u = \sum_{j=0}^\infty (\varphi_j u) \in W^{k,p}(\Omega)$

By Lemma 4, $\forall$ small $h > 0$

$(\partial^\alpha(\varphi_j u))_h = \partial^\alpha((\varphi_j u)_h)$, $\forall |\alpha| \leq k$

By Lemma 3, $(\partial^\alpha(\varphi_j u))_h \xrightarrow{h \rightarrow 0} \partial^\alpha(\varphi_j u)$ in $L^p(\Omega)$

$\Rightarrow (\varphi_j u)_h \xrightarrow{h \rightarrow 0} \varphi_j u$ in $W^{k,p}$

Now $\forall \varepsilon > 0$, for each $j \geq 0$, choose h_j so small that $(\varphi_j u)_{h_j}$ is supported in $\Omega_{j+1} \setminus \bar{\Omega}_{j-1}$

$$\bullet \quad \| \varphi_j u - (\varphi_j u)_{h_j} \|_{W^{k,p}(\Omega)} < \frac{\varepsilon}{2^j}$$

Let $v_j = (\varphi_j u)_{h_j}$. $\forall \Omega' \subset \subset \Omega$, only finitely many v_j 's do not vanish on Ω'

Then $v \stackrel{\text{def}}{=} \sum_{j=0}^{\infty} v_j$ is well-defined on Ω
 $\& v \in C^0(\Omega)$ (b/c. $v_j \in C^0(\Omega)$)

Moreover, $\|u - v\|_{W^{k,p}} \leq \sum_{j=0}^{\infty} \|\varphi_j u - v_j\|_{W^{k,p}}$

$$= \sum_{j=0}^{\infty} \frac{\varepsilon}{2^j} = 2\varepsilon$$

$$\Rightarrow v \in$$

$$W_0^{1,p}(\Omega) = \text{closure of } C_0^\infty(\Omega) \text{ in } W^{1,p}(\Omega)$$

$$W^{1,p}(\Omega) = \{u \in L^p(\Omega) \mid \text{weak } \nabla u \in L^p(\Omega)\}$$

Sobolev inequality

$$1 \leq p < \infty$$

$$(i) \forall u \in W_0^{1,p}(\Omega) \quad \Omega \subset \mathbb{R}^n$$

$$\|u\|_{L^{\frac{np}{n-p}}(\Omega)} \leq C(n,p) \|\nabla u\|_{L^p(\Omega)} \quad \text{if } 1 \leq p < n.$$

(Ω may or may not be bdd)

$$(ii) \forall u \in W_0^{1,p}(\Omega)$$

$$\|u\|_{L^\infty(\Omega)} \leq C(n,p) |\Omega|^{\frac{1}{n}-\frac{1}{p}} \|\nabla u\|_{L^p(\Omega)} \quad \text{if } p > n \text{ \& } \Omega \text{ bdd}$$

Moreover $u \in C^0(\bar{\Omega})$. (Rk. In fact, $u \in C^{1-\frac{1}{p}}(\bar{\Omega})$)

Corollary 1 (Poincaré's ineq)

if Ω is bdd, then $\leftarrow$ constant

$$\|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}$$

not true if "b" is removed. cons
 $u \equiv 1$
 $\forall u \in W_0^{1,p}(\Omega)$

Proof

if $1 \leq p < n$, then this is just (i) of Sobolev

if $p = n$, $u \in W^{1,p}(\Omega) \subset W^{1,n-\varepsilon}(\Omega) \quad \forall \text{ small } \varepsilon > 0$
 (走不下去了)

if $p > n$,

$$\|u\|_{L^p(\Omega)} \leq \|u\|_{L^\infty(\Omega)} |\Omega|^{\frac{1}{p}} \stackrel{(ii)}{\leq} C(n,p) |\Omega|^{\frac{1}{n}} \|\nabla u\|_{L^p(\Omega)}$$

Corollary 2 (Suppose Ω is bdd)

$$(i) \text{ if } 1 \leq p < \infty, \text{ then } W_0^{1,p}(\Omega) \hookrightarrow L^{\frac{np}{n-p}}(\Omega)$$

($X \hookrightarrow Y$ means: $\bullet$ X subspace of Y)

$$x \in X \xrightarrow{i} x \in Y$$

$\bullet$ identity map is continuous, i.e.

$$\|ix\|_Y \leq C \|x\|_X$$

why: (i) of Sobolev ineq.

$$(ii) \text{ if } p = n, \text{ then } W_0^{1,p}(\Omega) \hookrightarrow L^q(\Omega), \quad \forall 1 \leq q < \infty$$

$$W_0^{1,p}(\Omega) \hookrightarrow W^{1,n-\varepsilon} \hookrightarrow L^q(\Omega) \quad \text{by (i) of Cor 2. that } q < \infty$$

$$u \in C^1(\bar{\Omega}).$$

$$\cdot u \in C^2(\bar{\Omega})$$

$$\cdot \exists \text{ const. } C \text{ s.t.}$$

$$|u(x) - u(y)| \leq C|x - y|$$

$$\forall x, y \in \bar{\Omega}$$

$$\sqrt{x} \text{ on } [0, 1]$$

$$\sqrt{x} \in C^{\frac{1}{2}}([0, 1])$$

$$1 \leq p_1 < p_2$$

$$\Rightarrow L^{p_2}(\Omega) \subset L^{p_1}(\Omega)$$

$$\forall u \in L^{p_2}$$

$$\int_{\Omega} |u|^{p_1} dx \leq \left(\int_{\Omega} (|u|^{p_1})^{\frac{p_2}{p_1}} dx \right)^{\frac{p_1}{p_2}} \left(\int_{\Omega} 1^{\left(\frac{p_2}{p_1}\right)'} dx \right)^{\frac{1}{p_2}}$$

$$\leq \|u\|_{L^{p_2}}^{p_1} C(p_1, p_2, |\Omega|)$$

$$\|u\|_{L^{p_1}} \leq \|u\|_{L^{p_2}} \cdot C$$

(iii) if $n < p < \infty$, then $W_0^{1,p}(\Omega) \subset C^0(\bar{\Omega})$

(In fact $C^{1-\frac{n}{p}}(\bar{\Omega})$) ← 不证

why: by (ii) of Sobolev ineq.

Proof of Sobolev ineq:

i) Special case: $p=1$, $u \in C_0^\infty(\Omega)$. Extend $u \equiv 0$ on $\mathbb{R}^n$

$\Rightarrow u \in C_0^\infty(\mathbb{R}^n)$ → 有效

$$|u(x)| \leq \int_{-\infty}^{x_1} |\partial_{x_1} u(x_1, \dots, x_n)| dx_1 \quad z=1, \dots, n$$

$$|u(x)|^n \leq \prod_{i=1}^n \int_{-\infty}^{\infty} |\partial_{x_i} u| dx_i$$

$$|u(x)|^{\frac{n}{n-1}} \leq \left(\prod_{i=1}^n \int_{-\infty}^{\infty} |\partial_{x_i} u| dx_i \right)^{\frac{1}{n-1}}$$

$$\int fg \, dx \leq \|f\|_p \|g\|_{p'}$$

Generalized Hölder: $\frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_n} = 1$

$$\left| \int_X f_1 \dots f_n \, d\mu \right| \leq \|f_1\|_{p_1} \|f_2\|_{p_2} \dots \|f_n\|_{p_n}$$

$$\Rightarrow \int_{-\infty}^{+\infty} |u(x)|^{\frac{n}{n-1}} dx_1 \leq \int_{-\infty}^{+\infty} \left(\prod_{i=1}^n \int_{-\infty}^{+\infty} |\partial_{x_i} u| \, dx_i \right)^{\frac{1}{n-1}} dx_1$$

$$\leq \left(\int_{-\infty}^{+\infty} |\partial_{x_1} u| \, dx_1 \right) \left(\prod_{i=2}^n \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |\partial_{x_i} u| \, dx_i \, dx_1 \right)^{\frac{1}{n-1}}$$

$\uparrow$
 $\frac{n}{n-1}$ Generalized Hölder.
 $p_1 = p_2 = \dots = p_{n-1} = n-1$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |u|^{\frac{n}{n-1}} dx_1 \, dx_2 \leq \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |\partial_{x_2} u| \, dx_2 \, dx_1 \right)^{\frac{1}{n-1}}$$

$$\prod_{i=1,2}^n \left(\int \int \int_{-\infty}^{+\infty} |\partial_{x_i} u| \, dx_i \, dx_1 \, dx_2 \right)^{\frac{1}{n-1}}$$

$$\left(\int \int_{-\infty}^{+\infty} |\partial_{x_1} u| \, dx_1 \, dx_2 \right)^{\frac{1}{n-1}}$$

$$\dots \dots \dots$$

$$\left(\int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} \, dx \right)^{\frac{n-1}{n}} \leq \left(\prod_{i=1}^n \int_{\mathbb{R}^n} |\partial_{x_i} u| \, dx \right)^{\frac{1}{n-1} \cdot \frac{1}{n}}$$

$$\leq \frac{1}{n} \sum_{i=1}^n \int_{\mathbb{R}^n} |\partial_{x_i} u| \, dx$$

$$\|u\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \leq \frac{1}{n} \left(\sum_{i=1}^n \int_{\mathbb{R}^n} |\partial_{x_i} u| \, dx \right)$$

$$\|u\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} = \|u\|_{L^{\frac{n}{n-1}}(\Omega)}$$

$$\leq \frac{1}{n} \int_{\mathbb{R}^n} \sqrt{|x_1 u|^2 + \dots + |x_n u|^2} \cdot \sqrt{1^2 + \dots + 1^2} \, dx$$

$$= \frac{1}{\sqrt{n}} \int_{\mathbb{R}^n} |\nabla u| \, dx \quad \in C_0^1(\Omega)$$

Now if $1 < p < n$, replace u by $|u|^\gamma$
 ($\gamma > 1$ to be chosen)

$$\Rightarrow \| |u|^\gamma \|_{L^{\frac{n}{n-1}}(\Omega)} \leq \frac{1}{\sqrt{n}} \| \nabla (|u|^\gamma) \|_{L^1(\Omega)}$$

$$= \frac{1}{\sqrt{n}} \| \gamma |u|^{\gamma-1} \nabla |u| \|_{L^1(\Omega)}$$

$$\leq \frac{\gamma}{\sqrt{n}} \| |u|^{\gamma-1} \nabla |u| \|_{L^1(\Omega)}$$

$$\stackrel{\text{Hölder}}{\leq} \frac{\gamma}{\sqrt{n}} \| \nabla |u| \|_{L^p(\Omega)} \| |u|^{\gamma-1} \|_{L^{p'}(\Omega)}$$

choose γ s.t. $\gamma \frac{n}{n-1} = (p-1)p' = (p-1) \frac{p}{p-1}$

$$\frac{p}{p-1} = \gamma \left(\frac{p}{p-1} - \frac{n}{n-1} \right) = \gamma \frac{pn-p-np+n}{(p-1)(n-p)}$$

$$\gamma = \frac{p(n-p)}{n-p} > p$$

$$\left(\int_{\Omega} |u|^{\frac{pn}{n-p}} \, dx \right)^{\frac{n-1}{n}} \leq \frac{p(n-1)}{(n-p)\sqrt{n}} \| \nabla |u| \|_{L^p(\Omega)}$$

$$\left(\int_{\Omega} |u|^{\frac{pn}{n-p}} \, dx \right)^{\frac{p-1}{p}}$$

$$\left(\int_{\Omega} |u|^{\frac{pn}{n-p}} \, dx \right)^{\frac{n-p}{pn}} \leq \frac{p(n-1)}{(n-p)\sqrt{n}} \| \nabla |u| \|_{L^p(\Omega)} \| \nabla |u| \|_{L^p(\Omega)}$$

$$\forall u \in C_0^\infty(\Omega)$$



(Coming back to proof of Poincaré inequality)
 If $p=n$, take $1 < q < n$ s.t. $\frac{qn}{n-q} = n$
 (i) of Sobolev.

$$\|u\|_{L^p(\Omega)} = \|u\|_{L^{\frac{qn}{n-q}}(\Omega)} \leq C(n, q) \|\nabla u\|_{L^q(\Omega)}$$

Now $\forall u \in W_0^{1,p}(\Omega)$, by def,

$$\exists \{u_k\} \subset C_0^\infty(\Omega) \text{ s.t. } \|u_k - u\|_{W_0^{1,p}(\Omega)} \xrightarrow{k \rightarrow \infty} 0$$

$$\Rightarrow \cdot \{\nabla u_k\} \text{ is Cauchy in } L^p(\Omega) \quad \downarrow \quad u_k - u \text{ in } L^p$$

$$\cdot \|\nabla u_k\|_{L^p(\Omega)} \xrightarrow{k \rightarrow \infty} \|\nabla u\|_{L^p}$$

$$\text{By } \textcircled{1} \quad \|u_k - u_l\|_{L^{\frac{pn}{n-p}}} \leq C(p, n) \|\nabla u_k - \nabla u_l\|_{L^p(\Omega)}$$

$$\forall 1 \leq k, l$$

$$\Rightarrow \cdot \{u_k\} \text{ is Cauchy in } L^{\frac{pn}{n-p}}$$

$$\Rightarrow u_k \xrightarrow{k \rightarrow \infty} \textcircled{2} \text{ in } L^{\frac{pn}{n-p}} \text{ \& hence in } L^p$$

In $\textcircled{1}$, replace u by u_k , send $k \rightarrow \infty$

$$\Rightarrow \|u\|_{L^{\frac{pn}{n-p}}} \leq C(n, p) \|\nabla u\|_{L^p(\Omega)}$$

Proof of (i) is complete

Proof of (ii) Special case $|\Omega|=1$ $p > n$

$$\text{Let } \bar{u} = \frac{\sqrt{n} u}{\|\nabla u\|_{L^1(\Omega)}}$$

$$(W.T.S. \quad \|\bar{u}\|_\infty \leq C \|\nabla u\|_p)$$

Assume $u \in C_0^\infty(\Omega)$

Recall

$$\| |u|^r \|_{L^{\frac{n}{n-1}}(\Omega)} \leq \frac{r}{\sqrt{n}} \| |u|^{r-1} \|_{L^{p'}(\Omega)}$$

$$\lim_{q \rightarrow \infty} \|u\|_q$$

Replace the "u" by $\tilde{u} \Rightarrow$

$$\| |\tilde{u}|^r \|_{L^{\frac{n}{n-1}}} \leq \frac{r}{\sqrt{n}} \left\| \frac{\sqrt{n} \nabla u}{\|\nabla u\|_{L^p}} \right\|_{L^p(\Omega)} \| |\tilde{u}|^{r-1} \|_{L^{p'}},$$

$$\Rightarrow \| \tilde{u} \|_{L^{\frac{rn}{n-1}}} \leq \frac{r}{\sqrt{n}} \left\| \frac{\sqrt{n} \nabla u}{\|\nabla u\|_{L^p}} \right\|_{L^p(\Omega)} \| |\tilde{u}|^{r-1} \|_{L^{p'}}$$

$$\Rightarrow \| \tilde{u} \|_{L^{\frac{rn}{n-1}}} \leq r^{\frac{1}{p'}} \| |\tilde{u}|^{r-1} \|_{L^{p'}}^{\frac{1}{p'}} = r^{\frac{1}{p'}} \left(\int_{\Omega} |\tilde{u}|^{(r-1)p'} \right)^{\frac{1}{p'}}$$

$$= r^{\frac{1}{p'}} \left(\int_{\Omega} |\tilde{u}|^{(r-1)p'} \right)^{\frac{1}{p'}}$$

$$= r^{\frac{1}{p'}} \| \tilde{u} \|_{L^{p'(r-1)}}^{(r-1)}$$

$$\leq r^{\frac{1}{p'}} \| \tilde{u} \|_{L^{p'}}^{(1-\frac{1}{p'})}$$

$$\left(\int_{\mathbb{R}} |f|^p d\mu \right)^{\frac{1}{p}} \leq \left(\int_{\mathbb{R}} |f|^q dx \right)^{\frac{1}{q}}$$

$$\text{if } \mu(\mathbb{R}) = 1$$

Take $r = 2^v$, $v = 1, 2, \dots$

$$\delta = \frac{n'}{p'} = \frac{\frac{n}{n-1}}{\frac{p}{p-1}} > 1$$

$$\begin{matrix} p > n \\ \downarrow \\ p' < n' \end{matrix}$$

$$\| \tilde{u} \|_{\delta n'}^r \leq \delta^{r\delta^{-v}} \| \tilde{u} \|_{\delta^{r-1} n'}^{(1-\delta^{-v})}$$

$$\leq \delta^{r\delta^{-v}} \left(\| \tilde{u} \|_{\delta^{r-2} n'}^{\delta^{r-1} n'} \right)^{(1-\delta^{-v})}$$

$$\leq \delta^{r\delta^{-v} + (v-1)\delta^{-(v-1)}} \| \tilde{u} \|_{\delta^{v-2} n'}$$

Sobolev inequality, $1 \leq p < \infty$

(i) $\|u\|_{L^{\frac{np}{n-p}}(\Omega)} \leq C(n, p) \|\nabla u\|_{L^p(\Omega)}, \forall u \in W_0^{1,p}(\Omega)$

if $1 \leq p < n$

(ii) $\|u\|_{L^\infty(\Omega)} \leq C(n, p) |\Omega|^{\frac{1}{n}-\frac{1}{p}} \|\nabla u\|_{L^p(\Omega)}, \forall u \in W_0^{1,p}(\Omega)$

if Ω is bdd & $\infty > p > n$; moreover, $u \in C^0(\bar{\Omega})$

(in fact; $u \in C^\alpha(\bar{\Omega}), \alpha = 1 - \frac{n}{p} \in (0, 1)$)

Corollary 1 Poincaré ineq. $1 \leq p < \infty, \Omega$ bdd

We have $\|u\|_{L^p(\Omega)} \leq \|\nabla u\|_{L^p(\Omega)} C(p, n, |\Omega|), \forall u \in W_0^{1,p}(\Omega)$

Corollary 2 Sobolev imbedding $1 \leq p < \infty, \Omega$ bdd

X, Y Banach,

$X \subset Y$ subspace

We say X continuously imbedded into Y if $X \xrightarrow{i} Y$ is continuous, i.e.

$\|x\|_Y \leq C \|x\|_X \quad \forall x \in X$

$W_0^{1,p}(\Omega) \hookrightarrow \begin{cases} L^{\frac{np}{n-p}}(\Omega) & \text{if } 1 \leq p < n \\ L^\infty(\Omega) & \text{if } p = n \\ C^0(\bar{\Omega}) & \text{if } p > n \end{cases}$

(in fact, $C^\alpha(\bar{\Omega}), \alpha = 1 - \frac{n}{p}$)

Rk 1. This is true if replace $W_0^{1,p}(\Omega)$ by $W^{1,p}(\Omega)$ when $\partial\Omega \in C^1$

Rk 2. $k \geq 1, 1 \leq p < \infty, \Omega$ bdd

$$W_0^{k,p}(\Omega) \hookrightarrow \begin{cases} L^{\frac{np}{n-kp}}(\Omega) & \text{if } kp < n \\ L^{\frac{n}{n-k}}(\Omega), \forall q \in [1, \infty) & \text{if } kp = n \\ C^\alpha(\bar{\Omega}) & \text{if } kp > n, \alpha = k - \frac{n}{p} > 0 \end{cases}$$

$$C^{1.5}(\bar{\Omega}) = \{u \in C^1(\bar{\Omega}) \mid \frac{\partial u}{\partial x_i} \in C^{0.5}(\bar{\Omega}), i=1, \dots, n\}$$

$$\|u\|_{C^{1.5}(\bar{\Omega})} = \|u\|_{C^1(\bar{\Omega})} + \sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_{C^{0.5}(\bar{\Omega})}$$

eg. $u \in W^{1,p}(\mathbb{R}^n)$, $n < p < \infty$, Then $\lim_{|x| \rightarrow \infty} u(x) = 0$

By Sobolev imbedding applied on $B_1(x)$, we have

$$|u(x)| \leq \|u\|_{C^0(B_1(x))} \leq C(n, p, B_1(x)) \|u\|_{W^{1,p}(B_1(x))} \\ = C(\|u\|_{L^p(B_1(x))} + \|\nabla u\|_{L^p(B_1(x))})$$

as $|x| \rightarrow \infty$
by L.D.C.T

$$0 \leftarrow \left(\int_{\mathbb{R}^n} |u(y)|^p \chi_{B_1(x)}(y) dy \right)^{\frac{1}{p}} \downarrow 0$$

$$\wedge \quad u \in L^p$$

$$|u(y)|^p \in L^1(\mathbb{R}^n) \quad \text{RED}$$

- $f \geq 0$ uniformly on $[0, \infty)$
- $\lim_{x \rightarrow \infty} f(x) = 0$

$$\Rightarrow \lim_{x \rightarrow \infty} \int_x^\infty f(x) = 0$$

$$\int_0^\infty f(x) \in L^1([0, \infty))$$

→ Hölder space.

Banach.

$$|u(x) - u(y)| \leq C|x - y|^\alpha \quad \forall x, y \in \bar{\Omega}$$

$$\|u\|_{C^\alpha(\bar{\Omega})} = \|u\|_{C^0(\bar{\Omega})} + \sup_{x, y \in \bar{\Omega}} \frac{|u(x) - u(y)|}{|x - y|^\alpha}$$

Proof of (ii) of Sobolev ineq.

2 W.T.S: $\|u\|_{\infty} \leq C \|\nabla u\|_{L^p}$, $\infty > p > n$, $u \in W_0^{1,p}(\Omega)$

Let $\tilde{u}(x) = \frac{\sqrt{n} u(x)}{\|\nabla u\|_{L^p(\Omega)}}$ Assume $|\Omega| = 1$

$$\delta = \frac{n'}{p'} > 1 \quad (n' = \frac{n}{n-1}, p' = \frac{p}{p-1})$$

$$\|\tilde{u}\|_{n'\delta^r} \leq \delta^r \delta^{-r} \|\tilde{u}\|_{n'\delta^{r-1}}, \quad r=1,2,\dots$$

(reversed Hölder ineq.)

$$\begin{aligned} &\leq \delta^r \delta^{-r} \left[\delta^{(r-1)} \|\tilde{u}\|_{n'\delta^{r-2}} \right] \delta^{-(r-1)} \\ &\leq \delta^{r\delta^{-r} + (r-1)\delta^{-(r-1)}} \|\tilde{u}\|_{n'\delta^{r-2}} \end{aligned}$$

$$\begin{aligned} &\leq \delta^{r\delta^{-r} + (r-1)\delta^{-(r-1)} + \dots + \delta^{-1}} \\ &\leq \delta^{r\delta^{-r} + (r-1)\delta^{-(r-1)} + \dots + \delta^{-1}} \|\tilde{u}\|_{n'\delta^{r-2}} \leq 1 \end{aligned}$$

By proof of (i) of Sobolev ineq.

$$\|\tilde{u}\|_{n'} \leq \frac{1}{\sqrt{n}} \int_{\Omega} |\nabla \tilde{u}| dx$$

$$\begin{aligned} \Rightarrow \|\tilde{u}\|_{n'} &\leq \frac{1}{\sqrt{n}} \left\| \frac{\sqrt{n} \nabla u}{\|\nabla u\|_{L^p(\Omega)}} \right\|_{L^1(\Omega)} \leq \left\| \frac{\nabla u}{\|\nabla u\|_{L^p(\Omega)}} \right\|_{L^p(\Omega)} = 1 \\ &\leq \delta^{\sum_{r=1}^{\infty} r\delta^{-r}} = \chi < \infty \quad \chi = \chi(n,p) \end{aligned}$$

$$\Rightarrow \lim_{r \rightarrow \infty} [\|\tilde{u}\|_{n'\delta^r} \leq \chi]$$

$$\Rightarrow \|\tilde{u}\|_{L^{\infty}(\Omega)} \leq \chi$$

$$\frac{\sqrt{n}}{\|\nabla u\|_{L^p(\Omega)}} \|u\|_{L^{\infty}(\Omega)} \leq \chi$$

$$\Rightarrow \|u\|_{L^{\infty}(\Omega)} \leq \frac{\chi}{\sqrt{n}} \|\nabla u\|_{L^p(\Omega)}$$

Q: what if $|\Omega| \neq 1$?

Let $v(y) = u(x)$, $y = |\Omega|^{-\frac{1}{n}} x$, $x \in \Omega$, $y \in \underbrace{|\Omega|^{-\frac{1}{n}} \Omega}_D$

$$|D| = \int_D 1 dy = \int_{\Omega} |\Omega|^{-\frac{n}{n}} dx = 1$$

$$u \in W_0^{1,p}(\Omega) \Rightarrow v \in W_0^{1,p}(D)$$

Now applying (i) with " Ω " = D , " u " = v

$$\|v\|_{L^\infty(D)} \leq \frac{\lambda}{\sqrt{n}} \|\nabla v\|_{L^p(D)}$$

$$\begin{aligned} \frac{\nabla v(y)}{Dy} &= \frac{\nabla u(x)}{Dy} \\ &= \frac{\nabla u(x)}{D(|\Omega|^{-\frac{1}{n}} x)} \\ &= \frac{1}{|\Omega|^{-\frac{1}{n}}} \nabla u(x) \end{aligned} \quad \begin{aligned} &= \frac{\lambda}{\sqrt{n}} \left(\int_D |\nabla v(y)|^p dy \right)^{\frac{1}{p}} \\ &= \frac{\lambda}{\sqrt{n}} \left(\int_{\Omega} |\nabla u(x)|^p |\Omega|^{-1} dx \right)^{\frac{1}{p}} \frac{1}{|\Omega|^{-\frac{1}{n}}} \\ &= \frac{\lambda}{\sqrt{n}} |\Omega|^{\frac{1}{n} - \frac{1}{p}} \|\nabla u\|_{L^p(\Omega)} \end{aligned}$$

why $u \in C^0(\bar{\Omega})$? $\because u \in W_0^{1,p}(\Omega)$, $\therefore \exists \{u_k\} \subset C_0^\infty(\Omega)$
s.t. $u_k \xrightarrow{k \rightarrow \infty} u$ in $W^{1,p}(\Omega)$ -norm

$$\Rightarrow \|u_k - u_l\|_{L^\infty(\bar{\Omega})} \leq C \|\nabla(u_k - u_l)\|_{L^p(\Omega)} \xrightarrow{\text{as } k, l \rightarrow \infty} 0$$

$\Rightarrow \{u_k\}$ is Cauchy in $C^0(\bar{\Omega})$.

$\Rightarrow \exists u_\infty \in C^0(\bar{\Omega})$ s.t. $u_k \xrightarrow{k \rightarrow \infty} u_\infty$ in $C^0(\bar{\Omega})$

$\Rightarrow u_\infty = u$

$\Rightarrow u \in C^0(\bar{\Omega})$ QED

Rellich-Kondrakov Compact Embedding Theorem

Suppose Ω is bded. Then $W_0^{1,p}(\Omega)$ is compactly imbedded with $L^q(\Omega)$. $\forall q \in [1, \frac{np}{n-p})$ if $1 \leq p < n$

($\overset{\text{cpt}}{X} \hookrightarrow Y$ means: $i: X \rightarrow Y$ is cpt)
 $\swarrow \searrow$
 Banach

RK 1. $\forall p = n$, $\forall q \in [1, \infty)$, $\exists p_1 \in [1, \frac{n}{n-1})$ s.t. $\frac{np_1}{n-p_1} = q$
 Then $W_0^{1,p_1}(\Omega) \overset{\text{cpt}}{\hookrightarrow} L^q(\Omega) \Rightarrow W_0^{1,p}(\Omega) \overset{\text{cpt}}{\hookrightarrow} L^q(\Omega)$
 $W_0^{1,p} \hookrightarrow W_0^{1,p_1}$ (b.c. $p_1 < p$)

RK 2. $\forall p > n$, $W_0^{1,p}(\Omega) \overset{\text{cpt}}{\hookrightarrow} C^0(\bar{\Omega}) \overset{\text{cpt}}{\hookrightarrow} C^0(\bar{\Omega})$
 $\Rightarrow W_0^{1,p}(\Omega) \overset{\text{cpt}}{\hookrightarrow} C^0(\bar{\Omega})$ (in fact, ... by Arzela-Ascoli
 $C^\beta(\bar{\Omega}) \forall \beta < \alpha$)

X, Y Banach

$X \subset Y$

$\overset{\text{cpt}}{X} \hookrightarrow Y$

if $i: X \rightarrow Y$

$C^1(\bar{\Omega}) \hookrightarrow L^1(\Omega)$ Ω bded

Proof of Rellich-Kondrakov

consider case of $q=1$ first. Suppose A is bded set in $W_0^{1,p}(\Omega)$ W.T.S. A is precompact in $L^1(\Omega)$
 $\forall u \in W_0^{1,p}(\Omega)$, $(\exists u_k \rightarrow u, u_k \in C_c^\infty)$

extend u by letting $u=0$ in Ω^c

$\Rightarrow u \in W_0^{1,p}(\mathbb{R}^n)$ So can think of $A \subset W_0^{1,p}(\mathbb{R}^n)$

Define $A_h = \{u_h | u \in A\}$, $u_h =$ mollification of u

$$\int_{\mathbb{R}^n} \chi_h(x-y) u(y) dy$$

Claim A_h is precompact in $L^1(\Omega)$, $\forall$ fixed $h > 0$
 $\forall u \in A$

$$\begin{aligned} |u_h(x)| &\leq \int_{\Omega} |j_h(x-y) u(y)| dy \leq \|j_h\|_{L^\infty(\mathbb{R}^n)} \|u\|_{L^1(\Omega)} \\ &\leq \|j_h\|_{L^\infty} \|u\|_{L^p(\Omega)} |\Omega|^{1-\frac{1}{p}} \\ &\leq |\Omega|^{1-\frac{1}{p}} M \|j_h\|_{L^\infty(\mathbb{R}^n)} \\ &\quad (M \text{ upper bound for } \|u\|_p, \forall u \in A) \end{aligned}$$

$\{u_h\}_{h>0}^{u \in A}$ uniformly bdd on $\bar{\Omega}$
 $\bullet$ $|\nabla u_h(x)| \stackrel{\text{Lemma?}}{=} \left| \int_{\mathbb{R}^n} \nabla_x (j_h(x-y) u(y)) dy \right|$

$$\begin{aligned} &\leq \frac{1}{h} \|\nabla j_h\|_{L^\infty(\mathbb{R}^n)} \|u\|_{L^1(\Omega)} |\Omega|^{1-\frac{1}{p}} \\ &\leq \frac{1}{h} M |\Omega|^{1-\frac{1}{p}} \|\nabla j_h\|_{L^\infty} \end{aligned}$$

$\Rightarrow \{u_h\}_{h>0}^{u \in A}$ equicontin. on $\bar{\Omega}$

Now by Arzela-Ascoli, A_h is precompact in $L^1(\Omega)$,
 $\forall u \in A$.

$$u_h(x) = \int_{\mathbb{R}^n} j_h(x-y) u(y) dy \stackrel{\substack{z = \frac{x-y}{h} \\ y = x-hz}}{\int_{\mathbb{R}^n} j(z) u(x-hz) dz}$$

$\forall x \in \mathbb{R}^n \quad dy = h^n dz$

$$u(x) = u(x) \int_{\mathbb{R}^n} j(z) dz = \int_{\mathbb{R}^n} j(z) u(x) dz$$

$\Rightarrow$

$$|u_h(x) - u(x)| = \left| \int_{\mathbb{R}^n} j(z) (u(x) - u(x-hz)) dz \right|$$

$\stackrel{\substack{u \text{ is} \\ \text{smooth}}}{\leq} \int_{\mathbb{R}^n} j(z) \left(\int_0^1 \frac{d}{ds} u(x-sz) ds \right) dz$

$$= \left| \int_{\mathbb{R}^n} j(z) \int_0^1 \nabla u(x-sz) \cdot (-z) ds dz \right|$$

Generalized

$|z| dz dz$

$$\int_{\mathbb{R}^n} |u(x) - u_h(x)| dx \leq \text{Minkowski} \left(\int_{\mathbb{R}^n} |f(z)|^p dz \right)^{1/p} \int_0^h \left(\int_{\mathbb{R}^n} |\nabla u(x-sz)| dx \right)^{1/p} dz$$

$$= \|\nabla u\|_{L^p(\mathbb{R}^n)}^p h$$

$$\text{Supp } f(z) = \text{open } B_{1/2} \leq \|\nabla u\|_{L^p(\Omega)}^p |\Omega|^{1-\frac{1}{p}} h$$

$$\leq M |\Omega|^{1-\frac{1}{p}} h$$

(M upper bound for $\|u\|_{W_0^{1,p}(\Omega)}$
 $\forall u \in A$)

If $u \in W_0^{1,p}(\Omega)$ is Not Smooth, $\hookrightarrow$ take $\{u_k\} \subset C_0^\infty(\Omega)$, s.t. $u_k \xrightarrow{k \rightarrow \infty} u$ in $W_0^{1,p}(\Omega)$

$\Rightarrow (u_k)_h \rightarrow u_h$ in $L^p(\Omega)$ (HW)
 $\forall$ fixed $h > 0$

By what have been proven, we have

$$\|u_k - (u_k)_h\|_{L^1(\mathbb{R}^n)} \leq |\Omega|^{1-\frac{1}{p}} \|\nabla u_k\|_{L^p(\Omega)}^p h$$

Sending $k \rightarrow \infty$
 $\downarrow$
 $\|u - u_h\|_{L^1(\mathbb{R}^n)}$

$\downarrow$
 $\|\nabla u\|_{L^p(\Omega)}^p$

$\Rightarrow \forall u \in A$

$$\|u - u_h\|_{L^1(\Omega)} \leq M |\Omega|^{1-\frac{1}{p}} h$$

$\Rightarrow A_h$ & A are closed in L^1

$\forall \delta > 0$, since A_h is pre-compact, we can cover A_h by finitely many balls in $L^1(\Omega)$ with radius $= \delta/2$. Take $h = 2\delta$ ($\delta > 0$, to be determined)

$$\text{Take } \alpha \text{ s.t. } \alpha \delta M |\Omega|^{1-\frac{1}{p}} = \delta/2 \Rightarrow \alpha = \frac{1}{2M |\Omega|^{1-\frac{1}{p}}}$$

$\forall u \in A$, $u_{2\delta} \in \text{some ball } B_{\delta/2}$

$$\text{By } \odot, \|u - u_{2\delta}\|_{L^1(\Omega)} \leq M |\Omega|^{1-\frac{1}{p}} \alpha \delta = \delta/2 \Rightarrow B_\delta$$

i.e. A can be covered by finitely many balls with radius $= \delta$

$\Leftrightarrow A$ precompact in $L^1(\Omega)$

Now suppose $1 < q < \frac{np}{n-p}$

$$0 < p_1 < q < p_2 \Rightarrow \frac{1}{p_1} > \frac{1}{q} > \frac{1}{p_2} \Rightarrow \exists \lambda \in (0, 1), \text{ s.t.}$$

$$\frac{q\lambda}{p_1} + \frac{(1-\lambda)q}{p_2} = \frac{1}{q} \quad \frac{1}{\left(\frac{p_1}{q\lambda}\right)} + \frac{1}{\left(\frac{p_2}{q(1-\lambda)}\right)} = 1$$

$$\textcircled{2} \quad \|u\|_{L^q(\Omega)} \leq \|u\|_{L^{p_1}(\Omega)}^\lambda \|u\|_{L^{p_2}(\Omega)}^{1-\lambda} \quad \frac{1}{\frac{p_2}{q(1-\lambda)}} \\ \int_{\Omega} |u|^q dx = \int_{\Omega} |u|^{q\lambda} |u|^{q(1-\lambda)} dx \\ \stackrel{\text{Holder}}{\leq} \left(\int_{\Omega} |u|^{q\lambda} \cdot \frac{p_1}{q\lambda} \right)^{\frac{1}{\frac{p_1}{q\lambda}}} \left(\int_{\Omega} |u|^{q(1-\lambda)} \cdot \frac{p_2}{q(1-\lambda)} \right)^{\frac{1}{\frac{p_2}{q(1-\lambda)}}}$$

$$\text{By } \textcircled{2}, \quad \|u\|_{L^q(\Omega)} \leq \|u\|_{L^1(\Omega)}^\lambda \|u\|_{L^{\frac{np}{n-p}}(\Omega)}^{1-\lambda} \\ \leq \|u\|_{L^1(\Omega)}^\lambda \left(C(n, p, |\Omega|) \|u\|_{L^{p_1}(\Omega)} \right)^{1-\lambda} \\ \uparrow \\ \text{Sobolev inequality} \quad \text{then } \textcircled{2}$$

$\Rightarrow \forall u, v \in A$, we have

$$\|u - v\|_{L^q(\Omega)} \leq \|u - v\|_{L^1(\Omega)}^\lambda \left(\tilde{C}(n, p, |\Omega|) (2M) \right)^{1-\lambda}$$

Since A is precompact in $L^1(\Omega)$, A is precompact in $L^q(\Omega)$. QED

L^2 - elliptic theory

Ω bdd domain in $\mathbb{R}^n$

operator divergence of vector field $A(x) \cdot \nabla u$

$$Lu = -(\text{div}(x) u_{x_i})_{x_j} + b_i(x) u_{x_i} + c(x) u$$

"Divergence form" $a_{ij}, b_i, c \in L^\infty(\Omega)$

Def We say L is strictly elliptic on Ω if
 $(a_{ij}(x))_{n \times n} \geq \lambda_0 I_{n \times n}$ a.e. on Ω
 $\lambda_0 > 0$ const.

W.T. Solve

$$\begin{array}{l} \text{(DBVP)} \quad \left\{ \begin{array}{l} Lu = f(x) \quad \text{in } \Omega \\ \text{B.C.} \rightarrow u = 0 \quad \text{on } \partial\Omega \end{array} \right. \end{array}$$

Notion of weak sol's

- $u \in H_0^1(\Omega) (= W_0^{1,2}(\Omega))$
takes care of B.C.

- If $u \in H_0^1(\Omega)$
 $Lu = -(\text{div}(x) u_{x_i})_{x_j} + L^{ij} \cdot u_{x_i} + L^{ij} \cdot u_{x_j}$
 $= \text{distribution}$

distributional
derivative

Q: Existence & Uniqueness of weak sol.

$$Lu = -(\text{div}(x) u_{x_i})_{x_j} + b_i u_{x_i} + c u$$

f is allowed to be a distribution

$Lu = f$ is interpreted in the distributional sense

i.e. $\forall \varphi \in \mathcal{D}(\Omega) = C_0^\infty(\Omega)$

$$\langle Lu, \varphi \rangle = \langle f, \varphi \rangle$$

||

$$\int_{\Omega} \text{div}(x) u_{x_i} \varphi_{x_j} + b_i(x) u_{x_i} \varphi(x) + c(x) u \varphi \, dx$$

The case of f being a general distribution is too hard!

Assume $f \in (H_0^1(\Omega))^* \stackrel{\text{def}}{=} H^{-1}(\Omega)$

Natural Q: $\forall u \in H_0^1(\Omega), Lu \in H^{-1}(\Omega)$?

Yes: For fixed $u \in H_0^1(\Omega)$, define

$$\langle Lu, v \rangle = \int_{\Omega} a_{ij}(x) u_{x_i} v_{x_j} + b_i(x) u_{x_i} v + c(x) u v \, dx$$

• It is linear in v

• bdd in v ? Yes!

$$|\langle Lu, v \rangle| \leq C \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + C \|w\|_{L^2} \|\nabla u\|_{L^2} + C \|u\|_{L^2} \|v\|_{L^2}$$

$$\leq C \|u\|_{H_0^1} \|v\|_{H_0^1(\Omega)}$$

Def. Suppose $f \in H^{-1}(\Omega)$

u is said to be a weak sol of

$$(DBVP) \quad \begin{cases} Lu = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

iff $u \in H_0^1(\Omega)$

$\forall v \in H_0^1(\Omega)$

$$\langle Lu, v \rangle = \langle f, v \rangle$$

$$\int_{\Omega} a_{ij}(x) \underset{\uparrow}{u_{x_i}} \underset{\uparrow}{v_{x_j}} + b_i(x) \underset{\uparrow}{u_{x_i}} v + c(x) u v \, dx$$

u 的=阶导在distribution
意下存在.

u 的导数属于 L^2

$$= \langle f, v \rangle_{H^{-1}(\Omega), H_0^1(\Omega)}$$

Rk. $\forall u \in H_0^1(\Omega)$ Lu is interpreted as operator language
 $\langle Lu, v \rangle \stackrel{\text{def}}{=} L(u, v)$ as element in $H^{-1}(\Omega)$

Issue of existence & uniqueness in operator language

- Existence $\Leftrightarrow f \in R(L) \leftarrow \text{Range } (L: H_0^1(\Omega) \rightarrow H^{-1}(\Omega))$
- Uniqueness $\Leftrightarrow \text{null } N(L) = \{0\}$

L_2 - Regularity

$$f \in L^2 \Rightarrow u \in H^2$$

$$f \in H^m \Rightarrow u \in H^{m+2}(\Omega)$$

$$\forall m \geq 0 \quad W^{m+2,2}(\Omega)$$

$$W^{m+2,2}(\Omega) \hookrightarrow C^{m+2-\frac{n}{2}}(\bar{\Omega})$$

Fact: $L: H_0^1(\Omega) \rightarrow H^{-1}(\Omega)$ is linear & bdd

Proof $|\langle Lu, v \rangle| = \left| \int_{\Omega} (a_{ij} u_{x_i} v_{x_j} + b_i u_{x_i} v + c u v) dx \right|$

$$\leq C \left[\|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + \|\nabla u\|_{L^2} \|v\|_{L^2} + \|u\|_{L^2} \|v\|_{L^2} \right]$$

(Recall $a_{ij}, b_i, c \in L^\infty(\Omega)$) $\leq C \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)}$

$$\Rightarrow \|Lu\|_{H^{-1}(\Omega)} \leq C \|u\|_{H_0^1(\Omega)}$$

$$\Rightarrow L \text{ is bdd. QED}$$

eg. $\begin{cases} -\Delta u = f(x), & x \in \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$

$$f \in L^2(\Omega) \quad \forall v \in H_0^1(\Omega)$$

weak sol.

$$\bullet u \in H_0^1(\Omega)$$

$$\bullet \int_{\Omega} u_{x_i} v_{x_i} dx = \int_{\Omega} f(x) v(x) dx$$

" $\langle f, v \rangle$ "

Can we use $(\nabla u, \nabla v)_{L^2(\Omega)}$ as an equivalent inner product of $H_0^1(\Omega)$?

Recall: norm in $H^1(\Omega) = \left(\int_{\Omega} (|\nabla u|^2 + u^2) dx \right)^{\frac{1}{2}}$
 inner product: $(u, v)_{H^1(\Omega)} = (\nabla u, \nabla v)_{L^2} + (u, v)_{L^2}$
 H_0^1 has the same norm & inner product

Recall Poincaré inequality. $\forall 1 \leq p < \infty$

$$\|u\|_{L^p(\Omega)} \leq \|\nabla u\|_{L^p(\Omega)}, \quad \forall u \in W_0^{1,p}(\Omega)$$

$$\Rightarrow \int_{\Omega} u^2 dx \leq C \int_{\Omega} |\nabla u|^2 dx, \quad \forall u \in H_0^1(\Omega)$$

$$\Rightarrow \|u\|_{H_0^1(\Omega)} \geq \|\nabla u\|_{L^2(\Omega)} \geq C \|u\|_{H^1(\Omega)}, \quad \forall u \in H_0^1(\Omega)$$

$$\Rightarrow \|\nabla u\|_{L^2(\Omega)} \text{ is an equivalent norm of } H_0^1(\Omega)$$

$$\Rightarrow (\nabla u, \nabla v)_{L^2} \text{ is an equivalent inner}$$

$$\text{product } ((u, v))_{H_0^1(\Omega)}$$

$$((u, v))_{H_0^1(\Omega)} = \langle f, v \rangle_{H^1, H_0^1} \quad (\text{smiley face})$$

By Riesz - Representation Theorem,

$$\exists ! u \in H_0^1(\Omega) \text{ s.t. } (\text{smiley face})$$

$\Rightarrow$ existence & uniqueness of (DBVP)

Lax-Milgram Lemma Suppose $a(u, v)$ is linear & bdd quadratic form defined on Hilbert space real H :

- $a(u, c_1 v_1 + c_2 v_2) = c_1 a(u, v_1) + c_2 a(u, v_2) \quad \forall v_1, v_2 \in H$
- $a(c_1 u_1 + c_2 u_2, v) = c_1 a(u_1, v) + c_2 a(u_2, v) \quad c_1, c_2 \in \mathbb{R}$
- $\exists \text{ const. } M > 0$

$$|a(u, v)| \leq M \|u\| \|v\|, \quad \forall u, v \in H$$

Assume $a(u, v)$ is coercive (强正定)

$$a(u, v) \geq \underbrace{\lambda_0}_{>0 \text{ const.}} \|u\|^2$$

Then $\forall f \in H^*$, $\exists ! u \in H$, s.t.

$$a(u, v) = \langle f, v \rangle \quad \forall v \in H$$

Proof: Observe $\forall$ fixed $u \in H$, the functional
 $v \in H \rightarrow a(u, v) \in \mathbb{R}$ is linear & bdd
 By Rieze, $\exists!$ $\hat{u} \in H$, s.t.
 $a(u, v) = (\hat{u}, v)$. Write $\hat{u} = Au \in H$

Claim: $A: H \rightarrow H$ is linear & bdd ($\|A\| \in M$)
"1-1-ness": Recall $a(u, u) \geq \lambda_0 \|u\|^2$

$$\boxed{\square} \quad (Au, u) \leq \|Au\| \cdot \|u\|$$

$$\Rightarrow \|Au\| \geq \lambda_0 \|u\| \Rightarrow A \text{ is 1-1}$$

"Onto-ness": $R(A)$ is closed

why: Suppose $Au_k \xrightarrow{k \rightarrow \infty} \text{some } \hat{u} \in H$

$$\Rightarrow \lim \|Au_k - \hat{u}\| = 0 \xRightarrow{\boxed{\square}} \{u_k\} \text{ is Cauchy}$$

$$\Rightarrow u_k \xrightarrow{k \rightarrow \infty} \checkmark \text{ s.t. } A \text{ bdd} \Rightarrow Au_k \rightarrow A\checkmark$$

$$\Rightarrow \hat{u} = A\checkmark \in R(A)$$

(R) If $R(A) \neq H$, then $\exists w \in H$ s.t. $w \perp R(A)$ $w \neq 0$
 i.e. $(Au, w) = 0, \forall u \in H$

$$\text{In particular, } (Aw, w) = 0 \Rightarrow w = 0 \leftarrow$$

$$\checkmark \lambda_0 \|w\|^2$$

We have shown $A: H \rightarrow H$ is 1-1 & onto

Now $\forall f \in H^*$, by Rieze, $\exists! \tilde{f} \in H$ s.t.

$$(\tilde{f}, v) = \langle f, v \rangle, \forall v \in H$$

Because A is 1-1 & onto, $\exists! u \in H$ s.t. $Au = \tilde{f}$

$$\Rightarrow \langle f, v \rangle = (\tilde{f}, v) = (Au, v) = a(u, v)$$

QED

$$\text{Rk} \quad a(u, v) = \langle f, u \rangle \leq \|f\|_{H^*} \|u\|$$

$$\lambda_0 \|u\|^2$$

$$\Rightarrow \|u\| \leq \frac{1}{\lambda_0} \|f\|_{H^*}$$

$$\|A^{-1} \hat{f}\| \leq \frac{1}{\lambda_0} \|\hat{f}\|_{H^*}$$

$$\Rightarrow \|A^{-1}\| \leq \frac{1}{\lambda_0}$$

Existence Uniqueness Th $\exists \delta_0 \geq 0$ (const.) depending only on a_{ij}, b_i, c s.t. for each $\delta \geq \delta_0$, and each $f \in H^{-1}(\Omega)$

$$(\text{DBVP}) \quad \begin{cases} Lu + \delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

has one & only one weak sol u , moreover,

$$\|u\|_{H_0^1(\Omega)} \leq \frac{2}{\lambda_0} \|f\|_{H^{-1}(\Omega)}$$

Rk Why need $\delta \geq \delta_0$. $\begin{cases} -u'' - u = 0 & \text{on } (0, \pi) \\ u(x) = C \sin x & \\ u = 0 & \text{on } \partial(0, \pi) \end{cases}$ $L u$
Uniqueness fails!

Proof

Let $a_\delta(u, v) = \int_\Omega (a_{ij} u_{x_i} v_{x_j} + b_i u_{x_i} v + c u v) dx$
 $\Rightarrow a_\delta(u, v)$ is linear & bdd bilinear form on $H_0^1(\Omega)$

$$a_\delta(u, u) \geq \int_\Omega (a_{ij} u_{x_i} u_{x_j} + b_i u_{x_i} u + c u^2 + \delta u^2) dx$$

$$\geq \lambda_0 \int_\Omega |\nabla u|^2 dx - \int_\Omega \frac{1}{2\epsilon^2} |b|^2 u^2 + \frac{\epsilon^2}{2} |\nabla u|^2$$

$$- \|c\|_\infty \int_\Omega u^2 dx + \delta \int_\Omega u^2 dx$$

Take $\epsilon^2 = \lambda_0$

$$\left| \epsilon a \cdot \frac{b}{\epsilon} \right| \leq \frac{1}{2} \epsilon^2 |a|^2 + \frac{1}{2\epsilon^2} |b|^2$$

$$\geq \frac{\lambda_0}{2} \int_{\Omega} |\nabla u|^2 dx + \int_{\Omega} \left(6_0 - \|C\|_{\infty} - \frac{1}{2\lambda_0} \|T\|_{\infty}^2 \right) u^2 dx$$

$$\geq \dots + 0 \quad \text{if} \quad b_0 = \|c\|_\infty + \frac{1}{2\lambda_0} \|b\|_\infty^2$$

$$\Rightarrow a_6(u, u) \geq \frac{1}{2} \|u\|_{H_0^1}^2 \quad \forall u \in H_0^1(\Omega)$$

i.e. A_6 is coercive in H_0^1

Now Lax-McGragam implies

s.t. $a_b(u, v) = \langle f, v \rangle$

$$\|u\|_{H^1_0(\Omega)} \leq \frac{1}{\frac{\lambda_0}{2}} \|f\|_{H^{-1}}$$

$$Lu = - \underbrace{(a_{ij}(x))}_{\substack{\text{matrix} \\ \text{of order } n}} u_{x_i} + \underbrace{b(x)}_{\substack{\text{vector} \\ \text{of order } n}} u_{x_j} + \underbrace{c(x)}_{\substack{\text{scalar} \\ \text{function}}} u, \quad x \in \Omega \subset \mathbb{R}^n$$

Any, $b, c \in L^{\infty}(\Omega)$; strictly elliptic

$$(A_{\lambda} f(x))_{n \times n} \geq \lambda_0 I_{n \times n} \text{ a.e. on } \Omega \quad \lambda_0 > \text{const.}$$

Theorem $\exists \epsilon_0 = \epsilon_0(\lambda_0, \|b\|_\infty, \|C\|_\infty) > 0$ s.t. $\epsilon > \epsilon_0$

then $\int Lu \pm bu = f$ in Ω $f \in H^{-1}(\Omega) = (H_0^1(\Omega))^*$

$$(DBVP) \quad \begin{cases} \mu = 0 & \text{on } \partial\Omega \end{cases}$$

Dirichlet has one & only one weak sol $u \in H_0^1(\Omega)$

Moreover, $\|u\|_{H^1_0(\Omega)} \leq \frac{2}{\lambda_0} \|f\|_{H^{-1}}$

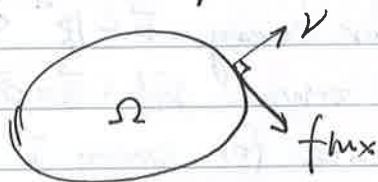
$$Lu_1 + bu_1 = f_1$$

$$Lu_2 + 6u_2 = f_2$$

$$\Rightarrow L(u_1 - u_2) + b(u_1 - u_2) = f_1 - f_2$$

$$\|u_1 - u_2\|_{H_0^1(\Omega)} \leq \|f_1 - f_2\|_{H^{-1}}$$

$$v_A = (u_{ij})_{n \times n} v$$



$$\text{Flux} = \text{density} \cdot \text{velocity} \xrightarrow{\text{Fourier's Law}} \frac{\text{Fourier's Law}}{\text{Fick's Law}} = - (u_{ij})_{n \times n} \nabla u$$

If $\partial\Omega$ is well-insulated,
then $\text{flux} \cdot \vec{v} = 0$

thermal + en
diffusion
各向异性.

$$\parallel$$

$$- (u_{ij})_{n \times n} \nabla u \cdot \vec{v}$$

$$\parallel$$

$$- (u_{ij})_{n \times n} \vec{v} \cdot \nabla u$$

$$\parallel$$

$$- \nabla u \cdot v_A$$

$$\parallel$$

$$\Rightarrow \frac{du}{dr} = 0$$

It is necessary to have " $\delta \geq \delta_0$ "
(DBVP) $\begin{cases} Lu = -u'' - u = f(x), & x \in (0, \pi) \\ u(0) = 0 = u(\pi) \end{cases}$

No uniqueness: $u(x) = C \sin x$

($\nabla \cdot \vec{v}$), $\sin x \in H_0^1(0, \pi)$?)

$$u \in W^{1,p}(\Omega) \cap C(\bar{\Omega}) \Rightarrow u \in W_0^{1,p}(\Omega)$$

$$1 \leq p < \infty \quad u|_{\partial\Omega} = 0$$

Recall

- $A_{n \times n} \vec{x} = \vec{b}$ has a sol for every $\vec{b} \in \mathbb{R}^n \Leftrightarrow A_{n \times n} \vec{x} = \vec{0}$ has only the trivial sol $\vec{x} = \vec{0}$
i.e. "Uniqueness $\Leftrightarrow$ existence for every $\vec{b}$ "
"Fredholm property"
- $A \vec{x} = \vec{b}$ has a sol $\Leftrightarrow \vec{b} \perp \vec{v}, \forall \vec{v} \text{ s.t. } A^* \vec{v} = \vec{0}$ (Real $A^* = A^T$, Complex $A^* = \overline{A^T}$)
- $\dim N(A) = \dim N(A^*)$

Preparation Fredholm Theory

Def Let X, Y be Banach spaces, $T: X \rightarrow Y$ be linear & bdd. We say T is Fredholm if

- $\dim N(T) < \infty$
- $R(T)$ is closed
- $\exists$ closed linear subspace Y_1 of Y s.t.
 $Y = R(T) \oplus Y_1, \dim Y_1 < \infty$

Index of T : $\text{ind}(T) = \dim N(T) - \dim Y_1$

e.g. $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$
 $\text{ind}(A) = 0$

Theorem If T is Fredholm & $K: X \rightarrow Y$ cpt. then $T+K$ is Fredholm, $\text{ind}(T+K) = \text{ind}(T)$

More preparation: Def The formal adjoint L^* of L is defined as

$$L^* u = -(a_{ij} u_{x_j})_{x_i} - (b_i(x) u)_{x_i} + c_i(x) u$$

★
(L)

Motivation Suppose everything is C^∞ -smooth, and
 $u, v \in C_0^\infty(\Omega)$

$$\begin{aligned} (Lu, v)_{L^2(\Omega)} &= \int_{\Omega} v Lu \, dx = \int_{\Omega} v (-(a_{ij} u_{x_i})_{x_j} + b_i u_{x_i} + cu) \, dx \\ &\stackrel{\text{integration by parts}}{=} \int_{\Omega} [-u (a_{ij} v_{x_i})_{x_j} - (b_i v)_{x_i} u + cvu] \, dx \\ &= \int_{\Omega} u L^* v \, dx = \underline{(L^* v, u)}_{L^2(\Omega)} \end{aligned}$$

Def $\forall v \in H_0^1(\Omega)$, $L^* v$ is understood as element in H^{-1}
 $\langle L^* v, u \rangle_{H^{-1}(\Omega), H_0^1(\Omega)} = \int_{\Omega} [a_{ij} u_{x_i} v_{x_j} + b_i(x) v_{x_i} u + cvu] \, dx$
Recall $\langle Lu, v \rangle_{H^{-1}, H_0^1}$

Fredholm Alternatives Theorem

(i) Either the homogeneous
 (DBVP) $\begin{cases} Lu = 0 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$ ☺

has a weak sol $u \neq 0$ (no uniqueness), or

$\forall f \in H^{-1}(\Omega)$, the inhomogeneous

(DBVP) $\begin{cases} Lu = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$ ☹

has a unique weak sol

(ii) the subspace $N(L)$ of all sol's of ☺

has finite dimension = dim of subspace $N(L^*)$

of all sol's of $\begin{cases} L^* v = 0 & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega \end{cases} \quad v \in H_0^1$

$N(L) = \text{sol of } \odot$

(13) $\odot$ has a weak sol iff $\langle f, v \rangle = 0$
 $\forall v \in N(L^*)$

Proof. Recall $\exists b_0$ s.t.

$$(DBVP) \begin{cases} Lu + b_0 u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

has one & only one sol

$$\Rightarrow L + b_0 I : H_0^1(\Omega) \rightarrow H^{-1}(\Omega) \text{ is 1-1 \& onto}$$

$$L = (L + b_0 I) - b_0 I$$

where $I : H_0^1 \rightarrow H^{-1}$ inclusion map

defined as : $\forall u \in H_0^1$,

$$\langle Iu, v \rangle_{H^{-1}, H_0^1} = \int_{\Omega} uv \, dx$$

$$\forall v \in H_0^1$$

Claim $I : H_0^1 \rightarrow H^{-1}$ cpt.

Inclusion map $H_0^1 \xrightarrow{I_1} L^2 \xrightarrow{I_2} H^{-1}$. I_2 defined as

$$\forall u \in L^2(\Omega), \langle I_2 u, v \rangle_{H^{-1}, H_0^1} = \int_{\Omega} uv \, dx$$

Observe $I = I_2 I_1$,

bdcl cpt. by Rellich-Kondrakov

$$\Rightarrow I \text{ cpt.}$$

$$W_0^{1,p} \xrightarrow{\text{cpt}} L^q$$

$$1 \leq q < \frac{np}{(n-p)^+}$$

Now

$$L = (L + b_0 I) - b_0 I : H_0^1 \rightarrow H^{-1}$$

$$\begin{array}{ccc} \text{isomorphism} & \nwarrow & \text{cpt} \\ \downarrow & & \\ \text{ind} = 0 & & \end{array}$$

$$\text{ind}(L) = \text{ind}(L + b_0 I) = 0$$

$$\Rightarrow L \text{ is 1-1} \Leftrightarrow L \text{ is onto} \quad (L : H_0^1 \rightarrow H^{-1}) \Rightarrow (i)$$

Δ
 $V(L)$

and $\dim N(L) < \infty$. Similarly, $\dim N(L^*) < \infty$

$$L^*: H_0' \rightarrow H^{-1} \text{ bdd}$$

Claim $\dim N(L) \leq \dim N(L^*)$

Suppose $N(L^*) = \text{span} \{e_1, \dots, e_k\} \subset H_0'$

By Hahn-Banach, $\exists f_1, \dots, f_k \in H^{-1}$ s.t.

$$\langle f_i, e_j \rangle = \delta_{ij}$$

Sub Claim: $\text{span} \{f_i\}_{i=1}^k \cap R(L) = \{0\}$

Otherwise, $\exists u \in H_0'$ & consts. $c_1, \dots, c_k$ s.t.

$$\langle Lu = c_1 f_1 + \dots + c_k f_k, e_i \rangle \Rightarrow$$

$$\langle Lu, e_i \rangle = c_i \Rightarrow Lu = 0 \rightarrow \leftarrow$$

$$\langle L^* e_i, u \rangle$$

Since $\text{ind}(L) = 0$, $\exists$ closed subspace $Y_1 \subset H^{-1}$ s.t.

$$H^{-1} = R(L) \oplus Y_1, \dim Y_1 = \dim N(L)$$

$$\Rightarrow f_i = \underbrace{Lu_i}_{\in R(L)} + \underbrace{y_i}_{\in Y_1}$$

$$u_i \in H_0'$$

Sub claim $\{y_i\}_{i=1}^k$ lin. indep. (& hence, $k \leq \dim Y_1$)

$$\dim N(L)$$

If not, $\exists c_1, \dots, c_k$ not all of them $= 0$

$$\text{s.t. } c_1 y_1 + \dots + c_k y_k = 0$$

$$c_1 f_1 + \dots + c_k f_k = c_1 Lu_1 + \dots + c_k Lu_k$$

$$\text{span} \{f_i\}$$

$$R(L)$$

$\Rightarrow LHS = 0$ contradicts $\{f_i\}_{i=1}^k$ lin. indep.

$$\Rightarrow \dim N(L^*) \leq \dim N(L)$$

The same argument $\Rightarrow \dim N(L^*) \geq \dim N(L)$
 $\Rightarrow (i)$

$$Lu = -(a_{ij}(x)u_{x_i})_{x_j} + b_i(x)u_{x_i} + c(x)u, \quad x \in \underset{\substack{\uparrow \\ \text{bdd.}}}{\Omega} \subset \mathbb{R}^n$$

$$(DBVP) \quad \begin{cases} Lu = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

Q: Solvability of (DBVP)?

Fredholm Alternatives Theorem

Suppose L is strictly elliptic on Ω

$$a_{ij}, b_i, c \in L^\infty(\Omega)$$

(i) either the homogeneous DBVP

$$\textcircled{\smiley} \quad \begin{cases} Lu = 0 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

has a nontrivial weak sol $u \neq 0$

or $\forall f \in H^{-1}(\Omega), \exists!$ weak $u \in H_0^1(\Omega)$

(ii) Let $N(L)$ be the set of all sol's of $\textcircled{\smiley}$

$(L: H_0^1 \rightarrow H^{-1})$; let $N(L^*)$ be the set of all weak sol's of

$$\begin{cases} L^*v = 0 & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega \end{cases} \quad L^*v = -(a_{ij}v_{x_i})_{x_j} + (b_i v)_{x_i} + cv$$

$(L^*: H_0^1 \rightarrow H^{-1})$ Then $\dim N(L) = \dim N(L^*) < \infty$

$$\text{e.g. } \begin{cases} -u'' - u = \sin x & \text{on } (0, \pi) \\ u(0) = 0 = u(\pi) \end{cases}$$

No sol!

(iii) (DBVP) has a sol $\Leftrightarrow \langle f, v \rangle_{H^{-1}, H_0^1} = 0, \forall v \in N(L)$
 " $f \in {}^\perp N(L)$ "

Proof of (iii):

In operator language. W.T.S:

$$R(L) = {}^\perp N(L)$$

First, $R(L) \subset {}^\perp N(L)$

Why: $\forall u \in H_0^1(\Omega), \forall v \in N(L)$

$$\langle Lu, v \rangle_{H^{-1}, H_0^1} = \langle L^* v, u \rangle_{H^{-1}, H_0^1} = 0$$

$$\Rightarrow Lu \in {}^\perp N(L) \Rightarrow R(L) \subset {}^\perp N(L)$$

2nd, $R(L) \supset {}^\perp N(L)$

$\forall f \in {}^\perp N(L)$. Recall $H^{-1}(\Omega) = R(L) \oplus \text{span} \{f_i\}_{i=1}^k$

$$k = \dim N(L) = \dim N(L^*)$$

$$N(L^*) = \text{span} \{e_j\}_{j=1}^k, \langle f_i, e_j \rangle = \delta_{ij}$$

$$\Rightarrow f = c_1 f_1 + \dots + c_k f_k + r, r \in R(L)$$

$$\langle f, e_j \rangle = \sum_{i=1}^k \underbrace{c_i}_{=0} \underbrace{\langle f_i, e_j \rangle}_{=\delta_{ij}} + \underbrace{\langle r, e_j \rangle}_{=0}$$

$$\Rightarrow c_j = 0, j = 1, \dots, k$$

$$\Rightarrow f \in R(L)$$

$$\Rightarrow R(L) \supset {}^\perp N(L)$$

$$v = c \sin x$$

$$v_j = c_1 \sin x + c_2 \cos x$$

$$v'' + v = 0$$

$$\lambda^2 + 1 = 0$$

$$\begin{cases} Lv = 0 & \text{on } \Omega \\ v = 0 & \text{on } \partial\Omega \\ L^* v = Lv \end{cases}$$

e.g. $Lu = -u'' - u, \Omega = (0, \pi)$

(DBVP) $\begin{cases} Lu = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$

Q: For what f does it have a sol?

A: $\langle f, \sin x \rangle_{H^{-1}, H_0^1} = 0$

$\wedge f = x \sin x \times f$ 必须变号

"On the street": $\int_0^{\pi} (-u'' - u) \sin x = f(x) \sin x \, dx$

"

$$\int_0^{\pi} u(x) [L(\sin x)] \, dx = \int_0^{\pi} [u'(\sin x)' - u \sin x] \, dx - (-u' \sin x) \Big|_0^{\pi}$$

$$\Rightarrow \int_0^{\pi} f(x) \sin x \, dx = 0$$

"In the Zury tower":

A: $f \in H^{-1}(\Omega)$. $\langle f, \sin x \rangle_{H^{-1}, H_0^1} = 0$

$N(L^*) = \text{span}\{\sin x\}$? $\sin x \in H_0^1(0, \pi)$??

$\downarrow$
 L

Regularity Theory $\rightarrow \begin{cases} Lv = 0 & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega \end{cases}$
classical sol ?

Theorem $1 \leq p < \infty$ if $u \in W^{1,p}(\Omega) \cap C(\bar{\Omega})$, $u|_{\partial\Omega} = 0$
then $u \in W_0^{1,p}(\Omega)$

Preparation Lemma $1 \leq p < \infty$. If $u \in W^{1,p}(\Omega)$,
 $\text{Supp } u \subset\subset \Omega$, then $u \in W_0^{1,p}(\Omega)$

Proof. Consider mollification u_ε of u :

$$u_\varepsilon(x) = \int_{\Omega} \zeta_\varepsilon(x-y) u(y) \, dy$$

$$\because u \in L^p(\Omega), \therefore u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} u \text{ in } L^p(\Omega) \text{ (old le. 3)}$$

$\Omega_{100\varepsilon}$. Recall

$$\frac{\partial u_\varepsilon}{\partial x_k}(x) = \left(\frac{\partial \zeta}{\partial x_k} \right)_\varepsilon(x), \quad \forall x \in \Omega_\varepsilon = \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > \varepsilon\}$$

— weak derivative



Take small $\varepsilon > 0$, s.t. $\text{supp } u \subset \Omega_{100\varepsilon}$
Now $\forall x \in \Omega \setminus \Omega_\varepsilon$, $u_\varepsilon \equiv 0$,

$$u \equiv 0 \quad \text{in } \Omega \setminus \Omega_{100\varepsilon}$$

$$\Rightarrow \frac{\partial u}{\partial x_k} = 0 = \left(\frac{\partial u}{\partial x_k} \right)_\varepsilon \quad \text{in } \Omega \setminus \Omega_{100\varepsilon}$$

$$\quad \quad \quad \uparrow \quad \text{in } \Omega \setminus \Omega_\varepsilon$$

$$\Rightarrow \frac{\partial u_\varepsilon}{\partial x_k} = \left(\frac{\partial u}{\partial x_k} \right)_\varepsilon \quad \text{on } \Omega$$

$$\downarrow \varepsilon \rightarrow 0^+$$

$$\frac{\partial u}{\partial x_k} \quad \text{in } L^p(\Omega)$$

$$\varepsilon \rightarrow 0 \Rightarrow u_\varepsilon \rightarrow u \quad \text{in } W^{1,p}_0(\Omega)$$

$$u_\varepsilon \in C_0^\infty(\Omega)$$

Now by def. of $W^{1,p}_0(\Omega)$, $u \in W^{1,p}_0(\Omega)$

• Proof of Theorem: $u = u^+ + \underbrace{u^-}_{\text{老王头}} = \min(0, u)$

W.T.S. $u^\pm \in W^{1,p}_0(\Omega)$

Let $v_\varepsilon = (u - \varepsilon)^+$

$$\Rightarrow v_\varepsilon \in W^{1,p}(\Omega) \quad \rightarrow \text{利用了 } u \text{ 在边界上的连续性}$$

$$v_\varepsilon \equiv 0 \quad \text{near } \partial\Omega \quad \text{i.e.} \quad \text{supp } v_\varepsilon \subset\subset \Omega$$

$\Downarrow$ Lemma

$$v_\varepsilon \in W^{1,p}_0(\Omega)$$

Just need to show $v_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} u^+ \quad \text{in } W^{1,p}(\Omega)$

$$\int_\Omega |v_\varepsilon(x) - u^+(x)|^p \leq \int_\Omega \varepsilon^p dx = \varepsilon^p |\Omega| \rightarrow 0$$

$$\int_\Omega \left| \frac{\partial v_\varepsilon}{\partial x_k}(x) - \frac{\partial u^+}{\partial x_k}(x) \right|^p dx \leq \int \left| \frac{\partial u}{\partial x_k}(x) \right|^p dx$$

$$\{0 < u(x) \leq \varepsilon\}$$

$$= \int \mathbf{1}_{\{0 < u \leq \varepsilon\}} \left| \frac{\partial u}{\partial x_k} \right|^p dx \quad \xrightarrow[\varepsilon \rightarrow 0]{\text{LDCT}} 0$$

$$\Rightarrow v_\varepsilon \rightarrow u^+ \quad \text{in } W^{1,p}(\Omega)$$

$$\Rightarrow u^+ \in W^{1,p}_0(\Omega) \quad \text{b.c.} \quad W^{1,p}_0(\Omega) \text{ is closed in } W^{1,p}(\Omega)$$

QED

Introduction to Regularity Theory

Ω

不考虑边界

L^2 -interior regularity

$$Lu = -(a_{ij}(x)u_{x_i x_j})_{x_j} + b_i(x)u_{x_i} + c(x)u \text{ in } \Omega$$

strictly elliptic on Ω

If $a_{ij} \in C^1(\Omega)$, $b_i, c \in L^\infty(\Omega)$, and u is a H^1 -weak sol of $Lu = f$ in Ω , where $f \in L^2(\Omega)$

Then $u \in H^2_{loc}(\Omega)$ ($H^2 = W^{2,2}$) ($-u'' = f$) for e.g.

Moreover $\|u\|_{H^2(\Omega')} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)})$

$\forall \Omega'' \subset \subset \Omega$

$C \sim n$, geometry of Ω' & Ω
ellipticity const. λ_0
 $(a_{ij})_{n \times n} \geq \lambda_0 I_{n \times n}$

Higher-order interior Regularity

$$\begin{aligned} \text{Formally: } & L - (a_{ij}u_{x_i x_j})_{x_j} + b_i u_{x_i} + cu = f \\ & - (a_{ij}(u_{x_k})_{x_i})_{x_j} - (a_{ij})_{x_k} u_{x_i x_j} + b_i (u_{x_k})_{x_i} \\ & + (b_i)_{x_k} u_{x_i} + c_{x_k} u + cu_{x_k} = f_{x_k} \end{aligned}$$

Let $v = u_{x_k} \Rightarrow H^1_{loc}(\Omega)$

$$-(a_{ij}v_{x_i})_{x_j} + b_i v_{x_i} + cv = f_{x_k} + \underbrace{(a_{ij})_{x_k} u_{x_i x_j}}_{a_{ij} \in C^2(\Omega)} - \underbrace{(b_i)_{x_k} u_{x_i}}_{b_i \in C^1(\Omega)} - c_{x_k} u$$

Assume $\cdot a_{ij} \in C^2(\Omega)$, $b_i, c \in C^1(\Omega)$

$\cdot f \in H^1_{loc}(\Omega)$

$\Rightarrow u \in H^3_{loc}(\Omega)$ (b/c. $v \in H^2_{loc}(\Omega)$)

more generally, assume $\cdot f \in H^m_{loc}(\Omega)$

$m \geq 1$

$\cdot a_{ij} \in C^{m+1}(\Omega)$, $b_i, c \in C^m(\Omega)$

$$\Rightarrow u \in H_{loc}^{m+2}(\Omega)$$

Special case: $a_{ij}, b_i, c, f \in C^\infty(\Omega)$

$$\Rightarrow u \in H_{loc}^k(\Omega) = W_{loc}^{k,2}(\Omega)$$

$$\hookrightarrow C_{loc}^{k-\frac{n}{2}}(\Omega)$$

$$\Rightarrow u \in C^\infty(\Omega)$$

Global Regularity Theory

Assume $\cdot$ $a_{ij} \in C^1(\bar{\Omega})$, $b_i, c \in L^\infty(\Omega)$
 $\cdot \partial\Omega \in C^2$

Then if u is a weak sol of

$$H_0^1 \quad \begin{cases} Lu = f(x) & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad f \in L^2(\Omega)$$

$$\Rightarrow u \in H^2(\Omega)$$

$$\|u\|_{H^2(\Omega)} \leq C [\|f\|_{L^2(\Omega)} +]$$

$\|u\|_{L^2(\Omega)}$
 C -geometry of Ω

Higher order:
 $\begin{cases} -\Delta u = f \\ u|_{\partial\Omega} = 0 \end{cases}$

More generally assume $m \geq 1$

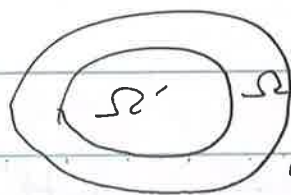
$$\cdot \partial\Omega \in C^{m+2}$$

$$\cdot f \in H^m(\Omega)$$

$$\cdot a_{ij} \in C^{m+1}(\bar{\Omega}), b_i, c \in C^m(\bar{\Omega})$$

$$\Rightarrow u \in H^{m+2}(\Omega) \text{ \& } \|u\|_{H^{m+2}(\Omega)} \leq C [\|f\|_{H^m} + \|u\|_{L^2}] :$$

Special case: $a_{ij}, b_i, c, f \in C^\infty(\bar{\Omega}) \Rightarrow C^\infty(\bar{\Omega})$
 $\partial\Omega \in C^\infty$



$$Lu = -(a_{ij}(x)u_{x_j})_{x_i} + b_i(x)u_{x_i} + c(x)u$$

Strictly elliptic: $(a_{ij}(x))_{n \times n} \geq \lambda I_{n \times n}$ a.e. in Ω

Suppose $b, c \in L^\infty(\Omega)$, $a_{ij} \in C^1(\Omega)$

u is a weak sol of

$$Lu = f(x), \quad x \in \Omega$$

Interior H^2 estimate

$f \in L^2(\Omega)$. Then $\forall \Omega' \subset\subset \Omega$, $u \in H^2(\Omega')$ (i.e. $u \in H^2_{loc}(\Omega)$)

$$\|u\|_{H^2(\Omega')} \leq C [\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}]$$

$C \sim n, \lambda, \|a_{ij}, b, c\|_{L^\infty}$, geometry of Ω' & Ω

$$-\Delta u + u = f(x), \quad x \in \mathbb{R}^n$$

$$f \in L^2(\mathbb{R}^n)$$

• existence of weak sol $u(x)$?

• behavior of $u(x)$ as $|x| \rightarrow \infty$

Existence: Do we have $u \in H^1(\mathbb{R}^3)$ s.t.

$$\textcircled{\smile} \int_{\mathbb{R}^3} [\nabla u \cdot \nabla v + uv] dx = \int_{\mathbb{R}^3} f(x) v(x) dx, \quad \forall v \in H^1(\mathbb{R}^3)$$

(This holds, in particular, for $v \in C_0^\infty(\mathbb{R}^3)$)

$$(u, v)_{H^1(\mathbb{R}^3)} = (f, v)_{L^2(\mathbb{R}^3)}$$

$$v \in H^1(\mathbb{R}^3) \rightarrow \int_{\mathbb{R}^3} f(x) v(x) dx \text{ is linear \& bdd}$$

$$|\int_{\mathbb{R}^3} f(x) v(x) dx| \stackrel{\text{Schwarz}}{\leq} \|f\|_{L^2} \|v\|_{L^2} \leq \|f\|_{L^2} \|v\|_{H^1}$$

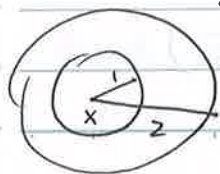
$\Rightarrow$ Yes $\exists!$ $u \in H^1(\mathbb{R}^3)$ s.t.

$\textcircled{\smile}$ holds (by Riesz Representation Th)

Now use H^2 -interior estimate with $\Omega = B_2(x)$

$\Rightarrow u \in H^2(B_1(x))$ ind. of x $\Omega' = B_1(x)$

$$\|u\|_{H^2(B_1(x))} \leq C [\|f\|_{L^2(B_2(x))} + \|u\|_{L^2(B_1(x))}]$$



$$\int_{B_2(x)} f^2(y) dy = \int_{\mathbb{R}^3} f^2(y) \chi_{B_2(x)}(y) dy \xrightarrow{|x| \rightarrow \infty} 0$$

(D.C.)

$$\hookrightarrow W^{2,2}(B_{1/2}(x)) \hookrightarrow C^{2-\frac{3}{2}}(\overline{B_{1/2}(x)})$$

Sobolev imbedding.

$$\Rightarrow \|u\|_{H^2(B_{1/2}(x))} \rightarrow 0$$

$$C\|u\|_{C^{\frac{1}{2}}(\overline{B_{1/2}(x)})} \geq C|u(x)|$$

More regularity of u ?

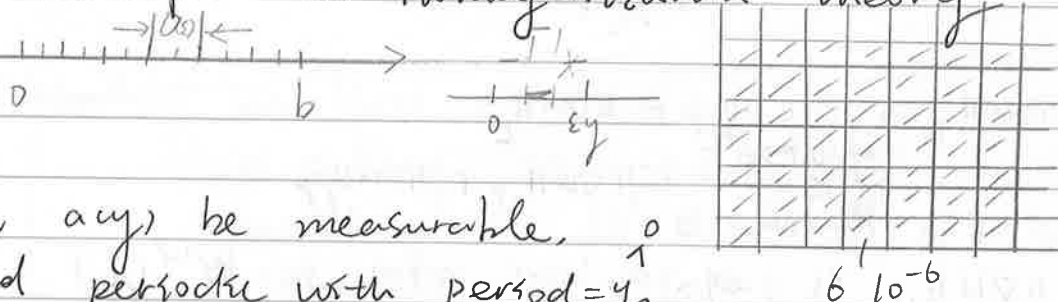
Assume $f \in L^2(\mathbb{R}^3) \cap H_{loc}^m(\mathbb{R}^3)$

$m \geq 1 \Rightarrow$ By higher order regularity theory

$$u \in H_{loc}^{m+2}(\mathbb{R}^3) \hookrightarrow C_{loc}^{m+2-\frac{3}{2}}(\mathbb{R}^3) \hookrightarrow C_{loc}^2(\mathbb{R}^3)$$

if $m \geq 2 \Rightarrow$ classical solution

An example in Homogenization Theory



Let $a(y)$ be measurable, $\int_0^1 a(y) dy = 1$
and periodic with period $= y_0$

$$\inf_0^{y_0} a(y) \leq a(y) \leq \frac{1}{y_0} \quad \text{a.e. } y \in \mathbb{R}^1$$

$y_0 = \text{const.}$

Let the thermal conductivity function of rod $[0, b]$ be given $a(\frac{x}{\epsilon})$ (period $= \epsilon y_0$)

Heat eq on the rod

$$\begin{cases} \frac{\partial u}{\partial t}(x, t) = (a(\frac{x}{\epsilon}) u_x(x, t))_x + f(x) & x \in (0, b) \\ u(0, t) = 0 = u(b, t) \end{cases}$$

Consider steady state

want: effective thermal condition

等效, 实效

Consider steady state

$$\begin{cases} -(a(\frac{x}{\varepsilon})u')' = f(x), & x \in (0, b) \\ u(0) = 0 = u(b) \end{cases}$$

Q : • existence ?

• $u_\varepsilon \rightarrow ?$ as $\varepsilon \rightarrow 0$

By Fredholm's Alternatives Theorem.

$\exists!$ weak sol $u_\varepsilon \in H_0^1(0, b)$ (b/c " $0=0$ ")

$$\int_0^b a(\frac{x}{\varepsilon}) u_\varepsilon' v' dx = \int_0^b f(x) v(x) dx, \quad \forall v \in H_0^1(0, b)$$

Take $v = u_\varepsilon \Rightarrow \int_0^b a(\frac{x}{\varepsilon}) (u_\varepsilon')^2 dx = \int_0^b f(x) u_\varepsilon(x) dx \leq \left(\int_0^b f^2 dx \right)^{\frac{1}{2}} \left(\int_0^b u_\varepsilon^2 dx \right)^{\frac{1}{2}}$

$$\lambda_0 \int_0^b (u_\varepsilon')^2 dx \leq C \|f\|_{L^2(0, b)} \left(\int_0^b u_\varepsilon^2 dx \right)^{\frac{1}{2}}$$

$$\|u\|_{W^{1,p}} = \|u\|_{L^p} + \|\nabla u\|_{L^p}$$

if $u \in W_0^{1,p}$
 $\leq C \|\nabla u\|_{L^p} + \|u\|_{L^p}$
 Poincaré

$\|\nabla u\|_{L^p}$ is equivalent norm in $W_0^{1,p}(\Omega)$

$$k = \frac{1}{\varepsilon \gamma} \int_0^{\gamma \varepsilon} a(\frac{x}{\varepsilon}) dx \quad \begin{cases} -(k v)'' = f(x) \\ v(0) = 0 = v(b) \end{cases}$$

$$\Rightarrow \int_0^b (u_\varepsilon')^2 dx \leq \tilde{C} \int_0^b f^2(x) dx, \quad \forall \varepsilon > 0$$

$$\Rightarrow \{u_\varepsilon\} \text{ is bdd } H_0^1(0, b) \Rightarrow$$

after passing to a subseq.

$$u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} u_0 \text{ weakly in } H_0^1(0, b)$$

$$+ \text{ strongly in } L^2(0, b)$$

Recall $H_0^1(0, b) \xrightarrow{\text{cpt}} L^2(0, b)$

So after passing to yet another subseq.

$u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \text{some } \bar{u}_0 \text{ in } L^2(0, b)$

Claim 1: $u_0 = \bar{u}_0$ why: Fix $w \in L^2(0, b)$. Consider
 $u \in H_0^1(0, b) \rightarrow \int_0^b w(x) u(x) dx$. It is lin. & bdd.
 $\Rightarrow \int_0^b w(x) u_\varepsilon(x) dx \xrightarrow{\varepsilon \rightarrow 0} \int_0^b w(x) u_0(x) dx$
 $\downarrow$
 $\int_0^b w(x) \bar{u}_0(x) dx \quad \parallel \quad \Rightarrow \text{done}$

Claim 2: $u'_\varepsilon \rightarrow u'_0$ in $L^2(0, b)$
 (i.e. $\forall w \in L^2(0, b)$, $\int_0^b u'_\varepsilon w(x) dx \rightarrow \int_0^b u'_0 w(x) dx$
 $u \in H_0^1 \rightarrow \int_0^b u' w(x) dx$ is
 lin. & bdd on H_0^1)

Claim 3: Let $v_\varepsilon(x) = a(\frac{x}{\varepsilon}) u'_\varepsilon(x)$. After passing to
 a subseq. $v_\varepsilon \rightarrow \text{some } v_0$
 weakly in $H^1(0, b) \xrightarrow{\text{cpt}} L^2(0, b)$
 Strongly in $L^2(0, b)$

Recall $\int_0^b a(\frac{x}{\varepsilon}) u'_\varepsilon(x) \varphi'(x) dx = \int_0^b f(x) \varphi(x) dx$. $\forall \varphi \in C_0^\infty$

$\Rightarrow$ weak $v'_\varepsilon = -f$ $\varphi \in C_0^\infty(0, b)$

$\Rightarrow v_\varepsilon \in H^1(0, b)$ & $\{v_\varepsilon\}$ is bdd. in $H^1(0, b)$

Recall

Fact $\forall h \in L^\infty(\mathbb{R})$, periodic in y with period

Then $h(\frac{x}{\varepsilon}) \xrightarrow{L^\infty\text{-weak}^*} \langle h \rangle = \frac{1}{y_0} \int_0^{y_0} h(y) dy$
 $\varepsilon \rightarrow 0$

3.e. $\forall g(x) \in L^1(0, b)$
 $\int_0^b h\left(\frac{x}{\varepsilon}\right) g(x) dx \xrightarrow{\varepsilon \rightarrow 0} \langle h \rangle \int_0^b g(x) dx$

Now $u'_\varepsilon(x) = v'_\varepsilon(x) / a(\frac{x}{\varepsilon})$
 $\downarrow \leftarrow \text{weakly in } L^2(a,b) \rightarrow \downarrow$ Claim 4
 u'_0 $v_0 \leftarrow \frac{1}{a}$

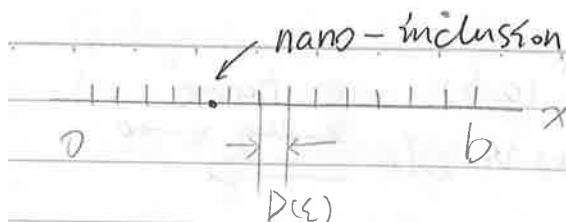
Proof of Claim 4. $\forall w \in L^2(a, b)$

$$\int_a^b w(x) V_\varepsilon(x) \frac{1}{a(\frac{x}{\varepsilon})} dx \xrightarrow{\text{W.T.S.}} \int_a^b w(x) V_0(x) \langle \frac{1}{a} \rangle dx$$

$$\int_0^b w(x) V_0(x) \frac{1}{a(\frac{x}{\varepsilon})} dx + \int_0^b w(x) (V_\varepsilon - V_0) \frac{1}{a(\frac{x}{\varepsilon})} dx \leq \frac{1}{\lambda_1}$$

$$\int_0^b w(x) V_0(x) dx \leq \frac{1}{h}$$

$$\Rightarrow u_0' = v_0 \left\langle \frac{1}{a} \right\rangle \left(- \left\langle \frac{1}{\frac{1}{a}} \right\rangle u_0' \right)' = v_0' = +f$$



Composite 复合材料

$$\begin{cases} -\frac{d}{dx} \left(a\left(\frac{x}{\varepsilon}\right) \frac{du}{dx} \right) = f(x), & x \in (a, b) \\ u(a) = 0 = u(b) \end{cases}$$

$$u \xrightarrow{\varepsilon \rightarrow 0} ? \quad a(y) \in L^\infty$$

$$\lambda_0 \leq h(y) \leq \frac{1}{\lambda_0}$$

$$\lambda_0 > 0 \text{ const.}$$

$$\text{period} = y_0 > 0$$

$$f \in L^2(0, b)$$

Recall

- existence of sol u_ε
- $u_\varepsilon \rightarrow$ same u_0 after passing to a subseq.
weakly in $H_0^1(0, b)$
strongly in $L^2(0, b)$
- $u'_\varepsilon \rightarrow u'_0$ weakly in $L^2(0, b)$
- let $v_\varepsilon(x) = a\left(\frac{x}{\varepsilon}\right) u'_\varepsilon(x)$
 $\Rightarrow v_\varepsilon \rightarrow$ same v_0
weakly in $H_0^1(0, b)$
and strongly in $L^2(0, b)$

Fact (to be proved).

$$\forall h \in L^\infty(\mathbb{R}), \text{ period} = y_0 > 0$$

$$\text{Then } h\left(\frac{x}{\varepsilon}\right) \text{ } L^\infty\text{-weak}^*, \langle h \rangle \stackrel{\text{def}}{=} \frac{1}{y_0} \int_0^{y_0} h(y) dy$$

$$\text{i.e. } \forall g \in L^1(0, b), \text{ we have}$$

$$\int_0^b h\left(\frac{x}{\varepsilon}\right) g(x) dx \rightarrow \langle h \rangle \int_0^b g(x) dx$$

By def of weak sol, $\forall v \in H_0^1(0, b)$, we have

$$\int_0^b a(\frac{x}{\varepsilon}) u_\varepsilon'(x) v'(x) dx = \int_0^b f(x) v(x) dx \xrightarrow{\text{sending } \varepsilon \rightarrow 0}$$

$$\int_0^b v_0(x) v'(x) dx = \int_0^b f(x) v(x) dx$$

But: $u_\varepsilon'(x) = \frac{v_\varepsilon(x)}{a(\frac{x}{\varepsilon})}$ "u(x)"

weakly in $L^2(0, b)$

$$u_0'(x) \xrightarrow{S_0} v_0(x) < \frac{1}{a} >$$

now by ☹

$$\int_0^b \frac{u_0'(x)}{\langle \frac{1}{a} \rangle} v'(x) dx = \int_0^b f(x) v(x) dx$$

$\forall v \in H_0^1(0, b)$ (Note $u_0 \in H_0^1(0, b)$!)

$\Rightarrow u_0$ is a weak sol of

$$\begin{cases} -\left(\frac{1}{\langle \frac{1}{a} \rangle} u_0'\right)' = f(x), & x \in (0, b) \\ u_0(0) = 0 = u_0(b) \end{cases} \quad "C \equiv 0"$$

Effective model

This problem has uniqueness of sol

$\Rightarrow u_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} u_0$ weakly in $H_0^1(0, b)$
strongly in $L^2(0, b)$

without passing to a subseq

$\frac{1}{\langle \frac{1}{a} \rangle}$ = effective thermal conductivity

Epilogue of L^2 -theory (De Giorgi-Nash theory)

$$Lu = -(a_{ij}(x)u_{x_i})_{x_j} + b_i(x)u_{x_i} + c(x)u, \quad x \in \Omega$$

strictly elliptic, $a_{ij}, b_i, c \in L^\infty(\Omega)$, Ω bdd in $\mathbb{R}^n$

Theorem

Suppose $u \in H^1(\Omega)$ is a weak sol of

$$Lu = f(x), \quad x \in \Omega$$

with $f \in L^q(\Omega)$, $q > \frac{n}{2}$. Then

- i) (interior regularity) $\forall \Omega' \subset\subset \Omega$, $\exists \alpha = \alpha(n, q, \lambda_0, \|a_{ij}, b_i, c\|_\infty)$
 s.t. $u \in C^\alpha(\bar{\Omega}')$

Moreover,

$$\|u\|_{C^\alpha(\bar{\Omega}')} \leq C [\|f\|_{L^q(\Omega)} + \|u\|_{L^2(\Omega)}]$$

$C = C(n, q, \lambda_0, \|a_{ij}, b_i, c\|_\infty, \text{geom of } \Omega' \& \Omega)$

- ii) (global boundedness) If $u \in H_0^1(\Omega)$, then $u \in L^\infty(\Omega)$

$$\|u\|_{L^\infty(\Omega)} \leq C [\|f\|_{L^q(\Omega)} + \|u\|_{L^2(\Omega)}]$$

$C = C(n, q, |\Omega|, \lambda_0, \|b_i, c\|_{L^\infty(\Omega)})$

- iii) (global regularity) Suppose $u \in H_0^1(\Omega)$, $\partial\Omega \in C^2$
 then $\exists \alpha = \alpha(n, q, \lambda_0, \|a_{ij}, b_i, c\|_{L^\infty(\Omega)}, \text{diam } \Omega)_{>0}$
 s.t. $u \in C^\alpha(\bar{\Omega})$

$$\|u\|_{C^\alpha(\bar{\Omega})} \leq C [\|f\|_{L^q(\Omega)} + \|u\|_{L^2(\Omega)}]$$

For Neumann/Robin B.C, see Tag de Prüfung
 thesis

L^p -theory $1 < p < \infty$

$$\Omega \text{ bdd in } \mathbb{R}^n \quad Lu = a_{ij}(x) u_{x_i x_j}(x) + b_i(x) u_{x_i} + c(x) u$$

$$a_{ij} \in C^0(\bar{\Omega}), b_i, c \in L^\infty(\Omega)$$

L strictly elliptic

Def We say u is a strong sol of

$$Lu = f(x), \quad x \in \Omega$$

if $u \in W^{2,p}(\Omega)$, and the PDE holds a.e. on Ω

$$(DBVP) \quad \begin{cases} Lu = f(x) \\ u|_{\partial\Omega} = 0 \end{cases}$$

Strong sol if $u \in W^{2,p}(\Omega) \cap W_0^{1,p}$ & PDE holds a.e. on Ω

$$u \in W_0^{2,p}$$

$\Downarrow$

$$u_k \rightarrow u \text{ in } W^{2,p}\text{-norm}$$

$$C_0^\infty(\Omega) \Downarrow$$

$$\frac{\partial u_k}{\partial x_i} \rightarrow \frac{\partial u}{\partial x_i} \text{ in } W^{1,p}\text{-norm}$$

Fredholm Alternatives Suppose $\partial\Omega \in C^2$

Then uniqueness of (DBVP) $\Leftrightarrow$ existence of (DBVP) for every $f \in L^p(\Omega)$

RK if $c(x) \leq 0$ a.e. on Ω , then uniqueness

Interior L^p -estimate: Let u be a strong sol of $Lu = f$ in Ω . Then $\forall \Omega' \subset\subset \Omega$, we have

$$\|u\|_{W^{2,p}(\Omega')} \leq C[\|f\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega)}]$$

$$C \sim \lambda_0, n, p, \|a_{ij}\|_{C^0(\bar{\Omega})}, \|b_i, c\|_{L^\infty(\Omega)}, \text{geom of } \Omega'$$

Global L^p -estimates. Suppose u is a strong sol of (DBVP) & $\partial\Omega \in C^2$. then

$$\|u\|_{W^{2,p}(\Omega)} \leq C[\|f\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega)}]$$

L^p -regularity theory

If $u \in W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega)$ is a strong sol of $Lu = f \in L^q(\Omega)$ w/ $q > p > 1$ & $\partial\Omega \in C^2$.

Then $u \in W^{2,q}(\Omega) \cap W_0^{1,q}(\Omega)$

Schauder theory

$Lu = a_{ij}(x)u_{x_i x_j} + b_i(x)u_{x_i} + c(x)u$. Strictly elliptic
 $a_{ij}, b_i, c \in C^\alpha(\bar{\Omega})$, $0 < \alpha < 1$, $\partial\Omega \in C^{2+\alpha}$

Fredholm Alternatives Consider

$$(DBVP) \quad \begin{cases} Lu = f(x) & x \in \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

Then uniqueness $\Leftrightarrow$ existence of $C^{2+\alpha}(\bar{\Omega})$ sol for every $f \in C^\alpha(\bar{\Omega})$

RK1 $C \leq 0 \Rightarrow$ Uniqueness

RK2 Existence under weak condition

Assume $a_{ij}, b_i, c \in C_{loc}^\alpha(\Omega) \cap L^\infty(\Omega)$, and $\partial\Omega$ satisfies exterior sphere condition

and $C \leq 0$ on $\bar{\Omega}$

Then $\forall \varphi \in C^0(\partial\Omega)$, $f \in C_{loc}^\alpha(\Omega) \cap L^\infty(\Omega)$

$$\begin{cases} Lu = f & \text{in } \Omega \\ u|_{\partial\Omega} = \varphi \end{cases}$$

has a unique sol ($u \in C_{loc}^{2+\alpha}(\Omega) \cap C^0(\bar{\Omega})$)

Schauder estimate

i) interior estimate Suppose $u \in C_{loc}^{2+\alpha}(\Omega)$

sol of $Lu = f \in C^\alpha(\bar{\Omega})$. Then $\forall \Omega' \subset\subset \Omega$ we have

$$\|u\|_{C^{2+\alpha}(\bar{\Omega}')} \leq C [\|f\|_{C^\alpha(\bar{\Omega})} + \|u\|_{C^0(\bar{\Omega})}]$$

(ii) Global Estimates

$$\begin{cases} Lu = f(x), & x \in \Omega \\ u|_{\partial\Omega} = 0 \end{cases} \quad f \in C^\alpha(\bar{\Omega}). \text{ Then}$$

$$\|u\|_{C^{2+\alpha}(\bar{\Omega})} \leq C [\|f\|_{C^\alpha(\bar{\Omega})} + \|u\|_{C^0(\bar{\Omega})}]$$

$$\begin{cases} -(a_{ij}(x) u_{x_i})_{x_j} + b_i u_{x_i} + cu = f \in C^\alpha(\bar{\Omega}) \\ u|_{\partial\Omega} = 0 \end{cases}$$

$$u \in H_0^1(\Omega) \quad \partial\Omega \in C^{2+\alpha}$$

$\Downarrow$ 满足 Schauder estimate.

$$u \in C^{2+\alpha}(\bar{\Omega})$$

L^p 收敛 $\Rightarrow$ L^1 收敛 $\xrightarrow{\text{No}}$ L^∞
(依测度收敛) $\nmid$ 处处

Hw 4

抽子引

收敛

Problem 2 $\Omega = (a, b)$, $u \in W_0^{1,p}(\Omega)$
 $1 < p < \infty$

(i) u is A.C. & $u(a) = 0 = u(b)$

Proof. By def of $W_0^{1,p}$, $\exists u_k \in C_0^\infty(\Omega)$ s.t.
 $u_k \xrightarrow{k \rightarrow \infty} u$ in $W_0^{1,p}(\Omega)$ -norm
 $\Rightarrow u_k \rightarrow u$ in $L^p(\Omega)$, $u_k' \rightarrow u'$ in $L^p(\Omega)$
 $\Rightarrow u_k \xrightarrow{k \rightarrow \infty} u$ a.e. on Ω
 ~~$u_k' \rightarrow u'$~~

W.T.S.

$$u(x) = \int_a^x u'(t) dt, \quad \forall x \in [a, b]$$

note

$$u_k(x) = \int_a^x u_k'(t) dt, \quad \forall x \in [a, b]$$

a.e. $\downarrow$

$$u(x) = \int_a^x u'(t) dt, \quad \forall x \in [a, b]$$

after passing
to a subseq

$$\Rightarrow u(x) \stackrel{\text{a.e.}}{=} \int_a^x u'(t) dt$$

But RHS is A.C. $\Rightarrow u$ is A.C.

$$\begin{aligned} \int_a^x u_k'(t) dt - \int_a^x u'(t) dt &\leq \int_a^b |u_k' - u'| dt \\ &\leq \left(\int_a^b |u_k' - u'|^p dt \right)^{\frac{1}{p}} \left(\int_a^b 1^p dt \right)^{\frac{1}{p}} \\ \therefore \int_a^x u_k' dt &\rightarrow \int_a^x u' dt \end{aligned}$$

HW3

Prob2

$u \in \mathcal{D}'(a, b)$ & $u' = 0 \Rightarrow u \equiv \text{const. } c$ in the sense of distribution

Proof Since $u' = 0$, we have $\forall \theta \in C_0^\infty(a, b)$, $\langle u, \theta' \rangle = 0$ 😊

Pick up $\varphi_0 \in C_0^\infty(a, b)$ s.t. $\int_a^b \varphi_0(x) dx = 1$
 $\forall \varphi \in C_0^\infty(a, b)$, define
 $\psi(x) = \varphi(x) - \varphi_0(x) \int_a^b \varphi(t) dt$
 $\Rightarrow \psi \in C_0^\infty(a, b)$

W.T.S. $\exists \theta \in C_0^\infty(a, b)$ s.t. $\theta' = \psi$
 Define $\theta(x) = \int_a^x \psi(t) dt \in C^\infty(a, b)$

$\because \varphi, \varphi_0 \equiv 0$ in a nbhd of $x=a$
 $\therefore \theta \equiv 0$

$\because \varphi, \varphi_0 \equiv 0$ in a nbhd of $x=b$,
 $\therefore \theta = \text{const.}$

$\theta(b) = 0 \Rightarrow \theta \equiv 0$ in a nbhd of $x=b$

By 😊 $0 = \langle u, \phi \rangle - \langle u, \phi_0 \rangle \int_a^b \varphi(t) dt$

$\Rightarrow \langle u, \phi \rangle = \langle c, \phi \rangle, \quad \forall \phi \in C_0^\infty(\Omega)$

$\Rightarrow u = c \quad \text{QED.}$

HW 3

 $n \geq 3$ $f \in L^1 \cap L^\infty$

Prob 4

$$u(x) = (P * f)(x) = \int_{\mathbb{R}^n} P(x-y) f(y) dy$$

$$P(x) = \frac{1}{(n-2)\omega_n |x|^{n-2}}$$

$\frac{1}{|B_1|}$

$$\int_{\mathbb{R}^n} \frac{1}{|x-y|^{n-2}} |f(y)| dy$$

$$= \int_{|x-y| < 1} + \int_{|x-y| \geq 1} \quad \text{电荷的质量}$$

$$= I + II$$

$$II \leq \int_{|x-y| \geq 1} |f(y)| dy = \|f\|_{L^1(\mathbb{R}^n)}$$

$$I \leq \|f\|_{L^\infty(\mathbb{R}^n)} \int_{|x-y| < 1} \frac{1}{|x-y|^{n-2}} dy$$

$$\frac{z=y-x}{dz=dy} \quad \|f\|_{L^\infty(\mathbb{R}^n)} \int_{|z| < 1} \frac{1}{|z|^{n-2}} dz$$



$$\int_0^1 \int_{|x|=r} \frac{1}{r^{n-2}} r^{n-1} dr d\sigma$$

$$\omega_n = C(n) \|f\|_{L^\infty(\mathbb{R}^n)} \int_0^1 r dr$$

$\Rightarrow u(x)$ is well-defined on $\mathbb{R}^n$

$$|u(x)| \leq C(n) [\|f\|_{L^1(\mathbb{R}^n)} + \|f\|_{L^\infty(\mathbb{R}^n)}]$$

$\Rightarrow (1) \& (3)$

$$dz = dr \cdot dbr$$

$$= dr \cdot r^{n-1} db_1$$

$$\langle -\Delta u, \varphi \rangle = \langle -u, \Delta \varphi \rangle$$

No

Date

(iii) u is a distributional sol of $-\Delta u = f$ in $\mathbb{R}^n$

$$\Leftrightarrow \forall \varphi \in C_0^\infty(\mathbb{R}^n)$$

$$\int_{\mathbb{R}^n} u(x) (-\Delta \varphi(x)) dx = \int_{\mathbb{R}^n} f(x) \varphi(x) dx$$

$$\int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} P(x-y) f(y) dy \right) (-\Delta \varphi(x)) dx$$

$$\stackrel{\text{Fubini}}{=} \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} P(x-y) (-\Delta \varphi(x)) dx \right] f(y) dy$$

$\parallel \varphi(y)$ by Result in class

$$= \int_{\mathbb{R}^n} \varphi(y) f(y) dy$$

$$(iv) \frac{u(x+he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \left(\frac{1}{|x+he_i-y|^{n-2}} - \frac{1}{|x-y|^{n-2}} \right) f(y) dy$$

$$g(t) = |x+te_i-y|^{2-n} \stackrel{\text{MVT}}{=} \int_{\mathbb{R}^n} g'(\bar{t}) f(y) dy, \bar{t} \in (0, h)$$

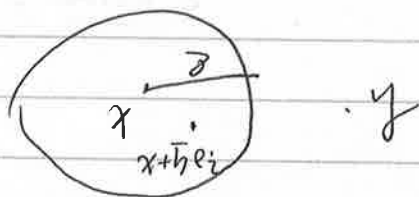
$$\begin{aligned} g'(t) &= (2-n) |x+te_i-y|^{1-n} \cdot \frac{x+te_i-y}{|x+te_i-y|} \cdot e_i \\ &= (2-n) |x+te_i-y|^{-n} \cdot (x_i+t-y_i) \\ &= (2-n) \left[\int_{B_2(x)} + \int_{(B_2(x))^c} \right] \end{aligned}$$

如果各阶偏导连续且相等

$f \in C^1$

如果各阶偏导连续且相等

$f \in C^1$



For γ $\checkmark O_n (B_b(x))^c$

$$\begin{aligned}
 |\text{integrand}| &\leq_{h \text{ small}} |x + \bar{h}e_i - y|^{1-n} |f(y)| \\
 &\leq \left(\frac{b}{2}\right)^{1-n} |f(y)| \\
 &\in L^1((B_b(x))^c)
 \end{aligned}$$

 $O_n B_b(x)$

$$\left| \int_{B_b(x)} \right| \leq \int_{|x + \bar{h}e_i - y| < b + \bar{h}} |x + \bar{h}e_i - y|^{1-n} dy \|f\|_\infty$$

$$\begin{aligned}
 \frac{z = y - (x + \bar{h}e_i)}{dz = dy} &\int_{|z| < b + \bar{h}} |z|^{1-n} dz
 \end{aligned}$$

$$= \int_{|z| < b + \bar{h}} |z|^{1-n} dz$$

$$= \int_0^{b+\bar{h}} \int_{|x|=1} r^{1-n} r^{n-1} dr dS$$

$$= W_n(b + \bar{h})$$

$$\Rightarrow \frac{u(x + \bar{h}e_i) - u(x)}{h} \leq (2-n) \int_{(B_b(x))^c} + (n-2)W_n(b + \bar{h})$$

$b > 0$ fixed

$$(2-n) \int_{(B_b(x))^c} - (n-2)W_n(b + \bar{h}) \Rightarrow$$

$$\leq \liminf \dots \leq \limsup \frac{u(x + \bar{h}e_i) - u(x)}{h} \leq$$

$$\begin{aligned}
 &(2-n) \int_{B_b(x)} |x - y|^{2-n} (x_s - y_s) \\
 &+ (n-2)W_n(b + \bar{h}) \|f\|_\infty
 \end{aligned}$$

f

$D'(\Omega)$

~~$D(\Omega) \subset C_c^\infty(\Omega)$~~
~~supply~~
 ~~$D(\Omega)$~~

By L.D.C.T., sending $\delta \rightarrow 0$, we have

$$\leq \lim \leq \lim \leq (2-n) \int |x-y|^{2n} (x_i - y_i) f(y) dy$$

$$\frac{\partial u}{\partial x_i}(x) \text{ exists.} = (2-n) \int_{\mathbb{R}^n} \frac{x_i}{|z|^{n+1}} f(x-z) dz$$

Continuity of $\frac{\partial u}{\partial x_i}$ at x^0

$$\left| \frac{\partial u}{\partial x_i}(x) - \frac{\partial u}{\partial x_i}(x^0) \right| \leq \left| \int_{\mathbb{R}^n} \left[\frac{x_i - y_i}{|x-y|^{n+1}} - \frac{x_i^0 - y_i^0}{|x^0-y|^{n+1}} \right] f(y) dy \right|$$

Supp f $\subset B_R(0)$

$\leq C$ for large x

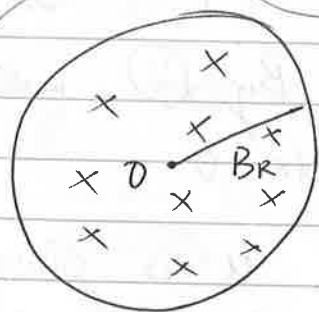
$$|x|^{n-2} u(x) = \int_{B_R(0)} \frac{|x|^{n-2}}{(n-2)W_n |x-y|^{n-2}} f(y) dy$$

$$\leq C f(y) \in L^1(B_R(0))$$

$$\xrightarrow{LDCI} \frac{1}{n(n-2)W_n} \int_{B_R(0)} f(y) dy$$

$$\Rightarrow u(x) \approx_{|x| \text{ large}} P(x) \int_{\mathbb{R}^n} f(y) dy \quad \text{质量}$$

点电荷 对应
产生的电势



$$\cdot x$$

$$n(n-2) / B_R(0)$$

$$Lu = -(a_{ij}(x) u_{x_i})_{x_j} + b_i(x) u_{x_i} + c(x) u$$

$$\begin{cases} Lu = f & \text{in } \Omega \\ \frac{\partial u}{\partial \nu_A} = 0 & \text{on } \partial\Omega \end{cases}$$

$$\text{or } (a_{ij}) \nabla \cdot \nabla u$$

$$(i) \int_{\Omega} [-(a_{ij} u_{x_i})_{x_j} v] + b_i u_{x_i} + c u v \, dx$$

$$= \int_{\Omega} f(x) u(x) \, dx \quad (\text{smiley face}) \Rightarrow \text{ii)}$$

$$- \int_{\partial\Omega} a_{ij} u_{x_i} v \nu_j \, ds + \int_{\Omega} a_{ij}$$

$$\text{or } \frac{\partial u}{\partial \nu_A}$$

$$v \in H^1(\Omega) \Rightarrow v_k \in C^1(\bar{\Omega}) \xrightarrow{k \rightarrow \infty} v \text{ in } H^1$$

$$\int_{\Omega} f_{x_j} g \, dx = - \int_{\Omega} f g_{x_j} \, dx + \int_{\partial\Omega} f g \nu_j \, ds \quad v \in C^1(\bar{\Omega})$$

Take $v \in C_0^\infty(\Omega)$, By (smiley face), we have

$$\int_{\Omega} (Lu - f) v \, dx = 0$$

$C_0^\infty(\bar{\Omega})$ b/c. u is classical

$$\exists v_k \in C_0^\infty(\Omega) \rightarrow Lu - f \text{ in } L^2(\Omega)$$

$$\Rightarrow \|Lu - f\|_{L^2(\Omega)} = 0$$

$$\Rightarrow Lu = f \text{ in } \Omega$$

Now take $v \in C^1(\bar{\Omega})$

$$(\text{smiley face}) \Rightarrow \int_{\partial\Omega} a_{ij} u_{x_i} \nu_j v \, ds = 0$$

$$\int_{\partial\Omega} \frac{\partial u}{\partial \nu_A} v \, ds = 0 \Rightarrow \frac{\partial u}{\partial \nu_A} = 0 \text{ on } \partial\Omega$$

$$n = 2$$

$$\begin{cases} \Delta u = 0 & \text{in } B_1 \setminus \{0\} \\ u = o(|\ln r|) & \text{as } r \rightarrow 0 \end{cases}$$

W.T.S.

$$\sup_{B_1 \setminus \{0\}} u = \sup_{\partial B_1} u$$

Recall $\Delta \ln |x| = 0$ if $x \neq 0$

Define for fixed $\varepsilon > 0$

$$V_\varepsilon(x) = u(x) + \varepsilon \ln |x| \Rightarrow V_\varepsilon = u \text{ on } \partial B_1$$

$$\Rightarrow \Delta V_\varepsilon = 0 \text{ in } B_1 \setminus \{0\}$$

As $|x| \rightarrow 0$

$$\begin{aligned} V_\varepsilon(x) &= o(\ln |x|) + \varepsilon \ln |x| \\ &= (\varepsilon + o(1)) \ln |x| \rightarrow -\infty \end{aligned}$$

$\Rightarrow$ Take $\delta > 0$ so small s.t.

$$V_\varepsilon|_{\partial B_\delta} < \sup_{\partial B_1} u$$

Now applying max principle on $\overline{B_1 \setminus B_\delta}$ to $V_\varepsilon \Rightarrow$

$$\max_{\overline{B_1 \setminus B_\delta}} V_\varepsilon = \max_{\partial B_1 \cup \partial B_\delta} V_\varepsilon = \sup_{\partial B_1} u$$

for small $\delta > 0$

Now sending $\delta \rightarrow \delta_1 \Rightarrow$

$$\sup_{\overline{B_1 \setminus \{0\}}} V_\varepsilon = \sup_{B_1} u$$

Now sending $\varepsilon \rightarrow 0$ Done.

(NBVP) $\begin{cases} -\Delta u = f & \text{in } \Omega, f \in L^2(\Omega) \\ \frac{\partial u}{\partial \nu} \big|_{\partial \Omega} = 0 \end{cases}$

Fredholm alternatives $\Rightarrow \int_{\Omega} f v \, dx \Rightarrow \int_{\Omega} f(x) \, dx = 0$

(NBVP) has a sol $\Leftrightarrow \langle f, v \rangle = 0$

$\forall v \in H^1(\Omega)$ s.t.

$$\begin{cases} -\Delta v = 0 & \text{in } \Omega \\ \frac{\partial v}{\partial \nu} = 0 & \text{on } \partial \Omega \end{cases}$$

$\forall w \in H^1(\Omega)$
 $\int_{\Omega} \nabla v \cdot \nabla w \, dx = 0$

Take $w = v$

$$\Downarrow$$

$$\int_{\Omega} |\nabla v|^2 \, dx = 0$$

$$\Downarrow$$

$$v \equiv \text{const. } c$$

$$\Delta v = 0$$

弱导在一条线上是友有意义

Hw 4

Prob 3

YBD on 1?



$$u_1 \in C^1(\bar{\Omega}_1)$$

$$u_2 \in C^2(\bar{\Omega}_2)$$

$$\text{Claim } \frac{\partial u}{\partial x_i} = \begin{cases} \frac{\partial u_1}{\partial x_i} & \text{in } \Omega_1 \\ \frac{\partial u_2}{\partial x_i} & \text{in } \Omega_2 \end{cases}$$

$$\text{Check: } \forall \varphi \in C_0^\infty(\Omega)$$

$$\int_{\Omega_1} \frac{\partial u_1}{\partial x_i} \varphi \, dx + \int_{\Omega_2} \frac{\partial u_2}{\partial x_i} \varphi \, dx$$

$$= - \int_{\Omega_1} u_1 \frac{\partial \varphi}{\partial x_i} \, dx + \int_{\partial \Omega_1} u_1 \varphi \nu_i \, ds$$

$$- \int_{\Omega_2} u_2 \frac{\partial \varphi}{\partial x_i} \, dx + \int_{\partial \Omega_2} u_2 \varphi (1 - \nu_i) \, ds$$

$$\int \nabla \cdot \vec{F} \, dV = \int \vec{F} \cdot \vec{n} \, ds = - \int_{\Omega} u \frac{\partial \varphi}{\partial x_i} \, dx$$

$$\vec{F} = \varphi \vec{e}_i$$

Drive

$$\int \frac{\partial u_i}{\partial x_i} \, dV = \int_{\partial \Omega} u_i \nu_i \, ds$$

$$\vec{F} = (0, \dots, u_i, \dots)$$

$$\text{再取 } \vec{F} = (0, \dots, u_i, \dots)$$

Newtonian potential for $\int \frac{\partial(u\varphi)}{\partial x_i} \, dV$

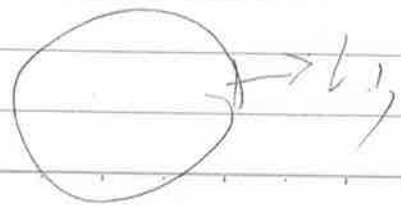
ellipsoid.

$$= \int_{\partial \Omega} u \varphi \nu_i \, ds$$

$$\vec{F} = (0, \dots, u_i, \dots)$$

$$\int_{\Omega} \nabla \cdot \vec{F} \, dx = \int_{\partial \Omega} \vec{F} \cdot \vec{\nu} \, ds$$

$$\vec{F} \cdot \vec{\nu} = u_i \nu_i$$



HW 5

$$1. \text{ (DBVP) } \begin{cases} Lu = f & \text{in } \Omega \\ u|_{\partial\Omega} = \varphi|_{\partial\Omega} \end{cases}$$

$$\text{Pf. (i)} \quad \begin{cases} Lv = L(u - \varphi) = Lu - L\varphi = f - L\varphi & \text{in } \Omega \\ (*) \quad v|_{\partial\Omega} = (u - \varphi)|_{\partial\Omega} = 0 \end{cases}$$

∴ v is a classical sol of $(*)$

$$\forall w \in C_0^\infty$$

$$\langle Lv, w \rangle = \int_{\Omega} - (a_{ij} v_{x_i})_{x_j} w + b_i v_{x_i} w + c v w \, dx$$

$$\star \parallel = \int_{\Omega} a_{ij} v_{x_i} w_{x_j} + b_i v_{x_i} w + c v w \, dx$$

$$\langle f - L\varphi, w \rangle = \int_{\Omega} f w \, dx - \int_{\Omega} a_{ij} \varphi_{x_i} w_{x_j} + b_i \varphi_{x_i} w + c \varphi w \, dx$$

$$\forall w \in H_0^1(\Omega), \exists \{w^k\} \subset C_0^\infty, w^k \rightarrow w \text{ in } H_0^1$$

$$\text{i.e. } w^k \rightarrow w \text{ in } L^2, w_{x_i}^k \rightarrow w_{x_i} \text{ in } L^2$$

$$\langle Lv, w^k \rangle = \langle f - L\varphi, w^k \rangle$$

$$\text{i.e. } \int_{\Omega} a_{ij} v_{x_i} w_{x_j}^k + \dots \, dx = \int_{\Omega} f w^k \, dx - \int_{\Omega} a_{ij} \varphi_{x_i} w_{x_j}^k + \dots \, dx$$

$$\int_{\Omega} a_{ij} v_{x_i} w_{x_j}^k + \dots \, dx - \int_{\Omega} a_{ij} v_{x_i} w_{x_j} + \dots \, dx$$

$$\leq C \cdot \left(\|v\| \cdot \|\nabla w^k - \nabla w\| + \|\nabla v\| \|w^k - w\| + \|v\| \cdot \|w^k - w\|_{L^2} \right) \rightarrow 0$$

$$\int_{\Omega} f(w^k - w) \, dx \leq \int_{\Omega} \|f\| \cdot \|w^k - w\|_{L^2} \, dx$$

2. $b_i \equiv 0$ $c(x) \geq 0$ a.e. on Ω .

(i) homogeneous $\begin{cases} Lu = 0 & \text{in } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$

$$\forall \phi \in H_0^1 \quad \langle Lu, \phi \rangle = \int_{\Omega} a_{ij} u_{x_i} \phi_{x_j} + cu\phi \, dx$$

$$\text{chose } \phi = u \quad \begin{aligned} \langle Lu, u \rangle &= \int_{\Omega} a_{ij} u_{x_i} u_{x_j} + cu^2 \, dx \\ &\geq \lambda_0 \|\nabla u\|_2^2 \geq 0 \end{aligned}$$

By Poincaré ineq.

$$\|u\|_2 \leq C \|\nabla u\| = 0$$

$$\Rightarrow u = 0 \text{ a.e.}$$

u is a trivial sol.

(ii) (DBVP) $\begin{cases} Lu = YBD \\ u|_{\partial\Omega} = 0 \end{cases}$

$$\text{b/c. } \begin{cases} Lu = 0 & \text{on } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

has only trivial sol

By Fredholm Alternativity
(DBVP) has unique sol.

$$3. (NBVP) \quad \begin{cases} Lu = f & \text{in } \Omega \\ \frac{\partial u}{\partial \nu_A} = 0 & \text{on } \partial\Omega \end{cases}$$

$$(i) \quad \forall v \in C^1(\bar{\Omega})$$

$$\langle Lu, v \rangle = \int_{\Omega} \left(- (a_{ij} u_{x_i})_{x_j} v + b_i u_{x_i} v + c u v \right) dx - \int_{\Omega} \nabla (a_{ij} u_{x_i}) \cdot \nabla v \, dx$$

$$= \int_{\Omega} a_{ij} u_{x_i} v_{x_j} + b_i u_{x_i} v + c u v \, dx -$$

$$= \int_{\Omega} f v \, dx$$

$$= \langle f, v \rangle$$

$$\forall v \in H^1(\Omega) = W^{1,2}(\Omega)$$

$C^\infty(\Omega) \cap W^{k,p}$ is dense in $W^{k,p}(\Omega)$

$$\exists \{v_k\} \subset C^\infty(\Omega), \quad v_k \rightarrow v \text{ in } H^1(\Omega)$$

$$\text{i.e. } v_k \rightarrow v \text{ in } L^2, \quad \nabla v_k \rightarrow \nabla v \text{ in } L^2$$

$$\langle Lu, v_k \rangle = \langle f, v_k \rangle$$

$$\int_{\Omega} a_{ij} u_{x_i} v_{k x_j} + \dots \, dx = \int_{\Omega} f v_k \, dx$$

$$\int_{\Omega} \dots v \dots v_k \stackrel{\text{Hölder's}}{\leq} C (\|\nabla u\|_2 \cdot \|v_k - v\| + \dots) \, dx$$

$$\int_{\Omega} \dots \leq C \|f\|_2 \|v_k - v\|$$

$$\Rightarrow \int_{\Omega} a_{ij} u_{x_i} v_{x_j} + \dots \, dx = \int_{\Omega} f v \, dx.$$

13) $u \in C^2(\bar{\Omega})$ satisfies $\odot$

$$\int_{\Omega} \underbrace{a_{ij} u_{x_i} v_{x_j}}_{\in C^1(\bar{\Omega})} + b_i u_{x_i} v + c(x) u v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H^1(\Omega)$$

~~thm~~ take $v \in C_0^\infty(\Omega)$.

we have.
$$\int_{\Omega} -(a_{ij} u_{x_i})_{x_j} v + b_i u_{x_i} v + c u v \, dx = \int_{\Omega} f v \, dx$$

$C_0^\infty(\Omega) \cap H^1(\Omega)$ is dense in $H^1(\Omega)$

$\Rightarrow \exists v_k \in C_0^\infty \rightarrow v$ in $H^1(\Omega)$

$$\int_{\Omega} a_{ij} u_{x_i} v_{x_j} + b_i u_{x_i} v + c u v \, dx$$

$$= \int_{\Omega} (a_{ij} u_{x_i} v)_{x_j} \, dx + \int_{\Omega} -(a_{ij} u_{x_i})_{x_j} v + b_i u_{x_i} v + c u v \, dx$$

$$= \int_{\Omega} v \cdot \frac{\partial u}{\partial \nu_A} \, dx + \cancel{\langle Lu, v \rangle}$$

$$= \int_{\Omega} f v \, dx = \cancel{\langle f, v \rangle}$$

(iii) Define $L: H^1(\Omega) \rightarrow (H^1(\Omega))^*$ by
 $\langle Lu, v \rangle_{(H^1(\Omega))^*, H^1(\Omega)} = \text{LHS of } \textcircled{1}$.
 $\forall u, v \in H^1(\Omega)$

Prove that:

• $\forall u \in H^1(\Omega), Lu \in (H^1(\Omega))^*$

$$\langle Lu, v \rangle = \int_{\Omega} a_{ij} u_{x_i} v_{x_j} + b_i u_{x_i} v + c u v \, dx$$

$$\leq C (\|\nabla u\|_2 \cdot \|\nabla v\|_2 + \|\nabla u\|_2 \cdot \|v\|_2 + \|u\|_2 \cdot \|v\|_2)$$

$$< m (\|\nabla u\|_2 + \|u\|_2) = m \|u\|_{H^1}$$

• $L: H^1(\Omega) \rightarrow (H^1(\Omega))^*$ is linear & bdd.

linear ✓

✓ bddness: $Lu = \int_{\Omega} - (a_{ij} u_{x_i})_{x_j} + b_i u_{x_i} + c u v \, dx$

$$\|Lu\|_{(H^1)^*} = \sup_{\|v\|_{H^1}=1} |\langle Lu, v \rangle|$$

$$\leq C (\|\nabla u\|_2 + \|u\|_2)$$

$$\leq m \|u\|_{H^1}$$

• u is a weak sol of (NBVP) $\Leftrightarrow$

$$Lu = f \text{ in } (H^1)^*$$

$$\forall v \in H^1, \langle Lu - f, v \rangle = \int_{\Omega} a_{ij} u_{x_i} v_{x_j} + \dots + c u v \, dx - \int_{\Omega} f v \, dx$$

①

(iv) Fredholm Alternatives Theorem for (NBVP)
 either (NBVP) $\begin{cases} Lu = 0 & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial\Omega \end{cases}$
 has nontrivial weak sol.

or (NBVP) $\begin{cases} Lu = f \\ \frac{\partial u}{\partial \nu} = 0 \end{cases}$
 has unique weak sol

② if $f \in (H^1)^*$

$$\dim N(L) = \dim N(L^*) < \infty$$

$$L^*u = -(a_{ij}u_{x_j})_{x_i} - b_i u_{x_i} + cu$$

③ (NBVP) has sol iff $\langle f, v \rangle = 0$
 where $\begin{cases} L^*v = 0 \\ \frac{\partial v}{\partial \nu} = 0 \end{cases}$

~~(v) NBVP $\begin{cases} -\Delta u = f & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} \Big|_{\partial\Omega} = 0 \end{cases} \quad f \in L^2(\Omega)$~~

~~$L^* = L = -\Delta$~~

~~$\begin{cases} -\Delta u = 0 \\ \frac{\partial u}{\partial \nu} = 0 \end{cases}$~~

~~has nontrivial sol $u = \text{const. } C$~~

~~(NBVP) has unique sol iff~~

~~$\langle f, v \rangle = C \int_{\Omega} f \, dx = 0$~~

~~$\Rightarrow \|f\|_{L^1} = 0$~~

$$\begin{aligned} 0 \\ \text{(NBVP)} \quad \begin{cases} -\Delta u = f & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} \Big|_{\partial \Omega} = 0 \end{cases} \end{aligned}$$

homo $\begin{cases} -\Delta u = 0 \\ \frac{\partial u}{\partial \nu} \Big|_{\partial \Omega} = 0 \end{cases}$ has nontrivial sol $u = \text{const.}$

$\Rightarrow$ Sol of (NBVP) has no uniqueness

$$L^* = L$$

$$\begin{cases} -\Delta v = 0 \\ \frac{\partial v}{\partial \nu} \Big|_{\partial \Omega} = 0 \end{cases} \Rightarrow v = \text{const.}$$

(NBVP) has sol iff $\langle f, v \rangle = 0$
i.e. $\int f \, dx = 0$

$$\begin{aligned} ② \\ \text{(NBVP)} \quad \begin{cases} -\Delta u + u = f & \text{in } \Omega \\ \frac{\partial u}{\partial \nu} \Big|_{\partial \Omega} = 0 \end{cases} \end{aligned}$$

homo $\begin{cases} -\Delta u + u = 0 \\ \frac{\partial u}{\partial \nu} = 0 \end{cases}$ has unique sol. ($c \leq 0$)
 $u \equiv 0$ C.P.

$\Rightarrow$ (NBVP) has exactly one sol

$$\begin{cases} -\Delta v + v = 0 \\ \frac{\partial v}{\partial \nu} \Big|_{\partial \Omega} = 0 \end{cases} \Rightarrow v \equiv 0 \quad \text{😊}$$

(NBVP) has sol iff $\langle f, v \rangle = 0$
 $\equiv 0$

$\Rightarrow$ NBVP ~~also~~ always
has unique weak sol.

$$\begin{aligned} \text{😊 } 0 &= \langle -\Delta v + v, v \rangle \geq \lambda_0 \|\nabla v\|^2 + c\|v\|^2 \\ &\Rightarrow \|v\|_{H^1} = 0 \quad v = 0 \text{ a.e.} \end{aligned}$$

$$\text{if } u \in W^{1,p}$$

HW 4 Poincaré. $\|u\|_{L^p} \leq \|u\|_{L^p}$

$$1. \quad \|u\|_{L^p} \leq \|u\|_{L^p} = 0$$

$$\Rightarrow u=0 \text{ a.e.}$$

$$2. \quad \Omega = (a, b) \quad u \in W_0^{1,p} \quad 1 < p < \infty$$

(i) u is ~~ab~~ A.C. on $[a, b]$

$$u(a)=0=u(b)$$

Pf. Only need to show $u(x) = \int_a^x u'(t) dt$

$$\text{Define } v(x) = \int_a^x \underset{\substack{\uparrow \\ \text{weak derivative}}}{u'(t)} dt$$

$$u \in W_0^{1,p}$$

C_0^∞ is dense in $W_0^{1,p}$

$$\exists \{u_k\} \subset C_0^\infty \quad u_k \rightarrow u \text{ in } W_0^{1,p}$$

$$\text{i.e. } u_k \rightarrow u \text{ in } L^p, \quad \nabla u_k \rightarrow \nabla u \text{ in } L^p$$

$$\text{Define } u_k(x) = \int_a^x u_k'(t) dt$$

$$\|u_k - v\| \leq \int_a^x |u_k' - u'| dt$$

$$\leq \left(\int_a^b |u_k' - u'|^p dt \right)^{\frac{1}{p}} \cdot \left(\int_a^b \frac{1}{|u_k' - u'|^{p'}} dt \right)^{\frac{1}{p'}}$$

$$u_k \rightarrow v \text{ a.e.}$$

2. $\Omega = (a, b)$ $u \in W_0^{1,p}(\Omega)$ $1 < p < \infty$, show.

(i) u is A.C. on $[a, b]$ $u(a) = 0 = u(b)$

Only need to show $u = \int_a^x u'(t) dt$.

C_0^∞ is dense in $W_0^{1,p}$

$\Rightarrow \exists \{u_k\} \subset C_0^\infty(\Omega)$ s.t. $u_k \rightarrow u$ in $W_0^{1,p}$

i.e. $u_k \rightarrow u$ in L^p

$u'_k \rightarrow u'$ in L^p

$$u_k(x) = \int_a^x u'_k(t) dt$$

$u_k \rightarrow u$ a.e. after passing to a subseq.

$$\int_a^x |u'_k - u'| dt \leq \left(\int_a^b |u'_k - u'|^p \right)^{\frac{1}{p}} \cdot \left(\int_a^b 1^p \right)^{\frac{1}{p'}} \rightarrow 0$$

$$\therefore \int_a^x u'_k dt \rightarrow \int_a^x u' dt \quad \text{a.e.}$$

$$\therefore u = \int_a^x u'(t) dt \quad \text{a.e.}$$

$\Rightarrow u$ is A.C.

$$u(a) = 0$$

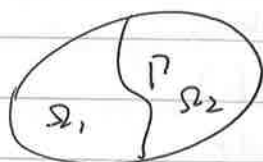
$$u(b) = \lim_{k \rightarrow \infty} \int_a^b u'_k(t) dt$$

$$= \lim_{k \rightarrow \infty} [u_k(b) - u_k(a)]$$

$$= 0$$

$$\begin{aligned} \text{(ii)} \quad u(x) - u(y) &= \int_y^x u'(t) dt \leq \int_a^b |u'(t)| dt \\ &\leq \|u'\|_{L^p} \cdot \|1\|_{L^{p'}} = C \cdot (y-x)^{1-\frac{1}{p}} \end{aligned}$$

3.



$$u = \begin{cases} u_1 & x \in \Omega_1 \\ u_2 & x \in \Omega_2 \end{cases}$$

① (Guess $\frac{\partial u}{\partial x_i} = \begin{cases} \frac{\partial u_1}{\partial x_i} & x \in \Omega_1 \\ \frac{\partial u_2}{\partial x_i} & x \in \Omega_2 \end{cases}$)

$$\langle \frac{\partial u}{\partial x_i}, \varphi \rangle = \int_{\Omega_1} \frac{\partial u_1}{\partial x_i} \varphi \, dx + \int_{\Omega_2} \frac{\partial u_2}{\partial x_i} \varphi \, dx$$

$$\begin{aligned} C_0^\infty(\Omega) &= \int_{\partial\Omega_1} u_1 \varphi \nu_i \, dx - \int_{\Omega_1} u_1 \varphi_{x_i} \, dx \\ &\quad + \int_{\partial\Omega_2} u_2 \varphi \nu_i \, dx - \int_{\Omega_2} u_2 \varphi_{x_i} \, dx \end{aligned}$$

$$= - \left(\int_{\Omega_1} u_1 \varphi_{x_i} \, dx + \int_{\Omega_2} u_2 \varphi_{x_i} \, dx \right)$$

$$= - (u, \varphi)$$

② $\frac{\partial u}{\partial x_i} \in L^p(\Omega) ?$

② C_0^∞ is dense in $W^{1,p}$

$$\exists \{u_k\} \quad u_k \rightarrow u \text{ in } W^{1,p}$$

$$\text{i.e. } u_k \rightarrow u \text{ in } L^p$$

$$u'_k \rightarrow u' \text{ in } L^p$$

$$\frac{\partial u_k}{\partial x_i} \rightarrow \frac{\partial u}{\partial x_i}$$

$$v_k \rightarrow Lu - f \text{ in } L^2$$

$$\int (Lu - f) v_k = 0 \quad \int (Lu - f) (Lu - f - v_k)$$

$$\int (Lu - f) (Lu - f) \leq \|Lu - f\| \cdot \|Lu - f\|$$

4. $u \in W_0^{1,p}(\Omega)$ $v \in W^{1,p'}(\Omega)$

Prove $\int_{\Omega} \frac{\partial u}{\partial x_i} v \, dx = - \int_{\Omega} u \frac{\partial v}{\partial x_i} \, dx$

Pf. ~~$C_0^\infty(\Omega)$ is dense~~

$C_0^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$

$\exists \{u_k\} \subset C_0^\infty(\Omega)$ s.t. $u_k \rightarrow u$ in $W_0^{1,p}$

~~$$\int_{\Omega} \frac{\partial u_k}{\partial x_i} v \, dx = \int_{\Omega} \frac{\partial v}{\partial x_i} u_k \, dx = - \int_{\Omega} v \frac{\partial u_k}{\partial x_i} \, dx$$~~

Def of weak v derivative

$$\int_{\Omega} \frac{\partial v}{\partial x_i} (u - u_k) \, dx \leq \left\| \frac{\partial v}{\partial x_i} \right\|_{p'} \cdot \|u - u_k\|_p \rightarrow 0$$

$$\int_{\Omega} v \left(\frac{\partial u_k}{\partial x_i} - \frac{\partial u}{\partial x_i} \right) \, dx \leq \|v\|_{p'} \cdot \|u_k - u\|_p \rightarrow 0.$$

$$u \in W^{1,p}, \quad v \in C^1(\bar{\Omega})$$

$$3. \quad v \in C^1(\bar{\Omega}) \subset W^{1,p}(\Omega)$$

$$\{v_k\} \in C_0^\infty \quad v_k \rightarrow v$$

$$\frac{\partial(uv_k)}{\partial x_1} = \frac{\partial u}{\partial x_1} v_k + u \frac{\partial v_k}{\partial x_1}$$

$$\left(\int_{C_0^\infty} \cdot, \varphi \right) = \int_{C_0^\infty} \frac{\partial u}{\partial x_1} v_k \varphi \, dx + \int u \frac{\partial v_k}{\partial x_1} \varphi \, dx$$

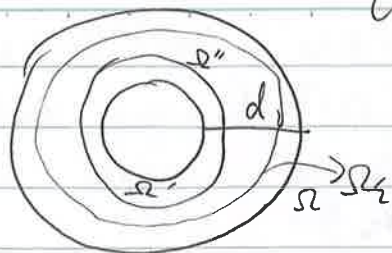
$$= - \int u \frac{\partial(v_k \varphi)}{\partial x_1} + \int u \frac{\partial v_k}{\partial x_1} \varphi \, dx$$

$$= - \int u \frac{\partial \varphi}{\partial x_1} v_k \, dx$$

$$= - (uv_k, \varphi)$$

$$\begin{array}{ccc} \downarrow & & \downarrow \quad \downarrow \\ \frac{\partial uv}{\partial x_1} & = & \frac{\partial u}{\partial x_1} v + u \frac{\partial v}{\partial x_1} \end{array}$$

6.



chose
 $\varepsilon < \frac{d}{3}$

$$\Omega'' = \{x \in \Omega \mid d(x, \Omega') > \varepsilon\}$$

$$\forall x \in \Omega_\varepsilon \setminus \Omega''$$

$$\exists \text{ small } \varepsilon < \varepsilon$$

$$u_\varepsilon(x) = \int_{\mathbb{R}^n} j_\varepsilon(x-y) u(y) dy = 0$$

$$= 0 \text{ on } \mathbb{R}^n \setminus \Omega''$$

$$\Rightarrow \text{supp } u_\varepsilon \subset \Omega''$$

$$u_\varepsilon \in C^\infty(\Omega)$$

$$u_\varepsilon \rightarrow u \text{ in } W^{k,p}(\Omega)$$

$$\Rightarrow u \in W^{k,p}(\Omega)$$

7

$$1. \text{ii)} f \in C'(a, c] \cap C'([c, b))$$

$$Df = f' + (f(c+) - f(c-)) \delta_c$$

$$\text{iii)} \frac{d}{dx} \int_{-\infty}^{+\infty} f(x-y) g(y) dy$$

$$= \int_{-\infty}^{+\infty} Df(x-y) g(y) dy$$

$$= \int_{-\infty}^{+\infty}$$

$$\frac{d}{dx} \int_{-\infty}^{+\infty} = \frac{d}{dx} \int_{-\infty}^{x-c} + \frac{d}{dx} \int_{x-c}^{+\infty}$$

$$f_+ = \begin{cases} f(x) & x < c \\ f(c+) + f'(c+)x & x > c \end{cases}$$

$$f_- = \begin{cases} f(c-) + f'(c-)x & x < c \\ f(x) & x > c \end{cases}$$

$$\psi(x) = \phi(x) - \phi_0(x) \int_a^b \phi(x) dx$$

$$\exists \theta \in \mathcal{D}(a, b)$$

$$\text{s.t. } \theta' = \psi(x)$$

$$\phi(x) = \theta'(x) + \phi_0(x) \int_a^b \phi(x) dx$$

$$\theta(x) \sim (x)$$